

COVARIANT STRETCHING–DIFFUSION RIGIDITY AND GLOBAL REGULARITY FOR THE THREE-DIMENSIONAL NAVIER–STOKES EQUATIONS

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ABSTRACT. We formulate global regularity as a target-relative flux–dissipation argument for the three-dimensional incompressible Navier–Stokes equations. If X is the volume-preserving material flow, $\mathbf{F} = \nabla X$, and $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, then vortex stretching and viscous diffusion are generated by the dual tensors $\mathbf{F}$ and $\mathbf{C}^{-1}$. Amplification in one material direction therefore forces compression, and hence an enhanced diffusive direction, in another. We prove the exact pullback identities expressing this stretching–diffusion duality and identify the corresponding equality geometry.

The global implication is proved conditional on two explicit analytic statements: construction of a compact target-retaining critical orbit from terminal loss, and an exhaustive target-relative alternative for its aggregate current after true exterior input is removed. On the flux-dominated branch, compactness and zero-production rigidity give a uniform positive net dissipation on every retained window, contradicting finiteness on infinitely many disjoint windows. A persistent current which repeatedly regenerates the target is treated separately by retaining the observer, scale, pressure gauge, and oriented current as marks. On the record-bounded row, every finite marked block is realized inside one uniformly bounded root profile. Exact composition multiplies its quotient amplitude by the product of the scale ratios, which tends to zero and contradicts the retained activity floor. Three parts of the proposed self-contained derivation are proved here. Ordered active cylinders admit one exact nonincreasing-scale chart, eliminating positive scale drift. Moreover, invariant averaging on an autonomous recurrent covariant class cancels convection, translation, and pressure and yields $\nu \langle |\nabla V|^2 \rangle - a_* \langle |V - \Pi_0 V|^2 \rangle = 0$; for $a_* \leq 0$ the recurrent class is spatially constant. In addition, the covariant heat scale cancels scalar scale modulation exactly in the Galilean-reduced kinetic–entropy balance, and the strain terms form a nonnegative square. The only unsigned interior term left by this calculation is pressure transfer. The remaining equation-specific input is the finite-stage pressure-transfer and concentration–remainder alternative, including the routing of every failure which occurs before bounded marked realization. No termwise pressure estimate is built into the covariant identity.

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1. INTRODUCTION

Consider the incompressible Navier–Stokes equations on $\mathbb{R}^3$,

$$(1.1) \quad \partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \operatorname{div} u = 0, \quad \nu > 0.$$

Three geometric observations organize the argument. Arnold’s description of Euler flow treats incompressible transport as conservative motion on the group of volume-preserving diffeomorphisms [2, 4]. Onsager’s variational description of irreversible evolution identifies a dissipative current with a gradient in a mobility metric [8]. For Navier–Stokes these statements fit together exactly:

$$(1.2) \quad \underbrace{\partial_t u + \mathbb{P}((u \cdot \nabla)u)}_{\text{Euler-covariant derivative}} = - \underbrace{\nu(-\mathbb{P}\Delta)}_{\text{Stokes mobility}} D\mathcal{E}(u), \quad \mathcal{E}(u) = \frac{1}{2}\|u\|_2^2.$$

The conservative term is orthogonal to $D\mathcal{E}(u) = u$, while the squared metric norm of the right-hand side is $\nu\|\nabla u\|_2^2$. Thus the energy inequality is the energy–dissipation identity associated with (1.2).

The third observation is specific to incompressible vorticity. In material coordinates the deformation tensor $\mathbf{F} = \nabla_a X$ transports vorticity, while viscosity is pulled back by the inverse Cauchy–Green metric $\mathbf{C}^{-1} = (\mathbf{F}^T \mathbf{F})^{-1}$. Since $\det \mathbf{F} = 1$, stretching and enhanced diffusion are dual aspects of the same volume-preserving deformation. This is the structural mechanism developed in Section 4.

The physical Stokes action alone does not exclude concentration because its dissipation on a normalized event at scale r is weighted by r . A normalized gap and a physical lower bound are therefore distinct assertions. In particular, cancellation of fluxes between nested cells cannot by itself replace this factor by a scale-independent constant. The required quantity is instead the net dissipation in a scale-critical localized-energy balance. The proof organizes cross-scale transfer in the following order.

1. A terminal loss of regularity generates a coherent forward orbit in repaired scale–translation coordinates. The orbit is normalized modulo translation, dilation, and pressure gauge. Its scale–center slice and observables remove Galilean modes by fixed-cutoff centering.

2. Every normalized crossing of a shell surrounding the singular target has unit signed progress. Decomposition of the repaired current identifies a persistent positively contributing component without using a dissipation bound.
3. The signed local-energy currents are combined on the coherent orbit. Refinement by a partition of unity leaves the full local-energy balance unchanged. Cancellation of internal convection, pressure, and chart currents requires, in addition, a disjoint spacetime decomposition with compatible oriented flux traces. The resulting outer inward flux is then compared with gross dissipation in the same critical units.
4. Strict flux domination gives positive net dissipation directly. At the sharp equality value, compactness produces a saturated scale-relative limit; the exterior, scale-rigid, and negative-drift reductions force this limit into the zero-Dirichlet equality class, which is spatially constant in the Galilean quotient.
5. If positive critical current instead regenerates active centers at successively smaller scales, the complete successor marks are retained. A finite block is realized inside one record-point root. The terminal quotient activity is at most a constant times the cube of the product scale, and therefore tends to zero, contrary to the fixed activity floor.
6. Compactness therefore gives a uniform positive net dissipation on every target-retaining window. Countable additivity contradicts the finite total net dissipation.

This separation starts from the localized Navier–Stokes energy identity. Its convection, pressure, cutoff-motion, chart, dissipation, and defect terms all depend linearly on the localization test. A compatible parent–child partition therefore prevents double counting but does not, by itself, cancel only the signed terms. Selective cancellation is a statement about oriented normal traces on a disjoint spacetime partition. Viscous dissipation and the suitable-energy defect then retain their sign on the disjoint interiors. Section 11 develops this local energy decomposition entirely within the Navier–Stokes equation and isolates the target-visible zero-dissipation problem.

There is also a direct recurrent formulation of the same mechanism. Ordered active cylinders admit one exact chart with nonincreasing scale, so positive scale drift need not be estimated. On an autonomous compact recurrent limit, covariant spatial averaging removes convection, translation, and every nonzero spatial pressure mode. The remaining zero-mode calculation gives

$$\nu \langle |\nabla V|^2 \rangle - a_* \langle |V - \Pi_0 V|^2 \rangle = 0.$$

For $a_* \leq 0$ this forces spatial constancy and hence loss of the mean-subtracted singular target. This pressure-cancellation argument is proved in Section 6; it is the structural replacement for the compact negative-drift and scale-rigid equality calculations.

The global reduction requires one aggregate PDE alternative. Either the critical inward flux is dominated by dissipation and every saturation sequence is target-null, or the surviving critical current repeatedly regenerates the target. The corrected moving-cutoff energy realizes the first branch with production equal to net dissipation. The marked successor construction in Appendix C.3 excludes the record-bounded value of the second branch by exact finite-block composition, without replacing the rooted event law by a translation-invariant hull. Section 11 proves the abstract flux–dissipation implication. Appendices C–G isolate the remaining equation-specific inputs.

Theorem 1.1 (Covariant dissipation-or-reduction criterion for regularity). *Assume that the canonical critical-orbit hypothesis hypothesis 10.2 and the target-relative current-alternative hypothesis hypothesis 11.7 hold for the class of global finite-energy suitable weak solutions of (1.1). Then every such solution satisfies*

$$\Sigma_u(T) = \emptyset \quad \text{for every finite } T > 0.$$

Consequently every divergence-free datum $u_0 \in H^m(\mathbb{R}^3)$, $m > 5/2$, generates a unique global classical solution, smooth for positive time, and every Leray–Hopf solution with that datum agrees with it.

The proof of [Theorem 1.1](#) is given in [Section 14](#). Its logical content is the conversion of qualitative saturation rigidity into a positive gap for one finite scale-critical net-dissipation measure.

Remark 1.2 (Logical scope). The material pullback identities, monotone scale interpolation, recurrent pressure cancellation, scale-covariant kinetic–entropy identity, moving-cutoff monotonicity, zero-production rigidity, compactness-to-gap argument, and bounded marked-cascade exclusion are proved in this paper. The main theorem remains conditional on the two explicitly marked analytic hypotheses. The remaining input is the finite-stage singular-germ pressure-transfer and concentration–remainder alternative. On the record-point row, bounded finite realization follows from the exact observer maps and [\(C.3\)](#); it is not an additional compactness assumption. Once that row is selected, an infinite regenerated chain is impossible by the product-scale estimate and no translation-invariant probability law is required. Appendices [C–G](#) state the remaining work in equation-specific form. No regularity theorem from another manuscript is an input.

Analytic inputs.

Input	Status in this paper	Self-contained replacement
Weak covariant calculus	proved here	Appendix A
Material stretching–diffusion identities	proved for smooth flow	Appendix B
Canonical critical orbit	analytic hypothesis	Appendix C
Gauge, pressure, and adjoint control	partly proved under stated local estimates	Appendix D
Target-relative current alternative	analytic hypothesis	Appendices F–G
Moving-cutoff realization	conditional on the global radius schedule	Appendix E
Marked regenerated-current reduction	proved here	Appendix C.3
Monotone recurrent rigidity	proved here	Section 6
Target-visible	proved here	Appendix G
zero-dissipation rigidity		

Proof architecture of the current alternative. The diagram records the logical flow of the conditional regularity theorem. Rectangles are live states, the diamond is the exhaustive alternative in [hypothesis 11.7](#), solid ellipses are terminal outputs. First failures of the pressure, exterior, scale, or realization tests leave through the ordered local terminal [4]. Once a record-bounded marked cascade has been selected, its product-scale estimate closes the branch internally at [8].

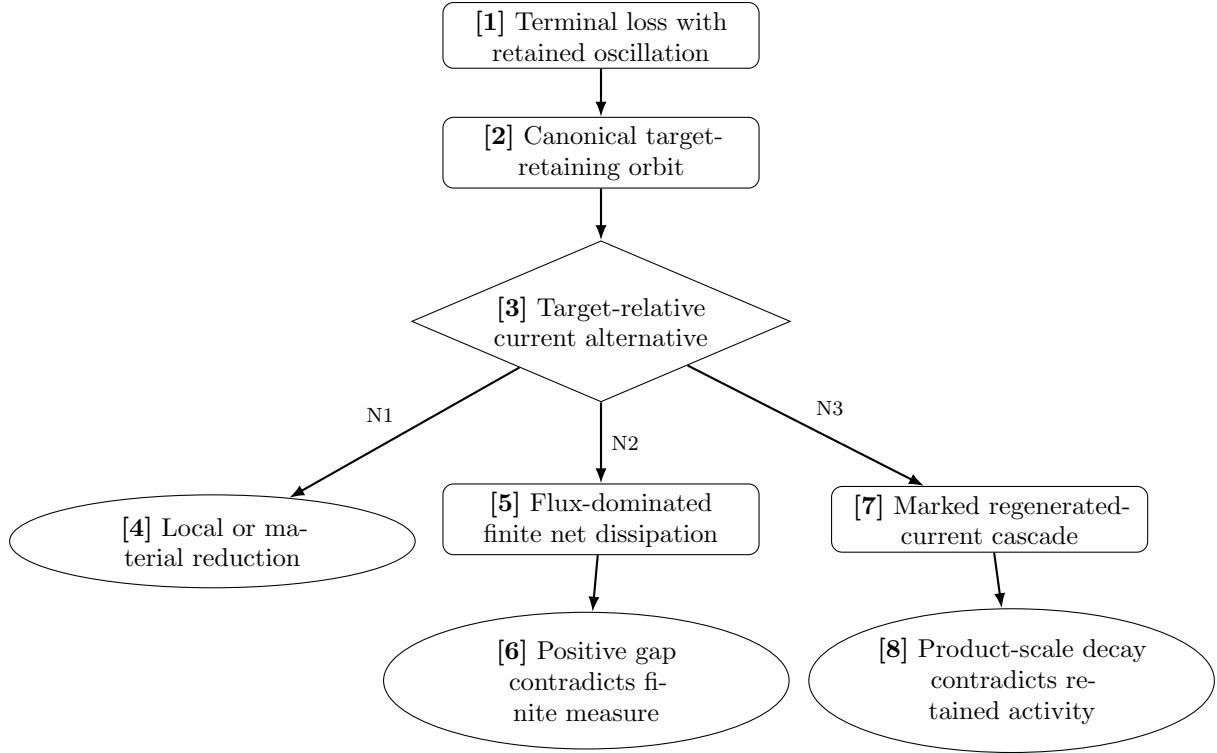
Proof-dependency diagram.

FIGURE 1. The repaired target-relative current alternative. The marked record-bounded branch closes by finite-block scale decay.

Node-by-node audit table.

Node	State or test	Exact source and successor
[1]	Terminal loss	Theorem 2.1 ; continue to [2].
[2]	Compact retained orbit	Hypothesis 10.2 ; continue to [3].
[3]	Exhaustive current split	Hypothesis 11.7 ; outputs [4], [5], [7].
[4]	Local terminal	Alternative (N1); terminal in the conditional theorem.
[5]	Dominated current	Alternative (N2) and Proposition 11.3 ; continue to [6].
[6]	Finite-measure contradiction	Theorems 11.8 and 11.9 ; terminal.
[7]	Regenerated current	Proposition C.16 and Definition C.15 ; continue to [8].
[8]	Bounded-cascade contradiction	Lemmas C.19 to C.21 and Theorem C.22 ; terminal.

The facts carried from [2] through every branch are the fixed oscillation floor, realization by the same physical terminal sequence, compatible pressure gauges, and the suitable local-energy identity. The marked branch additionally carries the observer increments and complete oriented current. Thus neither a subsequence nor a shift limit is permitted to forget the root or its pressure correction.

Branch-closure audit.

Branch	Retained output	Closure used by Theorem 1.1
N1	Local, material, exterior, or critical-tail reduction	The exclusion asserted in alternative (N1); this is part of the stated analytic input.
N2	Finite net-dissipation measure and target-visible window deficit	Theorems 11.8 and 11.9 .
N3	Persistent current supported on regenerated active centers	First failures route to N1 by Proposition C.16 ; the bounded residual is excluded by Theorems C.22 and C.26 .

2. SUITABLE SOLUTIONS AND THE SINGULAR TARGET

2.1. Suitable weak solutions. We use the standard finite-energy suitable class [\[6, 3, 7\]](#). Thus

$$(2.1) \quad u \in L^\infty(0, T; L_\sigma^2(\mathbb{R}^3)) \cap L^2(0, T; \dot{H}_\sigma^1(\mathbb{R}^3)), \quad p \in L_{\text{loc}}^{3/2}(\mathbb{R}^3 \times (0, T)),$$

the equation holds in distributions, the global Leray–Hopf energy inequality holds, and the local energy inequality holds for every nonnegative compactly supported smooth test function. Here the subscript σ denotes divergence-free fields. A terminal point (x_0, T) is regular when u is locally bounded in a backward parabolic neighborhood of that point. We write

$$(2.2) \quad \Sigma_u(T) := \{x_0 \in \mathbb{R}^3 : (x_0, T) \text{ is not regular}\}.$$

For $z_0 = (x_0, t_0)$ and $r > 0$, write $Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0)$ and define

$$(2.3) \quad A(z_0, r) := r^{-1} \operatorname{ess\,sup}_{t_0 - r^2 < t < t_0} \int_{B_r(x_0)} |u(x, t)|^2 dx,$$

$$(2.4) \quad E(z_0, r) := r^{-1} \int_{Q_r(z_0)} |\nabla u|^2 dx dt,$$

$$(2.5) \quad C(z_0, r) := r^{-2} \int_{Q_r(z_0)} |u|^3 dx dt,$$

$$(2.6) \quad D(z_0, r) := r^{-2} \int_{t_0 - r^2}^{t_0} \int_{B_r(x_0)} |p - (p)_{B_r(x_0)}(t)|^{3/2} dx dt.$$

These quantities are invariant under the Navier–Stokes scaling. The pressure mean in D fixes the local pressure gauge.

Theorem 2.1 (Velocity concentration at a singular point). *There is a threshold $\varepsilon_v = \varepsilon_v(\nu) > 0$ such that every $x_0 \in \Sigma_u(T)$ admits radii $r_j \downarrow 0$ for which*

$$(2.7) \quad C((x_0, T), r_j) \geq \varepsilon_v \quad \text{for every } j.$$

Proof. Use the one-scale velocity regularity criterion of Gustafson–Kang–Tsai [\[5\]](#) with spatial and temporal exponents equal to three. In the present normalization it gives a constant $\varepsilon_{\text{reg}} > 0$ such that $C(z_0, r) < \varepsilon_{\text{reg}}$ implies boundedness in $Q_{r/2}(z_0)$. The statement is invariant under [\(7.1\)](#). Therefore a singular point satisfies

$$\limsup_{r \downarrow 0} C((x_0, T), r) \geq \varepsilon_{\text{reg}}.$$

Take $\varepsilon_v = \varepsilon_{\text{reg}}/2$ and choose a sequence converging to zero on which $C \geq \varepsilon_v$. $\square$

Appendix [C.1](#) records the additional pressure normalization and oscillation-retention statements needed by the later compactness construction; these do not enter the preceding concentration lemma.

2.2. Scale-critical concentration and local kinetic energy. The target of the variational interpretation is the set of germs retaining the velocity concentration forced by singularity.

Definition 2.2 (Solution-specific singular concentration target). Let $\varepsilon_v(\nu) > 0$ be the threshold in the velocity ε -regularity theorem and put $\eta_* := \varepsilon_v/2$. For a fixed solution u and terminal time T , let $\mathfrak{S}_{\text{sing},T}(u)$ be the set of pairs $(x_0, \{r_j\}_{j \geq 1})$, with $r_j \downarrow 0$, such that

$$(2.8) \quad C((x_0, T), r_j) \geq \eta_* \quad \text{for every } j.$$

Every $x_0 \in \Sigma_u(T)$ occurs in at least one element of $\mathfrak{S}_{\text{sing},T}(u)$ by [Theorem 2.1](#). Conversely, if u is bounded in a backward neighborhood of (x_0, T) , then $C((x_0, T), r) = O(r^3)$, so no sequence based at x_0 belongs to $\mathfrak{S}_{\text{sing},T}(u)$.

The following estimate explains why localized kinetic probes are sufficient to measure the energy side of this target.

Lemma 2.3 (Local interpolation). *For every suitable weak solution and every interior cylinder $Q_r(z_0)$,*

$$(2.9) \quad C(z_0, r) \leq c(A(z_0, r)^{3/4}E(z_0, r)^{3/4} + A(z_0, r)^{3/2}),$$

where c is universal.

Proof. After translating the cylinder, the Sobolev and interpolation inequalities on B_r give, for almost every time,

$$\int_{B_r} |u|^3 \leq c \left(\int_{B_r} |u|^2 \right)^{3/4} \left(\int_{B_r} |\nabla u|^2 \right)^{3/4} + cr^{-3/2} \left(\int_{B_r} |u|^2 \right)^{3/2}.$$

Integrate over a time interval of length r^2 , apply Hölder's inequality to the first term, and divide by r^2 . Substitution of (2.3) and (2.4) yields (2.9). $\square$

Corollary 2.4 (Concentration requires store or dissipation). *Suppose $C(z_j, r_j) \geq \eta > 0$. Then*

$$\max\{A(z_j, r_j), E(z_j, r_j)\} \geq c\eta^{2/3} > 0.$$

If $E(z_j, r_j) \rightarrow 0$, then $\liminf_j A(z_j, r_j) \geq c\eta^{2/3}$.

Proof. Both assertions follow directly from (2.9). $\square$

Thus a singular target retains local kinetic energy, viscous dissipation, or both. The canonical orbit constructed in [Section 10](#) sharpens this alternative by subtracting the weighted velocity mean. Pressure remains part of the local regularity criterion and of the conservative flux; viscosity supplies the dissipative current.

3. EULER TRANSPORT AND THE STOKES GRADIENT

3.1. The solenoidal Gelfand triple. Let

$$\mathcal{V}_\sigma := \dot{H}_\sigma^1(\mathbb{R}^3), \quad \mathcal{H}_\sigma := L_\sigma^2(\mathbb{R}^3), \quad \mathcal{V}_\sigma^* := \dot{H}_\sigma^{-1}(\mathbb{R}^3).$$

The Stokes operator $A_S : \mathcal{V}_\sigma \rightarrow \mathcal{V}_\sigma^*$ is defined weakly by

$$(3.1) \quad \langle A_S v, \varphi \rangle = \int_{\mathbb{R}^3} \nabla v : \nabla \varphi \, dx.$$

On smooth fields, $A_S = -\mathbb{P}\Delta$. Put

$$(3.2) \quad K := \nu A_S.$$

This is the Riesz isomorphism associated with the Dirichlet form multiplied by ν . For $g, h \in \mathcal{V}_\sigma^*$, define

$$(3.3) \quad (g, h)_{K^{-1}} := \langle g, K^{-1}h \rangle, \quad \|g\|_{K^{-1}}^2 = (g, g)_{K^{-1}}.$$

The projected convection operator is

$$(3.4) \quad B(v, w) := \mathbb{P}((v \cdot \nabla)w), \quad \langle B(v, w), \varphi \rangle = - \int_{\mathbb{R}^3} v \otimes w : \nabla \varphi \, dx.$$

For sufficiently regular divergence-free v ,

$$(3.5) \quad \langle B(v, v), v \rangle = 0.$$

Definition 3.1 (Euler-covariant derivative). Along a divergence-free curve $u(t)$, define

$$(3.6) \quad \mathfrak{D}_t^E u := \partial_t u + B(u, u).$$

This is the material derivative after projecting away the pressure constraint.

Proposition 3.2 (Exact covariant gradient identity). *Let u be a smooth solution of (1.1). For $\mathcal{E}(u) = \frac{1}{2}\|u\|_2^2$,*

$$(3.7) \quad \mathfrak{D}_t^E u = -K D\mathcal{E}(u) = -Ku.$$

If $J_\nu := -Ku$, then

$$(3.8) \quad \|J_\nu\|_{K^{-1}}^2 = \langle Ku, u \rangle = \nu \|\nabla u\|_2^2,$$

and

$$(3.9) \quad \frac{d}{dt} \mathcal{E}(u(t)) = -\|J_\nu(t)\|_{K^{-1}}^2.$$

Proof. Apply $\mathbb{P}$ to (1.1). Since $\mathbb{P}\nabla p = 0$, this gives $\partial_t u + B(u, u) + Ku = 0$, which is (3.7). The definition of the inverse metric gives

$$\|Ku\|_{K^{-1}}^2 = \langle Ku, u \rangle = \nu \int_{\mathbb{R}^3} |\nabla u|^2 \, dx.$$

Pairing (3.7) with u and using (3.5) proves (3.9). $\square$

3.2. The finite energy inequality. For a Leray–Hopf solution, choose its standard weakly continuous representative and set

$$(3.10) \quad M(t) := \mathcal{E}(u(t)) + \nu \int_0^t \|\nabla u(s)\|_2^2 \, ds.$$

The global energy inequality says that M is nonincreasing after changing it on a null set. Its distributional derivative therefore defines a nonnegative Radon measure.

Theorem 3.3 (Energy dissipation for suitable weak solutions). *There is a nonnegative measure μ_{en} on $(0, T)$ such that, at every pair of times $0 \leq t_1 < t_2 < T$ for which the strong energy inequality is represented,*

$$(3.11) \quad \mathcal{E}(u(t_2)) + \int_{t_1}^{t_2} \|J_\nu(t)\|_{K^{-1}}^2 \, dt + \mu_{\text{en}}((t_1, t_2]) = \mathcal{E}(u(t_1)).$$

In particular,

$$(3.12) \quad \int_0^T \|J_\nu(t)\|_{K^{-1}}^2 \, dt \leq \mathcal{E}(u(0)).$$

For every classical solution, $\mu_{\text{en}} = 0$ on each compact interval of classical existence.

Proof. Define $\mu_{\text{en}} := -D_t M$ in distributions. Monotonicity of M gives $\mu_{\text{en}} \geq 0$, and the increment formula for M is exactly (3.11). The bound (3.12) follows by discarding the two nonnegative terminal terms. Smooth solutions satisfy the energy equality (3.9), so their defect measure vanishes. $\square$

Equation (3.11) is the finite dissipation bound used in the rest of the paper. The available quantity is the initial kinetic energy, and irreversible motion is measured by the Stokes action $\|J_\nu\|_{K^{-1}}^2$.

4. MATERIAL DEFORMATION AND STRETCHING–DIFFUSION DUALITY

Let u be smooth on a time interval and let $X(a, t)$ be its material flow,

$$(4.1) \quad \partial_t X(a, t) = u(X(a, t), t), \quad X(a, t_0) = a.$$

Write

$$(4.2) \quad \mathbf{F} := \nabla_a X, \quad \mathbf{C} := \mathbf{F}^T \mathbf{F}, \quad \mathbf{\Omega}(a, t) := \omega(X(a, t), t), \quad \omega := \nabla \times u.$$

The tensor $\mathbf{C}$ is the pullback of the Euclidean metric by the volume-preserving map X . Denote the Eulerian strain by $S = \frac{1}{2}(\nabla u + \nabla u^T)$ and put $S_L = S \circ X$.

Proposition 4.1 (Exact material pullback). *For every smooth solution and every time on which (4.1) is defined,*

$$(4.3) \quad \det \mathbf{F} = 1, \quad \partial_t \mathbf{F} = (\nabla u \circ X) \mathbf{F},$$

$$(4.4) \quad \partial_t \mathbf{C} = 2\mathbf{F}^T S_L \mathbf{F}, \quad \det \mathbf{C} = 1.$$

For every smooth scalar field q ,

$$(4.5) \quad (\Delta_x q) \circ X = \partial_{a_i} (\mathbf{C}^{ij} \partial_{a_j} (q \circ X)),$$

where $(\mathbf{C}^{ij}) = \mathbf{C}^{-1}$. The pulled-back vorticity satisfies

$$(4.6) \quad \partial_t \mathbf{\Omega} = S_L \mathbf{\Omega} + \nu \partial_{a_i} (\mathbf{C}^{ij} \partial_{a_j} \mathbf{\Omega}).$$

Equivalently, the viscous Cauchy invariant $\xi := \mathbf{F}^{-1} \mathbf{\Omega}$ satisfies

$$(4.7) \quad \partial_t \xi = \nu \mathbf{F}^{-1} \partial_{a_i} (\mathbf{C}^{ij} \partial_{a_j} (\mathbf{F} \xi)).$$

Proof. Differentiating (4.1) in a gives the evolution of $\mathbf{F}$. Jacobi's formula and incompressibility give

$$\partial_t \det \mathbf{F} = (\operatorname{div} u) \circ X \det \mathbf{F} = 0,$$

and the initial determinant is one. Differentiating $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ proves (4.4). Formula (4.5) is the Laplace–Beltrami formula with volume density $\sqrt{\det \mathbf{C}} = 1$.

The vorticity equation is

$$(\partial_t + u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta \omega.$$

The antisymmetric part of ∇u annihilates ω , so $(\omega \cdot \nabla) u = S \omega$. Composition with X and (4.5) give (4.6). Finally, $\partial_t \mathbf{F}^{-1} = -\mathbf{F}^{-1} (\nabla u \circ X)$; the inviscid terms cancel after substituting $\mathbf{\Omega} = \mathbf{F} \xi$, which proves (4.7). $\square$

Corollary 4.2 (Material enstrophy balance). *If the fields decay sufficiently at infinity, then*

$$(4.8) \quad \frac{1}{2} \frac{d}{dt} \int_{\mathbb{R}^3} |\mathbf{\Omega}|^2 da + \nu \int_{\mathbb{R}^3} \mathbf{C}^{ij} \partial_{a_i} \mathbf{\Omega} \cdot \partial_{a_j} \mathbf{\Omega} da = \int_{\mathbb{R}^3} \mathbf{\Omega} \cdot S_L \mathbf{\Omega} da.$$

Proof. Pair (4.6) with $\mathbf{\Omega}$, integrate in a , and integrate the divergence term by parts. The material volume element is $da = dx$ because $\det \mathbf{F} = 1$. $\square$

The duality in (4.6) is algebraic. If the singular values of $\mathbf{F}$ are $\sigma_1 \geq \sigma_2 \geq \sigma_3 > 0$, then

$$(4.9) \quad \sigma_1 \sigma_2 \sigma_3 = 1, \quad \lambda(\mathbf{C}^{-1}) = \{\sigma_1^{-2}, \sigma_2^{-2}, \sigma_3^{-2}\}, \quad \sigma_3^{-2} \geq \sigma_1.$$

Thus strong material stretching produces a strongly diffusive compressed direction. This observation alone is not a coercive estimate: the pulled-back vorticity could avoid variation in that direction. The equality mechanism is therefore geometric. Avoiding the enhanced direction requires simultaneous alignment of vorticity, strain, and the material eigenframe. The central rigidity problem is to show that a target-retaining orbit cannot sustain this alignment indefinitely unless it approaches a stretching-depleted, lower-dimensional state.

Definition 4.3 (Materially reduced state). A compact material limit is *materially reduced* on the retained core when its zero-dissipation equations determine a volume-preserving invariant distribution on which the pulled-back vorticity evolves without transverse variation, and the induced equation belongs to an explicitly classified regular lower-dimensional or shear class. Material reduction is a closure alternative for a putative terminal singularity; it does not assert that a regular reduced velocity is spatially constant on a unit normalized window. In the moving-cutoff realization, the stronger zero-dissipation equation $\nabla V = 0$ holds on the retained core and gives the oscillation-removable equality state of Definition 12.7.

The first sentence of Definition 4.3 is an analytic hypothesis, not an automatic consequence of (4.9). The classification required to justify it is stated in Appendix G. This separation prevents the geometric interpretation from being used as an unproved regularity criterion.

5. COVARIANCE UNDER SCALE–TRANSLATION CHARTS

5.1. The repaired equation. Let $X(\tau) \in \mathbb{R}^3$ and $\Lambda(\tau) > 0$ be absolutely continuous, and define

$$(5.1) \quad x = X(\tau) + \Lambda(\tau)y, \quad \frac{dt}{d\tau} = \Lambda(\tau)^2,$$

$$(5.2) \quad u(x, t) = \Lambda(\tau)^{-1}V(y, \tau), \quad p(x, t) = \Lambda(\tau)^{-2}P(y, \tau).$$

Put

$$(5.3) \quad a(\tau) = \frac{\Lambda'(\tau)}{\Lambda(\tau)}, \quad b(\tau) = \frac{X'(\tau)}{\Lambda(\tau)}.$$

Proposition 5.1 (Covariant equation in a repaired chart). *The physical equation (1.1) is equivalent, in distributions on the image of the chart, to*

$$(5.4) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(V + y \cdot \nabla V) + b \cdot \nabla V, \quad \operatorname{div} V = 0.$$

Consequently the chart-covariant derivative

$$(5.5) \quad \mathfrak{D}_\tau^{X, \Lambda} V := \partial_\tau V + B(V, V) - a(V + y \cdot \nabla V) - b \cdot \nabla V$$

satisfies

$$(5.6) \quad \mathfrak{D}_\tau^{X, \Lambda} V = -\nu A_S V.$$

Proof. The change of variables in (5.1)–(5.2) gives the same factor Λ^{-3} in every term of the physical equation. After removing that factor, the time derivative contributes $a(V + y \cdot \nabla V) + b \cdot \nabla V$, giving (5.4). Applying the Helmholtz projection gives (5.6). $\square$

For the centered Type I variables

$$t = -e^{-\tau}, \quad \Lambda(\tau) = e^{-\tau/2}, \quad X = 0,$$

one has

$$(5.7) \quad a = -\frac{1}{2}, \quad b = 0,$$

and hence

$$(5.8) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V - \frac{1}{2}(V + y \cdot \nabla V).$$

This is the standard centered Type I convention. In a repaired Type II chart, the required coefficient regularity is

$$(5.9) \quad a, b \in L^1_{\text{loc}}(d\tau).$$

Bounded modulation is a separate assumption on any compact-window subclass where it is used.

5.2. Covariant energy and pressure. On a whole-space chart, physical kinetic energy and action take the form

$$(5.10) \quad \mathcal{E}(u(t(\tau))) = \frac{\Lambda(\tau)}{2} \int_{\mathbb{R}^3} |V(y, \tau)|^2 dy,$$

$$(5.11) \quad \|J_\nu(t)\|_{K^{-1}}^2 dt = \Lambda(\tau) \nu \int_{\mathbb{R}^3} |\nabla V(y, \tau)|^2 dy d\tau.$$

Proposition 5.2 (Energy identity in moving coordinates). *For a smooth whole-space represented solution,*

$$(5.12) \quad \frac{d}{d\tau} \left(\frac{\Lambda}{2} \|V\|_2^2 \right) = -\Lambda \nu \|\nabla V\|_2^2.$$

Thus (3.9) and (5.12) are the same scalar identity in different coordinates.

Proof. Pair (5.4) with V . Convection, translation, and pressure give zero. Moreover,

$$\int_{\mathbb{R}^3} V \cdot (V + y \cdot \nabla V) dy = -\frac{1}{2} \|V\|_2^2.$$

Since $\Lambda' = a\Lambda$, the derivative of the prefactor in (5.10) cancels this dilation term. The remaining term is the right-hand side of (5.12). $\square$

On compact cylinders, derivatives of the cutoff produce convective, pressure, scale, and translation fluxes in the local energy inequality. Their exact form is recorded in Appendix D; these are the local terms placed explicitly in the moving error density (12.25).

The required fixed- and moving-cutoff bounds are formulated in Sections D.2 and D.4 and must be derived from the suitable local energy inequality on the normalized windows.

Pressure remains the multiplier for incompressibility. On an inner ball use the decomposition

$$(5.13) \quad P = P_{\text{loc}} + H, \quad -\Delta P_{\text{loc}} = \partial_i \partial_j (\zeta V_i V_j), \quad \Delta H = 0,$$

with a spatial mean fixing H . The harmonic term can depend on space and time.

The local Calderón–Zygmund estimate and the harmonic interior estimate, with constants uniform on the normalized compact orbit, are the required conclusion of Section D.2.

Proposition 5.3 (Same-current covariance). *Let two admissible scale–translation charts represent the same physical solution on a common cylinder. The fields, test weights, and Stokes pairings transform by the corresponding pullback, while the physical current*

$$(5.14) \quad J_\nu = -\nu A_S u$$

is unchanged. In particular, (5.11) is invariant after the time change. Optimizing a concentration probe over admissible charts changes the probe but does not change J_ν .

Proof. Both charts are obtained from (5.1) by composing translations and positive dilations. The tensorial change of variables gives (5.10)–(5.11) in each chart. Since both right-hand sides equal the same physical energy and physical dissipation, their values agree. The current (5.14) is determined by u before either chart is selected. $\square$

6. MONOTONE CHARTS AND RECURRENT PRESSURE CANCELLATION

The scale and translation terms in (5.4) have two different roles. Their oscillation along a particular normalization is a choice of coordinates, whereas their limiting value on a recurrent compact class is part of the equation. The first result below removes artificial positive scale motion. The second identifies the recurrent equality class without estimating pressure on individual windows.

6.1. A monotone interpolation through ordered active cylinders.

Proposition 6.1 (Monotone scale interpolation). *Let*

$$J_j = [s_j, t_j], \quad t_j < s_{j+1}, \quad |J_j| = r_j^2, \quad r_{j+1} < r_j,$$

be an infinite ordered family of physical time intervals, and let $x_j \in \mathbb{R}^3$. There are locally absolutely continuous functions

$$\widehat{\Lambda} : (s_1, T) \rightarrow (0, \infty), \quad \widehat{X} : (s_1, T) \rightarrow \mathbb{R}^3,$$

where $T := \lim_j t_j$, with the following properties:

- (i) $\widehat{\Lambda} = r_j$ and $\widehat{X} = x_j$ on J_j ;
- (ii) $\widehat{\Lambda}$ is nonincreasing;
- (iii) after the time change

$$(6.1) \quad \widehat{\tau}(t) := \int_{s_1}^t \widehat{\Lambda}(q)^{-2} dq,$$

the coefficients of the corresponding chart satisfy

$$(6.2) \quad \widehat{a} := \frac{d}{d\widehat{\tau}} \log \widehat{\Lambda} \leq 0 \quad \text{a.e.}, \quad \widehat{b} := \frac{d\widehat{X}/d\widehat{\tau}}{\widehat{\Lambda}} \in L^1_{\text{loc}};$$

- (iv) every J_j is a unit interval in the $\widehat{\tau}$ -clock and $\widehat{\tau}(t) \rightarrow \infty$ as $t \uparrow T$.

On each active interval the represented fields are exactly the fixed-scale parabolic fields

$$(6.3) \quad \widehat{V}_j(y, \sigma) = r_j u(x_j + r_j y, s_j + r_j^2 \sigma), \quad 0 \leq \sigma \leq 1,$$

and therefore satisfy (5.4) with $\widehat{a} = \widehat{b} = 0$ there.

Proof. Set $\widehat{\Lambda} = r_j$ and $\widehat{X} = x_j$ on J_j . On each gap $[t_j, s_{j+1}]$, interpolate $\log \widehat{\Lambda}$ affinely from $\log r_j$ to $\log r_{j+1}$, and interpolate $\widehat{X}$ affinely from x_j to x_{j+1} . The resulting functions are locally absolutely continuous. Since $r_{j+1} < r_j$, the scale is nonincreasing.

The map $t \mapsto \widehat{\tau}(t)$ is strictly increasing and locally absolutely continuous. Its inverse satisfies

$$\frac{dt}{d\widehat{\tau}} = \widehat{\Lambda}^2.$$

Consequently

$$\widehat{a} = \widehat{\Lambda} \frac{d\widehat{\Lambda}}{dt} \leq 0, \quad \widehat{b} = \widehat{\Lambda} \frac{d\widehat{X}}{dt},$$

which proves (6.2). Both coefficients are locally integrable because the interpolants are absolutely continuous and the scale is bounded above and below on every compact subinterval of (s_1, T) .

On J_j , the scale and center are constant, so

$$\widehat{\tau}(t) - \widehat{\tau}(s_j) = \frac{t - s_j}{r_j^2}.$$

Thus J_j has represented length one and the change of variables is exactly (6.3). Every active interval contributes one to (6.1). Since the intervals are disjoint and infinite in number, $\widehat{\tau}(t) \rightarrow \infty$ as $t \uparrow T$. Finally, Proposition 5.1 gives the represented equation. On J_j , both chart derivatives vanish, so $\widehat{a} = \widehat{b} = 0$. $\square$

Remark 6.2 (What monotone interpolation removes). Once the singular extraction has produced chronologically ordered active cylinders with decreasing scales, positive scale drift is unnecessary. It arises only if one insists on continuing a moment normalization through the inactive gaps. The monotone chart preserves the normalized fields on every active cylinder and represents the same physical solution in the gaps, but it does not by itself prove compactness of those gap segments or persistence of the target away from the active cylinders. Those are concentration–compactness questions, not modulation estimates.

6.2. The invariant mean identity. Fix constants $a_* \in \mathbb{R}$ and $b_* \in \mathbb{R}^3$, and consider entire smooth solutions of

$$(6.4) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a_*(V + y \cdot \nabla V) + b_* \cdot \nabla V, \quad \operatorname{div} V = 0.$$

For $c \in \mathbb{R}^3$ and $s \in \mathbb{R}$, define

$$(6.5) \quad (T_c V)(y, \tau) := V(y + e^{-a_* \tau} c, \tau), \quad (T_c P)(y, \tau) := P(y + e^{-a_* \tau} c, \tau),$$

$$(6.6) \quad (\Theta_s V)(y, \tau) := V(y, \tau + s), \quad (\Theta_s P)(y, \tau) := P(y, \tau + s).$$

Direct substitution shows that both maps preserve (6.4) and that

$$(6.7) \quad \Theta_s T_c \Theta_{-s} = T_{e^{-a_* s} c}.$$

They therefore generate the solvable, and hence amenable, group $G_{a_*} = \mathbb{R}^3 \rtimes \mathbb{R}$.

Lemma 6.3 (Pressure does no work outside the spatial zero mode). *Let $\mathcal{M}$ be a compact G_{a_*} -invariant family of bounded smooth entire solutions of (6.4), compact in a local topology in which the velocity and pressure-gradient evaluations below are continuous. Let μ be a G_{a_*} -invariant Borel probability measure on $\mathcal{M}$. In $L^2(\mathcal{M}, \mu)$, put*

$$u_i(V) := V_i(0, 0), \quad p_i(V) := \partial_i P(0, 0),$$

and let Π_0 be the orthogonal projection onto the subspace invariant under all T_c . If $u_i^0 := \Pi_0 u_i$ and $p_i^0 := \Pi_0 p_i$, then

$$(6.8) \quad \int_{\mathcal{M}} u_i p_i d\mu = \int_{\mathcal{M}} u_i^0 p_i^0 d\mu.$$

Proof. Let D_k be the skew-adjoint generator of the unitary action induced by T_{he_k} . Since the translation in (6.5) equals the ordinary translation by he_k at $\tau = 0$,

$$D_k u_i = \partial_k V_i(0, 0), \quad D_i u_i = 0.$$

Taking the divergence of (6.4) gives

$$(6.9) \quad -\Delta P = \partial_i \partial_j (V_i V_j).$$

The scale and translation terms have zero divergence. Hence, with $q_i := (I - \Pi_0)p_i$ and $A := -\sum_k D_k^2$, the commuting spectral resolution of the spatial translation action gives

$$q_k = D_k A^{-1} (I - \Pi_0) D_i D_j (u_i u_j)$$

on the nonzero spatial spectral subspace. More precisely, the right side is the L^2 -limit as $\varepsilon \downarrow 0$ of

$$q_{k,\varepsilon} := D_k (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j (u_i u_j).$$

The pressure curl relation determines the vector from its divergence and excludes an additional nonzero-frequency harmonic part. Skew-adjointness and incompressibility give

$$\int_{\mathcal{M}} u_k q_{k,\varepsilon} d\mu = - \int_{\mathcal{M}} (D_k u_k) (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j (u_i u_j) d\mu = 0.$$

Pass to the L^2 -limit. Since $u_i - u_i^0 \perp \text{ran } \Pi_0$, one also has $\int (u_i - u_i^0) p_i^0 d\mu = 0$. These two identities prove (6.8). Only ∇P occurs, so the conclusion is independent of the additive pressure gauge. $\square$

Theorem 6.4 (Rigidity of a recurrent nonexpanding covariant class). *Let $\mathcal{M}$ be a nonempty compact minimal G_{a_*} -invariant family satisfying the hypotheses of Lemma 6.3. If $a_* \leq 0$, every element of $\mathcal{M}$ is spatially constant. Equivalently, every continuous Galilean-invariant local oscillation observable vanishes on $\mathcal{M}$.*

Proof. Amenability of G_{a_*} gives an invariant probability measure μ on $\mathcal{M}$. Its support is a nonempty closed invariant subset, so minimality makes $\text{supp } \mu = \mathcal{M}$.

The spatially invariant subspace is preserved by time translation because the spatial subgroup is normal by (6.7); hence Π_0 commutes with the time-translation group. Evaluate (6.4) at $(0, 0)$, move every term to the left, and project onto that subspace. Spatial derivatives, including the nonlinearity written as $D_j(u_j u_i)$, vanish after projection. The term $y \cdot \nabla V$ vanishes at $y = 0$. Thus

$$(6.10) \quad \partial_\tau u_i^0 - a_* u_i^0 + p_i^0 = 0 \quad \text{in } L^2(\mathcal{M}, \mu).$$

Pairing with u_i^0 , summing in i , and using invariance under Θ_s gives

$$(6.11) \quad \int_{\mathcal{M}} u_i^0 p_i^0 d\mu = a_* \int_{\mathcal{M}} |u^0|^2 d\mu.$$

Next take the scalar product of the equation at $(0, 0)$ with u and average. The time derivative has zero mean. Integration by parts with the spatial generators gives

$$-\nu \int u \cdot \Delta V(0, 0) d\mu = \nu \int |\nabla V(0, 0)|^2 d\mu.$$

The nonlinear and constant-translation terms are spatial divergences and have zero mean. Using Lemma 6.3 and Eq. (6.11) for the pressure term yields

$$(6.12) \quad \nu \int_{\mathcal{M}} |\nabla V(0, 0)|^2 d\mu - a_* \int_{\mathcal{M}} |u - u^0|^2 d\mu = 0.$$

Here we used that multiplication by the spatially invariant constant a_* commutes with Π_0 . Both terms in (6.12) are nonnegative when $a_* \leq 0$; hence the first vanishes. Full support of μ and

continuity of the gradient evaluation imply $\nabla V(0, 0) = 0$ for every $V \in \mathcal{M}$. The group action moves $(0, 0)$ to every point of spacetime, so $\nabla V \equiv 0$ for every element of $\mathcal{M}$. Such velocities are spatially constant, and every mean-subtracted local oscillation is zero. $\square$

The constant-coefficient formulation is convenient for minimal autonomous hulls, but a monotone scale interpolation naturally produces a stationary law of coefficient paths rather than a single constant. The same identity holds at that level.

Definition 6.5 (Stationary monotone covariant ensemble). A stationary monotone covariant ensemble is a Borel probability space $(\mathcal{M}, \mu)$ of entire smooth represented solutions (V, P, a, b) of (5.4) with the following properties.

- (i) The local velocity, pressure-gradient, and coefficient observables used below are integrable, and the velocity observables are in the domains of the spatial and time generators. This holds, for example, on a compact locally smooth class with a uniform local coefficient bound.
- (ii) The law is invariant under time translation of the four coefficient paths.
- (iii) For $c \in \mathbb{R}^3$, let q_c solve

$$(6.13) \quad q'_c(\tau) = -a(\tau)q_c(\tau), \quad q_c(0) = c,$$

and set

$$T_c(V, P, a, b) := (V(\cdot + q_c(\tau), \tau), P(\cdot + q_c(\tau), \tau), a, b).$$

The class and the law are invariant under every T_c .

- (iv) The scale coefficient satisfies $a \leq 0$ almost everywhere for μ -almost every path.

The integrability in item (i) includes

$$\int_{\mathcal{M}} (|\nabla V(0, 0)|^2 + |a(0)| |V(0, 0)|^2) d\mu < \infty,$$

with time-zero values understood through the stationary measurable representative. Equivalently, one may formulate every identity below after Steklov averaging in time and then pass to almost every time.

Theorem 6.6 (Rigidity of a stationary monotone covariant ensemble). *Let $(\mathcal{M}, \mu)$ be a stationary monotone covariant ensemble. Let $u_i = V_i(0, 0)$, and let u_i^0 be its orthogonal projection in $L^2(\mathcal{M}, \mu)$ onto the spatially invariant observables. Then*

$$(6.14) \quad \nu \int_{\mathcal{M}} |\nabla V(0, 0)|^2 d\mu + \int_{\mathcal{M}} (-a(0)) |u - u^0|^2 d\mu = 0.$$

Consequently $\nabla V = 0$ at every spacetime point for μ -almost every represented solution. In particular, every mean-subtracted local oscillation observable vanishes μ -almost surely.

Proof. Equation (6.13) is precisely the condition that makes T_c preserve (5.4). At $\tau = 0$, its infinitesimal generators are the ordinary spatial derivatives. The pressure Poisson equation remains (6.9), because both modulation fields are divergence free. The spectral proof of Lemma 6.3 therefore gives

$$(6.15) \quad \int u_i p_i d\mu = \int u_i^0 p_i^0 d\mu.$$

Although time translation no longer conjugates T_c by a constant parameter, it preserves every spatial orbit. Indeed,

$$\Theta_s T_c(V, P, a, b) = T_{c_s} \Theta_s(V, P, a, b), \quad c_s = c \exp\left(-\int_0^s a(q) dq\right).$$

Thus the spatially invariant sigma-algebra is time invariant, and its orthogonal projection commutes with time translation. The functions $a(0)$ and $b(0)$ are themselves spatially invariant.

Projecting the equation at $(0, 0)$ onto the spatial zero mode now gives

$$\partial_\tau u_i^0 - a(0)u_i^0 + p_i^0 = 0.$$

Pair with u_i^0 , sum in i , and use stationarity to obtain

$$(6.16) \quad \int u_i^0 p_i^0 \, d\mu = \int a(0)|u^0|^2 \, d\mu.$$

Testing the full equation at the origin against u , averaging, and using spatial and time invariance removes the time derivative, convection, and the b -transport term. Diffusion gives $\nu \int |\nabla V(0, 0)|^2 \, d\mu$. Combining (6.15) and (6.16) gives

$$\nu \int |\nabla V(0, 0)|^2 \, d\mu - \int a(0)(|u|^2 - |u^0|^2) \, d\mu = 0.$$

Multiplication by the spatially invariant observable $a(0)$ preserves the zero-mode subspace. Orthogonality therefore gives

$$\int a(0)(|u|^2 - |u^0|^2) \, d\mu = \int a(0)|u - u^0|^2 \, d\mu,$$

which proves (6.14). Both terms are nonnegative. Hence the gradient vanishes almost surely at the origin. Time and covariant spatial invariance propagate this conclusion to every spacetime point. The Steklov formulation mentioned in Definition 6.5 justifies the same calculation when the coefficient paths are merely locally integrable. $\square$

Corollary 6.7 (No target-retaining recurrent limit). *Let $\mathcal{K}$ be a compact invariant class of autonomous limits with $a_* \leq 0$, and let $\mathcal{O}$ be a continuous nonnegative observable on $\mathcal{K}$ which vanishes on spatially constant velocities. If every minimal invariant component of $\mathcal{K}$ retains $\mathcal{O} \geq \eta > 0$, then $\mathcal{K}$ is empty.*

Proof. Every nonempty compact invariant class contains a compact minimal invariant component. Apply Theorem 6.4 on that component. It gives $\mathcal{O} = 0$, contrary to the retained lower bound. $\square$

The preceding rigidity statement becomes quantitative on a compact successor class through the following elementary ergodic lemma. This is the point at which recurrence replaces a list of increasingly sharp estimates.

Lemma 6.8 (Uniform block deficit or a saturation law). *Let K be a compact metric space, let $S : K \rightarrow K$ be continuous, and let $h \in C(K)$. Write $\mathcal{P}_S(K)$ for the set of S -invariant Borel probability measures and put*

$$(6.17) \quad \beta := \max_{\mu \in \mathcal{P}_S(K)} \int_K h \, d\mu.$$

Then exactly one of the following alternatives holds.

(i) $\beta < 0$, and there exist $N \geq 1$ and $\delta > 0$ such that

$$(6.18) \quad \sum_{k=0}^{N-1} h(S^k w) \leq -\delta \quad (w \in K);$$

(ii) $\beta \geq 0$, and an invariant probability measure μ_* attains β .

In particular, if a PDE balance gives $\int h \, d\mu \leq 0$ for every invariant measure and every equality law is target-null, while a continuous target observable satisfies $\mathcal{O} \geq \eta > 0$ on K , then alternative (i) holds.

Proof. The set $\mathcal{P}_S(K)$ is nonempty and weak-* compact, and the map $\mu \mapsto \int h \, d\mu$ is continuous. Hence the maximum in (6.17) is attained.

Assume $\beta < 0$ and that (6.18) fails for every N . Then for each n there is $w_n \in K$ such that

$$\sum_{k=0}^{n-1} h(S^k w_n) > -n^{-1}.$$

For the empirical measures

$$\mu_n := \frac{1}{n} \sum_{k=0}^{n-1} \delta_{S^k w_n}$$

one consequently has $\liminf_n \int h \, d\mu_n \geq 0$. Pass to a weak-* convergent subsequence. For every $f \in C(K)$,

$$\int (f \circ S - f) \, d\mu_n = \frac{f(S^n w_n) - f(w_n)}{n} \longrightarrow 0,$$

so the limit μ is S -invariant. It satisfies $\int h \, d\mu \geq 0$, contradicting $\beta < 0$. Thus a block length N exists; compactness of K makes the negative maximum of the continuous block sum a number $-\delta < 0$, proving (6.18).

For the final assertion, an invariant measure with zero average would be supported on the target-null equality class by the stated PDE rigidity, while $\mathcal{O} \geq \eta$ on all of K . Hence equality is impossible and the maximal invariant average is strictly negative. $\square$

Corollary 6.9 (Recurrent equality is the only zero-gap obstruction). *Let a compact target-retaining successor class have a continuous signed local-energy excess h , and suppose its finite block sums are the exact aggregate inward feeding minus viscous and suitable-defect production, up to bounded endpoint storage and true exterior input of sublinear block growth. If every invariant saturation law is realized either as a stationary monotone covariant ensemble in the sense of Definition 6.5 or as a zero-Dirichlet ordinary window, then the class has a uniform strictly dissipative block deficit.*

Proof. Signed additivity removes the internal currents before the invariant average is taken. The endpoint storage and exterior input vanish after division by the block length, so every invariant law has nonpositive mean excess. In the equality case, Theorem 6.6 makes the first realization spatially constant, while Theorem 12.8 makes the second oscillation-removable. Both contradict the retained target floor. Apply Lemma 6.8. $\square$

Remark 6.10 (Interface with noncompact extraction). The theorem closes an *ordinary compact recurrent* scale-collapse or scale-rigid limit, including the limiting case $a_* = 0$. It does not turn a rough, exterior, diffuse, or noncompact sequence into such a class. To use it in the global proof one must still show, directly from the suitable local energy inequality and pressure reconstruction, that repeated target-retaining descendants either lose the target, enter a separately removable exterior case, possess a compact autonomous recurrent limit to which the theorem applies, or form the marked regenerated-current cascade of Definition C.15. The last case is treated by Theorem C.26 and does not require a covariant hull retaining a fixed root.

7. SCALE-CRITICAL QUOTIENT GEOMETRY

The physical energy identity and the scale-critical concentration problem live at different levels. This section separates them and fixes the quotient structure used by the critical orbit.

7.1. Physical action and critical action. The Navier–Stokes scaling based at (x_0, t_0) is

$$(7.1) \quad u^{(\lambda)}(y, s) = \lambda u(x_0 + \lambda y, t_0 + \lambda^2 s), \quad p^{(\lambda)}(y, s) = \lambda^2 p(x_0 + \lambda y, t_0 + \lambda^2 s).$$

The quantities C, D are invariant under (7.1). The localized Dirichlet quantity is invariant after the parabolic factor is included:

$$(7.2) \quad r^{-1} \int_{Q_r(z_0)} |\nabla u|^2 dx dt = \int_{Q_1} |\nabla u^{(r)}|^2 dy ds.$$

This is the dissipation seen by a normalized unit window.

By contrast, the physical Stokes action transforms according to (5.11) and carries the factor Λ :

$$(7.3) \quad \int_{t(\tau_1)}^{t(\tau_2)} \nu \|\nabla u(t)\|_2^2 dt = \int_{\tau_1}^{\tau_2} \Lambda(\tau) \nu \|\nabla V(\tau)\|_2^2 d\tau.$$

Consequently, finite physical action does not control

$$(7.4) \quad \int_{\tau_1}^{\tau_2} \nu \int_{B_R} |\nabla V|^2 dy d\tau$$

uniformly along a collapsing orbit. For example, in centered Type I coordinates $\Lambda = e^{-\tau/2}$. A represented orbit with order-one Dirichlet dissipation on every unit interval has finite contribution to (7.3) but infinite contribution to (7.4). The global energy inequality alone therefore cannot supply the required contradiction.

The corrected moving-corrected localized energy is designed precisely at the level of (7.4). Its lower bound replaces the missing unweighted global bound in represented time. The production gap then says that a retained core must spend a fixed amount of this critical action, or of the other covariant production terms, on every target-retaining window.

7.2. Galilean and pressure quotients. For a constant vector $c \in \mathbb{R}^3$, the Galilean transform

$$(7.5) \quad u^c(x, t) := u(x - ct, t) + c, \quad p^c(x, t) := p(x - ct, t)$$

preserves (1.1). Addition of a function $\pi(t)$ to the pressure also preserves the equation. These modes have no bearing on regularity but can change an uncentered local kinetic integral. The variance (9.6) is invariant under the induced constant-velocity action:

$$(7.6) \quad \frac{1}{2} \int m |(V + c) - \overline{(V + c)}_m|^2 = \frac{1}{2} \int m |V - \bar{V}_m|^2.$$

For $c \neq 0$, (7.5) does not preserve the global finite-energy class on $\mathbb{R}^3$. It is used only as a symmetry of local profiles and observables; the physical finite-energy solution and its Stokes current are never replaced by u^c . Likewise, D and the pressure atlas depend only on the pressure class modulo functions of time.

Fix nonzero nonnegative radial cutoffs $\zeta_0, \zeta_1 \in C_c^\infty(B_R)$, a level $\Theta_0 > 0$, and write $V_{\lambda, q}(y) = \lambda V(q + \lambda y)$. For a nonzero cutoff ξ , set

$$(7.7) \quad \langle W \rangle_\xi := \frac{\int \xi W dy}{\int \xi dy}, \quad W_\xi^\circ := W - \langle W \rangle_\xi.$$

Translation and dilation are fixed by the four-component, Galilean-invariant gauge map

$$(7.8) \quad \begin{aligned} \mathfrak{F}_0(\lambda, q; V) &:= \int \zeta_0 |(V_{\lambda, q})_{\zeta_0}^\circ|^3 dy - \Theta_0, \\ \mathfrak{F}_j(\lambda, q; V) &:= \int y_j \zeta_1 |(V_{\lambda, q})_{\zeta_1}^\circ|^2 dy, \quad 1 \leq j \leq 3. \end{aligned}$$

Indeed, replacing V by $V + c$ adds the constant λc to $V_{\lambda,q}$, which is removed by both means in (7.7). Thus the gauge selects only scale and spatial center; it does not select a velocity representative. At a zero for which

$$(7.9) \quad D_{(\lambda,q)} \mathfrak{F}(1, 0; V) \in \mathbb{R}^{4 \times 4} \quad \text{is invertible,}$$

the implicit-function theorem selects the scale and center locally. Differentiating the four constraints produces (a, b) . Degeneration of (7.9) is an explicit alternative in the critical-orbit hypothesis; no smooth continuation through a degenerate point is assumed.

7.3. The quotient topology. On a fixed represented cylinder $B_R \times I$, put

$$(7.10) \quad \mathcal{X}_R(I) := (L_I^\infty L^2(B_R) \cap L_I^2 H^1(B_R) \cap L^3(B_R \times I)) \times L^{3/2}(B_R \times I) \times L^1(I; \mathbb{R}^4),$$

where the factors represent V , P , and (a, b) . The compactness topology consists of strong local L^3 convergence for velocity, weak local $L^2 H^1$ convergence for its gradient, strong local $L^{3/2}$ convergence for the normalized pressure, and L^1 convergence of modulation coefficients on the selected windows. Adjoint densities and cutoffs are added with the topology in (A.7) and (A.6).

Let $\mathcal{G}_{\text{ch}}$ denote the local chart action generated by translations, positive dilations, and pressure gauge. Near a zero satisfying (7.9), the normalized slice (10.3), together with (10.4), intersects the corresponding $\mathcal{G}_{\text{ch}}$ -orbit locally and transversely. Global continuation and consistency of these local slices are part of [hypothesis 10.2](#). Constant-velocity Galilean symmetry is not fixed by a time-dependent mean condition; instead, every scalar used below is invariant under that symmetry by construction. A compact family of normalized windows may therefore be studied on the chart slice and then quotiented by the residual Galilean action.

Proposition 7.1 (Covariance of the reduced data). *On the nondegenerate normalized slice, the following objects depend only on the quotient state:*

- (i) *the oscillatory kinetic energy $\mathcal{E}_\Gamma^\circ$ and concentration $\mathcal{C}_\Gamma^\circ$;*
- (ii) *the physical Stokes current and its inverse-metric norm;*
- (iii) *the ray distance after the target probe is pulled back with the chart;*
- (iv) *the normalized pressure class and, for a fixed padded terminal problem, the localized adjoint probability measure $m \, dy$.*

Proof. The first item follows from change of variables and (7.6). The second and third follow from [Proposition 5.3](#) and covariance of orthogonal projection in the Stokes metric. Pressure means are removed by (10.4). For a fixed padded terminal problem, the normalized adjoint equation pulls back covariantly when its domain, cutoff, and terminal datum are pulled back with the chart, and normalization fixes the mass. No comparison between distinct terminal problems is asserted or needed for the moving-corrected localized energy. $\square$

The proposition does not supply compactness or a summable moving-radius schedule. Those are the contents of [hypothesis 10.2](#) and [hypothesis 12.5](#), respectively.

8. BACKWARD ADJOINT LOCALIZATIONS AND ENTROPY

The adjoint density supplies scale-relative localization. Its entropy is kept separate from the kinetic-energy gradient identity; this separation prevents pressure and critical norms from acquiring artificial variational equations.

8.1. The normalized finite-horizon adjoint problem. Fix a repaired cylinder $B_R \times (\tau_0, \sigma)$, let

$$(8.1) \quad c(y, \tau) := V(y, \tau) - a(\tau)y - b(\tau), \quad \operatorname{div} c = -3a,$$

and choose $\chi \in C_c^\infty(B_R)$, $\chi \geq 0$, together with a nonnegative terminal datum $h \in C_c^\infty(B_R)$, $h \not\equiv 0$. First solve the linear terminal-boundary problem

$$(8.2) \quad -\partial_\tau \tilde{\rho} - \nu \Delta \tilde{\rho} - \operatorname{div}(c \tilde{\rho}) = 0, \quad \tilde{\rho}|_{\partial B_R} = 0, \quad \tilde{\rho}(\sigma) = h.$$

Define

$$(8.3) \quad Z(\tau) := \int_{B_R} \chi^2 \tilde{\rho} \, dy, \quad \rho := \frac{\tilde{\rho}}{Z}, \quad m := \chi^2 \rho.$$

We work on intervals on which $Z > 0$. Then $m \geq 0$ and

$$(8.4) \quad \int_{B_R} m(y, \tau) \, dy = 1.$$

For a retained unit window $I = [s, s+1]$, the adjoint problem is always solved on a padded interval ending at

$$(8.5) \quad \sigma_I = s+1 + \kappa_*, \quad \kappa_* > 0.$$

Consequently

$$(8.6) \quad \kappa_* e^{-2M_*} \leq \theta_I(\tau) \leq (1 + \kappa_*) e^{2M_*} \quad (\tau \in I).$$

Here M_* is a uniform bound for $\int_s^{\sigma_I} |a(r)| \, dr$ on the normalized padded windows, and θ_I is the unique solution of the covariant terminal-value problem

$$(8.7) \quad \theta'_I + 2a\theta_I = -1, \quad \theta_I(\sigma_I) = 0.$$

Indeed,

$$(8.8) \quad \theta_I(\tau) = \int_\tau^{\sigma_I} \exp\left(2 \int_\tau^r a(q) \, dq\right) \, dr,$$

which proves (8.6). The Gaussian entropy is evaluated only on the retained window, where $\theta_I > 0$, not at the terminal time. It is a covariant diagnostic on each padded window; the global monotonicity balance constructed below does not compare or reset finite-horizon adjoint densities.

Lemma 8.1 (Normalized adjoint equation). *Set*

$$(8.9) \quad \kappa(\tau) := \frac{Z'(\tau)}{Z(\tau)} = \int_{B_R} (-\nu \Delta \chi^2 + c \cdot \nabla \chi^2) \rho \, dy.$$

Then

$$(8.10) \quad -\partial_\tau \rho - \nu \Delta \rho - \operatorname{div}(c \rho) = \kappa \rho$$

in distributions, and (8.4) holds at every time at which the weak solution is represented continuously in L^2 .

Proof. Differentiate Z using (8.2) and integrate by parts. The support of χ is contained in B_R , so

$$Z' = \int_{B_R} (-\nu \Delta \chi^2 + c \cdot \nabla \chi^2) \tilde{\rho} \, dy.$$

Division by Z gives (8.9). Substituting $\tilde{\rho} = Z\rho$ into (8.2) gives (8.10). The last statement follows from the definition of Z . $\square$

This construction solves the linear equation before normalization. The coefficient κ is therefore determined by the solution and does not enter the linear terminal problem.

Lemma 8.2 (Weak adjoint construction). *Assume*

$$(8.11) \quad V \in L_\tau^2 H_y^1(B_R), \quad a, b \in L_\tau^1, \quad \operatorname{div} c = -3a.$$

Then (8.2) has a nonnegative weak solution with

$$\tilde{\rho} \in L_\tau^\infty L_y^2(B_R) \cap L_\tau^2 H_{0,y}^1(B_R).$$

It is obtained as the weak limit of solutions with smooth drifts c_ε . On every compact subinterval where $Z \geq z_0 > 0$, the normalized densities satisfy the same bounds, and $\sqrt{\rho} \in L_\tau^2 H_{\text{loc},y}^1$ whenever their Fisher information is uniformly bounded.

Proof. Reverse time in (8.2) and approximate V, a, b by smooth functions preserving $\operatorname{div} c_\varepsilon = -3a_\varepsilon$. Galerkin approximation gives smooth weak solutions. Testing by $\tilde{\rho}_\varepsilon$ makes the transport contribution $-\frac{1}{2} \int (\operatorname{div} c_\varepsilon) |\tilde{\rho}_\varepsilon|^2$, so Gronwall's inequality uses only $a_\varepsilon \in L^1$. This yields uniform $L^\infty L^2 \cap L^2 H_0^1$ bounds. Choose the approximating drifts to converge strongly in $L^2 L^6 + L^1 L^\infty$. The first summand follows from $V \in L^2 H^1$, and the affine part belongs to the second summand on B_R . The bounds on $\tilde{\rho}_\varepsilon$ then allow $c_\varepsilon \tilde{\rho}_\varepsilon$ to pass to the limit against every smooth test function. Weak compactness gives a solution of (8.2); testing the negative part proves nonnegativity. Division by $Z \geq z_0$ preserves the estimates. Finally,

$$\int \frac{|\nabla \rho|^2}{\rho + \varepsilon} dy = 4 \int |\nabla \sqrt{\rho + \varepsilon}|^2 dy,$$

and weak lower semicontinuity gives the last assertion as $\varepsilon \downarrow 0$. $\square$

8.2. Transport identity and the Gaussian equality model. For smooth positive ρ , write

$$(8.12) \quad \rho = (4\pi\nu\theta)^{-3/2} e^{-f}, \quad \theta > 0.$$

Equation (8.10) gives

$$(8.13) \quad \partial_\tau f + \nu \Delta f - \nu |\nabla f|^2 + c \cdot \nabla f = \kappa - \frac{3\theta'}{2\theta} + \operatorname{div} c.$$

Lemma 8.3 (Localized adjoint transport identity). *For every smooth scalar F ,*

$$(8.14) \quad \begin{aligned} \frac{d}{d\tau} \int_{B_R} \chi^2 F \rho dy &= \int_{B_R} \chi^2 (\partial_\tau F - \nu \Delta F + c \cdot \nabla F) \rho dy \\ &\quad + \int_{B_R} [F(-\nu \Delta \chi^2 + c \cdot \nabla \chi^2) - 2\nu \nabla \chi^2 \cdot \nabla F] \rho dy \\ &\quad - \kappa \int_{B_R} \chi^2 F \rho dy. \end{aligned}$$

Proof. Differentiate the integral, substitute (8.10), and integrate the Laplacian and divergence terms by parts. All spatial boundary terms vanish because $\chi \in B_R$. $\square$

The three lines of (8.14) separate interior evolution, cutoff transport, and normalization. Taking

$$(8.15) \quad W_f := \nu \theta |\nabla f|^2 + f - 3, \quad \mathcal{W}_{\text{ad},\chi}(\tau) := \int_{B_R} \chi^2 W_f \rho dy,$$

gives the exact localized derivative by setting $F = W_f$ in (8.14) and using (8.13). In particular, derivatives of χ and the scalar κ remain visible in the last two lines. The velocity drift contributes

through ∇c , and its compression contributes through $\operatorname{div} c = -3a$. No sign is assigned to those local terms without an additional estimate.

The equality model is most transparent without localization.

Proposition 8.4 (Adjoint entropy production with a variable heat scale). *Let ρ be a smooth positive probability density on $\mathbb{R}^3$ satisfying*

$$-\partial_\tau \rho - \nu \Delta \rho - \operatorname{div}(c\rho) = 0$$

with sufficient decay. With f as in (8.12), for every positive absolutely continuous θ one has

$$\begin{aligned} \frac{d}{d\tau} \int_{\mathbb{R}^3} (\nu\theta |\nabla f|^2 + f - 3)\rho \, dy &= 2\nu^2\theta \int_{\mathbb{R}^3} \left| \nabla^2 f - \frac{1}{2\nu\theta} I \right|^2 \rho \, dy \\ &\quad + \int_{\mathbb{R}^3} (\operatorname{div} c)\rho \, dy \\ &\quad + 2\nu\theta \int_{\mathbb{R}^3} (\nabla f \cdot \nabla \operatorname{div} c - (\nabla f)^T (\nabla c) \nabla f) \rho \, dy \\ &\quad + (\theta' + 1) \left(\nu \int_{\mathbb{R}^3} |\nabla f|^2 \rho \, dy - \frac{3}{2\theta} \right). \end{aligned} \tag{8.16}$$

For $c = 0$ and $\theta' = -1$, the derivative is nonnegative and vanishes precisely when $\nabla^2 f = (2\nu\theta)^{-1}I$ on the support of ρ .

Proof. Let $H(\rho) = -\int \rho \log \rho$ and $I(\rho) = \int |\nabla \log \rho|^2 \rho$. Direct differentiation with the backward adjoint equation and integration by parts give

$$\frac{dH}{d\tau} = -\nu I + \int (\operatorname{div} c)\rho$$

and

$$\frac{dI}{d\tau} = 2\nu \int |\nabla^2 \log \rho|^2 \rho - 2 \int (\nabla \log \rho \cdot \nabla \operatorname{div} c + (\nabla \log \rho)^T (\nabla c) \nabla \log \rho) \rho.$$

Differentiate

$$\nu\theta I + H - \frac{3}{2} \log(4\pi\nu\theta) - 3$$

in τ , use $\nabla \log \rho = -\nabla f$, and first collect the terms corresponding to $\theta' = -1$. Completing their square gives the first three lines of (8.16). The difference between general θ' and -1 is exactly its last line. $\square$

For the repaired Navier–Stokes drift $c = V - ay - b$, one has $\nabla \operatorname{div} c = 0$ and $\nabla c = \nabla V - aI$. Hence the choice (8.7) cancels every scalar scale term in (8.16) and gives

$$\begin{aligned} \frac{d}{d\tau} \mathcal{W}_{\text{ad}} &= 2\nu^2\theta \int_{\mathbb{R}^3} \left| \nabla^2 f - \frac{1}{2\nu\theta} I \right|^2 \rho \, dy \\ &\quad - 2\nu\theta \int_{\mathbb{R}^3} (\nabla f)^T (\nabla V)_{\text{sym}} \nabla f \rho \, dy. \end{aligned} \tag{8.17}$$

Here $\mathcal{W}_{\text{ad}}$ denotes the whole-space entropy in Proposition 8.4, and $(\nabla V)_{\text{sym}} := \frac{1}{2}(\nabla V + (\nabla V)^T)$. Formula (8.17) isolates the positive Gaussian square from velocity strain. The translation coefficient b drops out because it has no spatial derivative, while a drops out because the heat scale is transported by (8.7). The strain term has no sign by itself, but it completes a square with viscous production below.

In the unmodulated case $a = b = 0$, for which $\theta' = -1$, the Euclidean heat kernel is

$$(8.18) \quad \rho_*(y, \tau) = (4\pi\nu\theta)^{-3/2} \exp\left(-\frac{|y|^2}{4\nu\theta}\right), \quad f_* = \frac{|y|^2}{4\nu\theta},$$

and satisfies

$$\nabla^2 f_* = \frac{1}{2\nu\theta} I, \quad \int_{\mathbb{R}^3} (\nu\theta |\nabla f_*|^2 + f_* - 3) \rho_* \, dy = 0.$$

Hence both the entropy and its production vanish. This is the Euclidean normalization of Perelman's $\mathcal{W}$ -entropy [9].

For weak densities, the Fisher term is defined by

$$(8.19) \quad \int \chi^2 \frac{|\nabla \rho|^2}{\rho} \, dy := 4 \int \chi^2 |\nabla \sqrt{\rho}|^2 \, dy,$$

with value $+\infty$ when $\sqrt{\rho} \notin H_{\text{loc}}^1$. The entropy uses $0 \log 0 = 0$. At weak regularity the Fisher information has its usual lower-semicontinuous extension. The Shannon term in $\mathcal{W}_{\text{ad}, \chi}$ has the opposite semicontinuity orientation, so its passage to a limit requires an explicit strong-convergence and uniform-integrability argument, as recorded in Section D.3; no blanket lower-semicontinuity claim is made for the full entropy. No pointwise equation for $f = -\log \rho + \text{constant}$ is asserted on the zero set of ρ . Smooth positive adjoint approximations justify the localized transport formulae needed below.

9. STOKES-METRIC PROJECTION ONTO LOCALIZED KINETIC VARIATIONS

9.1. Scale-critical kinetic probes. On a compact represented-time interval I_τ , an admissible probe consists of a repaired scale-translation chart (X, Λ) and a nonnegative function

$$(9.1) \quad m \in W^{1,1}(I_\tau; W^{1,\infty}(B_R)) \cap L^\infty(I_\tau; W^{2,\infty}(B_R)),$$

$$\text{supp } m(\cdot, \tau) \subset B_{R_0} \quad (0 < R_0 < R), \quad \int_{B_R} m(y, \tau) \, dy = 1.$$

The smooth adjoint weights $m = \chi^2 \rho$ from Section 8, and regularized weak adjoint weights with the same normalization, are the principal examples. The physical multiplier is

$$(9.2) \quad w_\Gamma(x, t(\tau)) := \Lambda(\tau)^{-1} m\left(\frac{x - X(\tau)}{\Lambda(\tau)}, \tau\right).$$

Its physical mass and weighted velocity mean are

$$(9.3) \quad M_\Gamma(t) := \int_{\mathbb{R}^3} w_\Gamma \, dx = \Lambda(\tau)^2, \quad \bar{u}_\Gamma(t) := M_\Gamma(t)^{-1} \int_{\mathbb{R}^3} w_\Gamma u \, dx.$$

The Galilean-reduced localized kinetic probe is

$$(9.4) \quad F_\Gamma(u(t)) := \frac{1}{2} \int_{\mathbb{R}^3} w_\Gamma(x, t) |u(x, t) - \bar{u}_\Gamma(t)|^2 \, dx.$$

On the represented cylinder,

$$(9.5) \quad F_\Gamma(u(t(\tau))) = \frac{1}{2} \int_{B_R} m(y, \tau) |V(y, \tau) - \bar{V}_m(\tau)|^2 \, dy, \quad \bar{V}_m := \int_{B_R} m V \, dy.$$

Thus F_Γ is invariant under the Navier-Stokes scaling and is a Galilean-invariant version of the local quantity A . Denote its chart expression by

$$(9.6) \quad \mathcal{E}_\Gamma^\circ(V(\tau)) := \frac{1}{2} \int_{B_R} m |V - \bar{V}_m|^2 \, dy.$$

For variations of V with m fixed,

$$(9.7) \quad D\mathcal{E}_\Gamma^\circ(V)[\varphi] = \int_{B_R} m(V - \bar{V}_m) \cdot \varphi \, dy.$$

Indeed, the variation of $\bar{V}_m$ pairs with $\int m(V - \bar{V}_m) = 0$. Thus the target gradient below is the Stokes gradient of the Galilean-reduced variance in every representative. No time-dependent velocity-centering condition is imposed on the chart.

Let

$$(9.8) \quad q_\Gamma := \mathbb{P}(w_\Gamma(u - \bar{u}_\Gamma)).$$

For variations $\varphi \in \mathcal{H}_\sigma$,

$$DF_\Gamma(u)[\varphi] = (q_\Gamma, \varphi)_{L^2}.$$

Since $q_\Gamma \in \mathcal{V}_\sigma$ for almost every time, this derivative extends to currents $J \in \mathcal{V}_\sigma^*$ by $DF_\Gamma(u)[J] := \langle J, q_\Gamma \rangle$. The Stokes gradient current of the probe is

$$(9.9) \quad G_\Gamma := Kq_\Gamma.$$

Define the viscous action, the Stokes norm of the localized variation, and the signed localized power by

$$(9.10) \quad \mathcal{A}_\nu(u) := \|J_\nu\|_{K^{-1}}^2 = \nu \|\nabla u\|_2^2,$$

$$(9.11) \quad \mathcal{G}_\Gamma(u) := \|G_\Gamma\|_{K^{-1}}^2 = \nu \|\nabla q_\Gamma\|_2^2,$$

$$(9.12) \quad \mathcal{P}_\Gamma(u) := (J_\nu, G_\Gamma)_{K^{-1}} = DF_\Gamma(u)[J_\nu].$$

Lemma 9.1 (Explicit viscous power). *For smooth u and w_Γ ,*

$$(9.13) \quad \mathcal{P}_\Gamma = -\nu \int_{\mathbb{R}^3} w_\Gamma |\nabla u|^2 \, dx + \frac{\nu}{2} \int_{\mathbb{R}^3} |u - \bar{u}_\Gamma|^2 \Delta w_\Gamma \, dx.$$

The first term is local viscous dissipation. The second is the diffusive flux across the spatial variation of the probe.

Proof. Since $A_\sigma u$ is solenoidal and $\mathbb{P}$ is self-adjoint,

$$\mathcal{P}_\Gamma = -\nu \int \nabla u : \nabla q_\Gamma = -\nu \int \nabla u : \nabla (w_\Gamma(u - \bar{u}_\Gamma)).$$

Expand the last gradient and integrate $-\nu \int \partial_j u_i (u_i - \bar{u}_{\Gamma,i}) \partial_j w_\Gamma$ by parts. This gives (9.13). $\square$

In the represented variables, the power measure has the scale-free form

$$(9.14) \quad \mathcal{P}_\Gamma(u(t(\tau))) \, dt = \widehat{\mathcal{P}}_\Gamma(V(\tau)) \, d\tau, \quad \widehat{\mathcal{P}}_\Gamma(V) := -\nu \int_{B_R} m |\nabla V|^2 \, dy + \frac{\nu}{2} \int_{B_R} |V - \bar{V}_m|^2 \Delta m \, dy.$$

For smooth solutions the total derivative of the probe splits as

$$(9.15) \quad \frac{d}{dt} F_\Gamma(u(t)) = \mathcal{T}_\Gamma(u, p) + \mathcal{P}_\Gamma(u),$$

where

$$(9.16) \quad \mathcal{T}_\Gamma(u, p) := \frac{1}{2} \int |u - \bar{u}_\Gamma|^2 (\partial_t w_\Gamma + u \cdot \nabla w_\Gamma) \, dx + \int p (u - \bar{u}_\Gamma) \cdot \nabla w_\Gamma \, dx.$$

The term $\mathcal{T}_\Gamma$ contains conservative transport, pressure flux, and the motion of the scale-center coordinates. The term $\mathcal{P}_\Gamma$ contains viscosity alone. For suitable weak solutions, define

$$(9.17) \quad \mu_{\text{loc}} := - \left[\partial_t \frac{|u|^2}{2} + \operatorname{div} \left(\left(\frac{|u|^2}{2} + p \right) u \right) - \nu \Delta \frac{|u|^2}{2} + \nu |\nabla u|^2 \right].$$

The local energy inequality says precisely that this distribution is a nonnegative Radon measure. Combining it with the weak momentum identity for the weighted mean gives

$$(9.18) \quad F_\Gamma(t_2) - F_\Gamma(t_1) = \int_{t_1}^{t_2} (\mathcal{T}_\Gamma + \mathcal{P}_\Gamma) dt - \mu_{\text{loc}}(w_\Gamma; [t_1, t_2]),$$

where $\mu_{\text{loc}}(w_\Gamma; [t_1, t_2])$ denotes the measure in (9.17) paired with the nonnegative test weight on the indicated time interval. The identity holds at the corresponding local energy times; arbitrary compact subintervals follow by one-sided representatives.

9.2. The Stokes-metric projection inequality.

Definition 9.2 (Viscous alignment). Set

$$(9.19) \quad \text{Align}_\Gamma(u) := \begin{cases} \frac{(\mathcal{P}_\Gamma(u)_+)^2}{\mathcal{A}_\nu(u)\mathcal{G}_\Gamma(u)}, & \mathcal{A}_\nu(u)\mathcal{G}_\Gamma(u) > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Positive power means that the viscous current points toward increasing the localized kinetic probe. Negative power means that viscosity lowers that probe.

Theorem 9.3 (Stokes-metric projection inequality). *For almost every time,*

$$(9.20) \quad (\mathcal{P}_\Gamma(u)_+)^2 \leq \mathcal{A}_\nu(u)\mathcal{G}_\Gamma(u), \quad 0 \leq \text{Align}_\Gamma(u) \leq 1.$$

For every interval $I \Subset (0, T)$,

$$(9.21) \quad \int_I \mathcal{P}_\Gamma(u(t))_+ dt \leq \left(\int_I \mathcal{A}_\nu(u(t)) dt \right)^{1/2} \left(\int_I \mathcal{G}_\Gamma(u(t)) dt \right)^{1/2}.$$

The first factor is at most $\mathcal{E}(u(0))^{1/2}$.

Proof. The definitions give

$$\mathcal{P}_\Gamma = (J_\nu, G_\Gamma)_{K^{-1}}.$$

Cauchy–Schwarz in the K^{-1} -metric proves (9.20). A second application of Cauchy–Schwarz in time proves (9.21). The energy bound is (3.12). $\square$

The inequality separates the amount of irreversible motion from its orientation. Equal action can produce zero or maximal positive power, depending on the angle with the target gradient.

Definition 9.4 (Covariant ray-alignment defect). For a sufficiently regular divergence-free comparison curve v , set

$$(9.22) \quad R(v) := \mathfrak{D}_t^E v + K v.$$

For an admissible probe Γ and interval I , define

$$(9.23) \quad \begin{aligned} \mathcal{D}_{\Gamma, I}^{\text{ray}}(v) &:= \frac{1}{2} \int_I \|R(v)\|_{K^{-1}}^2 dt \\ &\quad + \frac{1}{2} \int_I \left(\mathcal{A}_\nu(v) - \frac{(\mathcal{P}_\Gamma(v)_+)^2}{\mathcal{G}_\Gamma(v)} \right) dt, \end{aligned}$$

where the quotient is defined as zero when $\mathcal{G}_\Gamma = 0$.

Proposition 9.5 (Metric meaning of the ray defect). *The functional $\mathcal{D}_{\Gamma, I}^{\text{ray}}$ is nonnegative. Pointwise in time,*

$$(9.24) \quad \mathcal{A}_\nu - \frac{(\mathcal{P}_{\Gamma+})^2}{\mathcal{G}_\Gamma} = \text{dist}_{K^{-1}}^2(J_\nu, \{\alpha G_\Gamma : \alpha \geq 0\}).$$

For an exact Navier–Stokes solution, the equation-defect term in (9.23) vanishes. Equality $\mathcal{D}_{\Gamma,I}^{\text{ray}}(u) = 0$ then means that the actual viscous current lies on the nonnegative target-gradient ray almost everywhere on I . Moreover,

$$(9.25) \quad 2\mathcal{D}_{\Gamma,I}^{\text{ray}}(u) = \int_I \mathcal{A}_\nu(u(t))(1 - \text{Align}_\Gamma(u(t))) \, dt.$$

Proof. Orthogonal projection of a vector J onto the closed ray generated by a nonzero vector G has coefficient $(J, G)_+ / \|G\|^2$. Its squared residual norm is $\|J\|^2 - (J, G)_+^2 / \|G\|^2$. Apply this identity in the K^{-1} -metric. When $G = 0$, the distance is $\|J\|$, in agreement with the convention in (9.23). The equation-defect statement is (3.7). Substitution of (9.19) gives (9.25). $\square$

Let $\mathfrak{G}_{\text{adm}}(I)$ denote repaired probes with scale-relative cutoff bounds and normalized adjoint weights as in Section 8. Let $\mathfrak{G}_{\text{adm}}(t)$ be their time- t slices. Define

$$(9.26) \quad \text{Align}_*(t) := \sup_{\Gamma \in \mathfrak{G}_{\text{adm}}(t)} \text{Align}_\Gamma(t), \quad \mathcal{D}_*^{\text{ray}}(I) := \inf_{\Gamma \in \mathfrak{G}_{\text{adm}}(I)} \mathcal{D}_{\Gamma,I}^{\text{ray}}(u).$$

Suprema and infima are understood through approximating sequences; no attainment is required. The Stokes mobility K and current J_ν are fixed throughout these optimizations.

Theorem 9.6 (Local energy identities and projection bound). *For every admissible probe and interval $[t_1, t_2] \subseteq (0, T)$, the following three statements hold simultaneously:*

- (i) *global energy decreases by viscous action and the nonnegative energy defect according to (3.11);*
- (ii) *the localized probe satisfies (9.18), separating coordinate/transport flux from viscous target power;*
- (iii) *positive viscous target power satisfies (9.21) and is therefore controlled by the finite global action.*

These conclusions remain valid along an optimizing sequence for (9.26).

Proof. Items (i)–(iii) are Theorem 3.3, the suitable local energy inequality tested by w_Γ , and Theorem 9.3, respectively. Each inequality holds for every admissible probe, so it holds termwise along any approximating sequence in (9.26). $\square$

9.3. Alignment, coercivity, and coherent probe selection. The alignment is an angle in the Stokes metric. It contains no information about the amount of action unless it is paired with $\mathcal{A}_\nu$, and it contains no compactness information unless the probe belongs to a normalized family. The following elementary observation fixes the equality orientation.

Lemma 9.7 (Positive-ray equality for a kinetic probe). *Let $w \geq 0$ have positive finite mass, put $\bar{u}_w = (\int w)^{-1} \int wu$, let $q = \mathbb{P}(w(u - \bar{u}_w))$, and suppose $u \in L_\sigma^2 \cap \dot{H}_\sigma^1$. If*

$$(9.27) \quad -Ku = \alpha Kq \quad \text{for some } \alpha \geq 0,$$

then $u = 0$.

Proof. Apply K^{-1} in the solenoidal Gelfand triple and pair the resulting identity $-u = \alpha q$ with u . Since u is solenoidal,

$$-\|u\|_2^2 = \alpha(q, u)_{L^2} = \alpha \int w|u - \bar{u}_w|^2 \, dx \geq 0.$$

Both sides vanish, so $u = 0$. $\square$

For a finite-energy whole-space state, vanishing ray defect is therefore a rigid equality. Along a singular rescaling, however, the physical ray defect is weighted by Λ exactly as in (7.3). It may have finite physical integral even when its scale-critical chart density remains positive on every unit window. This explains why the ray defect is only a covariant diagnostic; the exclusion proof instead controls the local Dirichlet chart density through the moving-cutoff energy.

Probe optimization also has a precise order. The physical solution first determines J_ν and the canonical chart. A fixed terminal-data rule then selects the finite-horizon adjoint problem on the padded window (8.5). Only after these choices are fixed may one take a supremum in Align or an infimum in $\mathcal{D}^{\text{ray}}$. Replacing the current by the gradient of an optimizing probe would change the equation and is not an admissible variation.

Definition 9.8 (Compatible family of localized probes). A compatible family of localized probes on a represented orbit is a family $\{\Gamma_j\}_{j \geq 1}$ on padded retained windows such that:

- (i) every Γ_j is constructed from the same physical solution by a fixed terminal-data rule and has the buffer (8.6);
- (ii) chart transitions satisfy the composition law in hypothesis 10.2;
- (iii) the current paired with every target is the pullback of $J_\nu = -\nu A_S u$;
- (iv) approximating probes converge in the compact topology required by the local adjoint construction.

The terminal adjoint equation is a testing device. Coherence here means that all probes test the same physical current. It does not identify distinct finite-horizon adjoint densities with one global density, and no such identification is used by the corrected localized energy below.

Proposition 9.9 (Scale-critical window form of the ray defect). *Let I_τ be a represented interval and I_t its physical image. For an exact Navier–Stokes solution, the pullback of the physical ray defect has the form*

$$(9.28) \quad 2\mathcal{D}_{\Gamma, I_t}^{\text{ray}}(u) = \int_{I_\tau} \Lambda(\tau) \widehat{\mathfrak{D}}_{\text{ray}, \Gamma}(V(\tau)) \, d\tau,$$

where $\widehat{\mathfrak{D}}_{\text{ray}, \Gamma}$ is invariant under a constant parabolic rescaling of the represented window. Equivalently, it is defined by the Radon–Nikodym relation

$$(9.29) \quad \widehat{\mathfrak{D}}_{\text{ray}, \Gamma}(\tau) \, d\tau := \Lambda(\tau)^{-1} \text{dist}_{K-1}^2(J_\nu, \mathbb{R}_+ G_\Gamma) \, dt.$$

Proof. Both vectors in the Stokes pairing transform by the pullback of the same chart. Their squared inverse-metric distance has the scaling of viscous action. Formula (5.11) shows that its physical measure is Λ times the dimensionless represented measure, which is exactly (9.29). $\square$

The hatted density exhibits the same physical-versus-critical distinction as (7.3) and (7.4). It is retained below as a covariant diagnostic, but the local coercive floor does not use it.

10. CANONICAL CRITICAL ORBITS

The covariant argument is carried by one normalized orbit rather than a list of asymptotic regimes. This section specifies the state space and the exact compactness conclusion required by the entropy proof.

10.1. Symmetry reduction and target-retaining windows. Fix a normalized probe radius $R_* > 1$ and a nonnegative inner cutoff $\zeta \in C_c^\infty(B_{R_*})$, $0 \leq \zeta \leq 1$, $\int \zeta^3 > 0$, for which $\{\zeta > 0\}$ is connected and whose support lies where the adjoint cutoff equals one. Put

$$(10.1) \quad \bar{V}_\zeta(\tau) := \frac{\int \zeta^3 V(y, \tau) \, dy}{\int \zeta^3 \, dy}.$$

For a unit represented interval $I = [s, s + 1]$, define

$$(10.2) \quad \mathcal{C}_\Gamma^\circ(V; I) := \int_I \int_{B_{R_*}} \zeta^3 |V - \bar{V}_\zeta|^3 dy d\tau.$$

This observable is independent of the adjoint terminal datum, continuous under strong local L^3 convergence, and invariant under constant velocity shifts.

The relevant symmetries are spatial translation, positive parabolic dilation, constant-velocity Galilean transformation, and addition of a function of time to the pressure. The repaired variables (X, Λ, a, b) encode scale and center. We fix those representatives by

$$(10.3) \quad \mathfrak{F}(1, 0; V(\tau)) = 0, \quad \det D_{(\lambda, q)} \mathfrak{F}(1, 0; V(\tau)) \neq 0$$

at almost every active time, with absolutely continuous continuation supplied by the orbit hypothesis. Galilean symmetry is not fixed by (10.3); the gauge itself is invariant under it, and the scalar observables remove it through (10.1) and (9.3). The pressure gauge is

$$(10.4) \quad \int_{B_1} P(y, \tau) dy = 0.$$

The nondegeneracy and absolute continuity of this gauge are conclusions of the critical-orbit hypothesis, not extra hypotheses imposed on an arbitrary weak solution.

Definition 10.1 (Normalized target-retaining window). Fix $\eta_0 > 0$ and a collection of compact-cylinder bounds $\mathbf{M} = (M_R)_{R \geq 1}$. A represented unit window $\mathfrak{w} = (V, P, a, b; I)$ is an $(\eta_0, \mathbf{M})$ -normalized target-retaining window when:

- (i) (V, P) is a suitable solution of (5.4) on a neighborhood of $B_R \times I$ for every fixed R ;
 - (ii) the gauges (10.3) and (10.4) hold, and the inverse of the gauge Jacobian in (7.9) is uniformly bounded on the window;
 - (iii) the local suitable norms satisfy, on each fixed $B_R \times I$,
- $$(10.5) \quad \|V\|_{L^\infty L^2(B_R)} + \|V\|_{L^2_t H^1(B_R)} + \|V\|_{L^3(B_R \times I)} + \|P\|_{L^{3/2}(B_R \times I)} + \|a\|_{L^1(I)} + \|b\|_{L^1(I)} \leq M_R;$$
- (iv) the oscillatory floor

$$(10.6) \quad \mathcal{C}_\Gamma^\circ(V; I) \geq \eta_0$$

holds for the selected inner probe.

The subtraction in (10.2) is essential. A spatially constant represented velocity can have positive uncentered L^3 mass, but it has no spatial oscillation and cannot by itself support a singular point; constant Galilean shifts are a special case. The self-contained orbit construction must therefore prove that singular concentration survives the variance reduction rather than assume this property.

10.2. The critical-orbit hypothesis.

Analytic hypothesis 10.2 (Canonical critical orbit from terminal loss). *Let (u, p) be a global finite-energy suitable weak solution of (1.1), and let $0 < T < \infty$. Suppose either*

- (a) $x_* \in \Sigma_u(T)$ for some finite point x_* , or
- (b) u is the suitable continuation of a smooth finite-energy solution whose maximal classical existence time is T .

Then there exist $\eta_0 = \eta_0(\nu) > 0$, compact bounds $\mathbf{M}$, an absolutely continuous repaired chart (X, Λ) , a represented suitable orbit

$$(V, P, a, b)(\tau), \quad \tau \in [0, \infty),$$

unit intervals $I_j = [\tau_j, \tau_j + 1]$ with pairwise disjoint interiors and $\tau_j \rightarrow \infty$, with the following properties:

- (i) after passing once to a subsequence and relabeling, with $x_j := X(\tau_j)$ and $r_j := \Lambda(\tau_j)$, one has $r_j \downarrow 0$; the physical time images of the windows converge to T ; in case (a) one may take $x_j \rightarrow x_*$, while case (b) also permits $|x_j| \rightarrow \infty$;
- (ii) every restricted state $\mathfrak{w}_j = (V, P, a, b; I_j)$ is an $(\eta_0, \mathbf{M})$ -normalized target-retaining window;
- (iii) in the product topology specified in [Section 7](#) and made explicit here, the sequential closure of the selected windows is sequentially compact. More explicitly, after translating each time interval to $[0, 1]$, every sequence in that closure has a subsequence for which

$$(10.7) \quad V_j \rightarrow V_\infty \text{ strongly in } L^3_{\text{loc}}, \quad \nabla V_j \rightharpoonup \nabla V_\infty \text{ in } L^2_{\text{loc}},$$

on every compact spatial cylinder in $\mathbb{R}^3 \times [0, 1]$, and in particular strongly in $L^3(\text{supp } \zeta \times [0, 1])$, with strong local $L^{3/2}$ convergence of normalized pressures and L^1 convergence of (a, b) ;

- (iv) chart changes obey the composition law and represent the same physical solution.

The displayed statement is an explicit hypothesis. [Appendix C](#) specifies the direct construction of the coherent orbit, and [Appendix D](#) specifies the gauge, pressure decomposition, and moving-cutoff estimates. A self-contained proof must establish all four clauses without invoking global regularity.

Remark 10.3 (Why one orbit is required). The corrected energy decrease is accumulated along one scalar function of represented time. Independent rescaling on each target-retaining window would provide unrelated endpoint values and would not support the telescoping argument. Clause (iv) is therefore part of the orbit hypothesis rather than a convention made after extraction.

10.3. The two classical scale regimes. Let $\ell(t) = \sqrt{T - t}$ denote the parabolic terminal scale. Along the canonical orbit, the dimensionless ratio

$$(10.8) \quad \mathfrak{r}(\tau) := \frac{\Lambda(\tau)}{\ell(t(\tau))}$$

records the classical distinction. A Type I window has $\mathfrak{r}$ bounded above and below after normalization; in the centered choice $t = -e^{-\tau}$, $\Lambda = e^{-\tau/2}$, one has $a = -1/2$. A Type II sequence contains windows on which $\mathfrak{r} \rightarrow 0$, and its coefficient $a = \Lambda'/\Lambda$ must be controlled through the repaired gauge [\(7.8\)](#).

The covariant equation [\(5.6\)](#) is identical in the two regimes. The difference is carried entirely by (a, b) and by the asymptotic path of the normalized orbit. The moving-cutoff alternative is consequently formulated for locally integrable a, b , without fixing $a = -1/2$ or assuming convergence of a . A centered ancient orbit, a rapid scale collapse, and a recurrent repaired orbit are either reduced to the corresponding local estimates or tested by the same moving production measure. The gap argument uses only the coherent orbit, compactness, the oscillatory floor, and the summable moving-cutoff case. Thus the argument uses precisely the compactness stated in the hypothesis and introduces no additional rescaled solution.

There remains one compatibility issue. A window selected at time τ_j must extend to the next selected window without replacing the underlying solution. Let $g_{j+1 \leftarrow j} \in \mathcal{G}_{\text{ch}}$ be the accumulated canonical transition from the terminal representative of I_j to the initial representative of I_{j+1} . Coherence means

$$(10.9) \quad g_{k \leftarrow j} = g_{k \leftarrow k-1} \circ \cdots \circ g_{j+1 \leftarrow j}, \quad j < k,$$

and every composition pulls back the same physical pair (u, p) . The moving cutoff in [hypothesis 12.5](#) is defined on this one orbit, so its corrected energy is one global scalar. There is no cross-window adjoint reset in the exclusion argument.

11. DISSIPATION AND DIMENSIONAL REDUCTION

The material identities identify the mechanism but do not by themselves produce a scale-critical finite measure. The required statement is a dissipation-or-reduction alternative: signed transport between nested regions is internal, whereas viscous dissipation and true exterior leakage are counted once. Its precise interface with the compact orbit is derived below from the localized energy equation.

Let $I = [s, t]$ be a represented interval and let $\psi \in C^\infty(I; C_c^\infty(\mathbb{R}^3))$, $\psi \geq 0$. Set

$$(11.1) \quad K_\psi(\tau) := \frac{1}{2} \int_{\mathbb{R}^3} |V(y, \tau)|^2 \psi(y, \tau) dy, \quad D_I(\psi) := \nu \int_I \int_{\mathbb{R}^3} |\nabla V|^2 \psi dy d\tau,$$

and define the signed localization current

$$(11.2) \quad J_I(\psi) := \int_I \left[\frac{1}{2} \int |V|^2 (\partial_\tau \psi + \nu \Delta \psi) + \int \left(\frac{1}{2} |V|^2 + P \right) V \cdot \nabla \psi - \frac{a}{2} \int |V|^2 (\psi + y \cdot \nabla \psi) - \frac{1}{2} b \cdot \int |V|^2 \nabla \psi \right] d\tau.$$

All spatial integrals in (11.2) are over $\mathbb{R}^3$. The pressure contribution is invariant under $P \mapsto P + c(\tau)$, because $\int V \cdot \nabla \psi = 0$.

Proposition 11.1 (Signed-current additivity). *For every represented suitable solution there is a nonnegative local energy-defect functional $M_I(\psi)$, linear on nonnegative tests, such that*

$$(11.3) \quad K_\psi(t) + D_I(\psi) + M_I(\psi) \leq K_\psi(s) + J_I(\psi).$$

For a smooth solution equality holds and $M_I(\psi) = 0$. If $\psi = \sum_{k=1}^N \psi_k$ with $\psi_k \geq 0$, then

$$(11.4) \quad K_\psi = \sum_{k=1}^N K_{\psi_k}, \quad D_I(\psi) = \sum_{k=1}^N D_I(\psi_k), \quad J_I(\psi) = \sum_{k=1}^N J_I(\psi_k), \quad M_I(\psi) = \sum_{k=1}^N M_I(\psi_k).$$

Consequently, for an exact parent–child partition $\psi_\alpha = \sum_{\beta \in \text{ch}(\alpha)} \psi_\beta$, the difference between the parent current and the sum of its child currents is identically zero. This is refinement invariance, not selective cancellation: adding the parent balance to all child balances double-counts every term, whereas subtracting the parent balance cancels every term, including D_I and M_I .

Proof. The suitable local energy inequality for the repaired equation, integrated between s and t , is precisely (11.3); the terms generated by the repaired drift are the last two terms of (11.2). The nonnegative distribution by which equality fails defines M_I . Every displayed term is linear in ψ and its derivatives, which proves (11.4). For smooth solutions the local energy identity holds with equality. $\square$

Definition 11.2 (Critical localized-energy balance). A critical localized-energy balance on a coherent orbit consists of a scalar $\mathcal{K} : [\tau_0, \infty) \rightarrow \mathbb{R}$, bounded below by K_- , and nonnegative Radon measures $\mathcal{D}$, $\mathcal{F}^+$, and $\mathcal{R}^+$ such that, for $\tau_0 \leq s < t$,

$$(11.5) \quad \mathcal{K}(t) + \mathcal{D}((s, t]) \leq \mathcal{K}(s) + \mathcal{F}^+((s, t]) + \mathcal{R}^+((s, t]).$$

Here $\mathcal{D}$ is gross viscous and suitable-defect expenditure, $\mathcal{F}^+$ is inward feeding across the moving scale-space boundary, and $\mathcal{R}^+$ is true exterior input. All three are expressed in the same represented, scale-critical clock. The inward flux is *dominated by dissipation* when

$$(11.6) \quad \mathcal{F}^+ \leq \mathcal{D} \quad \text{as Radon measures.}$$

Under this domination its net dissipation is

$$(11.7) \quad \mu_{\text{cov}} := \mathcal{D} - \mathcal{F}^+ \geq 0.$$

The currents entering $\mathcal{F}^+$ must be obtained either from one global localization or from a disjoint spacetime partition for which the energy-flux field has compatible oriented normal traces. On the latter formulation a common interface occurs once with each orientation and cancels, whereas dissipation and defect add over the disjoint interiors. Identity (11.4) supplies only refinement invariance; it does not construct those traces and does not imply (11.6).

Proposition 11.3 (Finite net dissipation). *If Eqs. (11.5) and (11.6) hold and $\mathcal{R}^+([\tau_0, \infty)) < \infty$, then μ_{cov} is a finite nonnegative Radon measure and*

$$(11.8) \quad \mu_{\text{cov}}([\tau_0, \infty)) \leq \mathcal{K}(\tau_0) - K_- + \mathcal{R}^+([\tau_0, \infty)) < \infty.$$

Proof. Flux domination makes the difference in (11.7) a nonnegative Radon measure. Subtract $\mathcal{F}^+((\tau_0, t])$ from (11.5), use $\mathcal{K}(t) \geq K_-$, and let $t \rightarrow \infty$. Continuity from below for the three nonnegative measures gives (11.8). $\square$

11.1. Normalized shell decomposition. Let

$$(11.9) \quad \mathfrak{C}_{\text{NS}} := \{\text{mat}, \text{vis}, \text{int}, \text{gauge}, \text{ext}\}.$$

The labels denote material deformation, viscous motion, signed interior transport including pressure, chart and material-relabeling motion, and true exterior flux. This is a decomposition of the actual repaired Navier–Stokes current, not a decomposition of its absolute value.

Definition 11.4 (Normalized singular-target shells). Let $\hat{\mathcal{X}}_{\text{sing}}$ be a Hausdorff state-or-germ space containing the regular orbit and the nonremovable singular germs. A normalized shell presentation consists of nested neighborhoods U_ε of the singular set and continuous functions $\mathcal{S}_\varepsilon : \hat{\mathcal{X}}_{\text{sing}} \rightarrow [0, 1]$ such that

$$(11.10) \quad \mathcal{S}_\varepsilon = 0 \quad \text{outside } U_{2\varepsilon}, \quad \mathcal{S}_\varepsilon = 1 \quad \text{on } U_\varepsilon.$$

On every regular precontact segment on which it is used, $\mathcal{S}_\varepsilon(u(\cdot))$ is absolutely continuous and its chain rule has the exact form

$$(11.11) \quad \frac{d}{dt} \mathcal{S}_\varepsilon(u(t)) = \sum_{c \in \mathfrak{C}_{\text{NS}}} q_\varepsilon^c(t) \quad \text{for a.e. } t.$$

A shell-crossing record is an interval $J_\varepsilon = [t_\varepsilon^-, t_\varepsilon^+]$ lying strictly before contact such that the endpoint values in (11.10) are zero and one, respectively.

Theorem 11.5 (Exact shell-crossing identity). *Every shell-crossing record for which the five functions in (11.11) are integrable satisfies*

$$(11.12) \quad 1 = \sum_{c \in \mathfrak{C}_{\text{NS}}} Q_\varepsilon^c, \quad Q_\varepsilon^c := \int_{J_\varepsilon} q_\varepsilon^c(t) dt.$$

Consequently

$$(11.13) \quad \max_{c \in \mathfrak{C}_{\text{NS}}} (Q_\varepsilon^c)_+ \geq \frac{1}{5}.$$

Neither conclusion uses an energy or entropy bound.

Proof. Integrate (11.11) over the crossing interval and use the endpoint values. Since the sum of five real numbers is one, the sum of their positive parts is at least one and one positive part is at least $1/5$. $\square$

Proposition 11.6 (Persistent contributing component). *Suppose contact supplies infinitely many shell-crossing records on pairwise disjoint precontact events. After passage to an infinite subsequence, one component $c_* \in \mathfrak{C}_{\text{NS}}$ satisfies*

$$(11.14) \quad Q_j^{c_*} \geq \frac{1}{5} \quad \text{for every index in the subsequence.}$$

Proof. Choose on each record a component satisfying (11.13). One of the five labels is selected infinitely often. $\square$

The shell layer identifies the current responsible for repeated progress toward the singular target. If the persistent component has zero pairing with every singular-target shell, this contradicts (11.14). A physical lower bound is a separate comparison with a finite measure. In material variables, signed interior transfer and pure gauge motion are expected to have zero target pairing; deformation either couples to enhanced diffusion or produces the regular invariant reduction described in Appendix G.

Proposition 11.1 proves the cancellation algebra. It does not by itself prove flux domination, because the assembled boundary may feed energy back into the retained region. Nor does it change the positive scaling exponent of the physical Stokes action. The Navier–Stokes content is therefore the following target-relative statement: after every already excluded exterior, scale, and modulation alternative has been removed, the aggregate feeding is dominated by gross dissipation, and equality is target-null. Appendices F and G give the precise realization.

Analytic hypothesis 11.7 (Target-relative current alternative). *For every terminal-loss orbit supplied by hypothesis 10.2, one of the following alternatives holds.*

- (N1) *A persistent shell component has zero target pairing, contradicting (11.14), or the orbit enters a material reduction or local alternative whose PDE rigidity theorem excludes terminal singularity.*
- (N2) *There exist an infinite index set $\mathcal{J} \subset \mathbb{N}$, pairwise disjoint half-open intervals $I_j \subset [\tau_0, \infty)$ whose closures are selected target-retaining windows, and a critical localized-energy balance with dominated inward flux and finite exterior input. Its net dissipation μ_{cov} is related to a sequentially lower semicontinuous window deficit*

$$(11.15) \quad \Delta_{\text{crit}} : \{\mathfrak{w}_j : j \in \mathcal{J}\} \longrightarrow [0, \infty],$$

such that

$$(11.16) \quad \mu_{\text{cov}}([\tau_0, \infty)) < \infty,$$

$$(11.17) \quad \mu_{\text{cov}}(I_j) \geq \Delta_{\text{crit}}(\mathfrak{w}_j), \quad j \in \mathcal{J},$$

$$(11.18) \quad \Delta_{\text{crit}}(\mathfrak{w}) = 0 \implies \mathfrak{w} \text{ is oscillation-removable and } \mathcal{C}_T^\circ(V; [0, 1]) = 0.$$

Here the closure and its topology are those of hypothesis 10.2, after translating each unit interval to $[0, 1]$.

- (N3) *The flux-dominated conclusion in (N2) fails, and the surviving positive critical current generates, from a uniformly record-bounded terminal extraction, a marked regenerated-current cascade in the sense of Definitions C.14 and C.15 and Proposition C.16. Every exterior, pressure-atlas-loss, affine, diffuse, and finite-descendant output is assigned to (N1) before this alternative is selected.*

The window deficit is a lower-semicontinuous target-visible part of the net critical dissipation after internal signed currents have been canceled. The first alternative eliminates a component without a dissipation estimate. On the second alternative, compactness supplies a numerical gap directly in the finite measure μ_{cov} ; no scale weight is inserted after normalization. The third alternative retains the current which prevents domination. It is subjected to the ordered

first-failure tests of [Proposition C.16](#). Every failed test is assigned to (N1); if no test fails, exact composition of the bounded descendant rebases contradicts the retained activity floor by [Theorem C.22](#). Thus no spatial averaging is used on this branch.

Theorem 11.8 (Normalized covariant rigidity gap). *Assume alternative (N2) and let $\eta_0 > 0$ be the oscillation floor of the canonical orbit. There exists $\delta_{\text{rig}} > 0$ such that every selected window indexed by $\mathcal{J}$ satisfies*

$$(11.19) \quad \Delta_{\text{crit}}(\mathfrak{w}_j) \geq \delta_{\text{rig}}.$$

Proof. The compact closure of the selected windows contains the closed subset

$$\mathcal{B}_{\eta_0} := \{\mathfrak{w} : \mathcal{C}_\Gamma^\circ(V; [0, 1]) \geq \eta_0\}.$$

It is closed because the velocities converge strongly in L^3 on the fixed retained core. If the infimum of Δ_{crit} on $\mathcal{B}_{\eta_0}$ were zero, compactness and lower semicontinuity would produce $\mathfrak{w}_\infty \in \mathcal{B}_{\eta_0}$ with $\Delta_{\text{crit}}(\mathfrak{w}_\infty) = 0$. This contradicts (11.18). Hence the infimum is a positive number δ_{rig} , which proves (11.19). $\square$

Theorem 11.9 (Critical net-dissipation exclusion). *Under alternative (N2), the pairwise disjoint target-retaining windows cannot form an infinite family.*

Proof. By [Theorem 11.8](#) and (11.17),

$$\mu_{\text{cov}}(I_j) \geq \delta_{\text{rig}} \quad (j \in \mathcal{J}).$$

Countable additivity on the infinitely many disjoint windows gives

$$(11.20) \quad \mu_{\text{cov}}([\tau_0, \infty)) \geq \sum_{j \in \mathcal{J}} \mu_{\text{cov}}(I_j) \geq \delta_{\text{rig}} \sum_{j \in \mathcal{J}} 1 = \infty,$$

contrary to (11.16). $\square$

Proposition 11.10 (Why the physical Stokes action is insufficient). *Let a normalized target-retaining window at physical scale r_j have Dirichlet defect at least $\delta_\nu > 0$. The physical Stokes action on the corresponding physical event is bounded below only by a constant times $r_j \delta_\nu$. Hence neither signed-current cancellation nor a normalized Dirichlet gap implies divergence of the physical action unless $\sum_j r_j = \infty$. The scale-independent lower bound in (11.17) must therefore come from the critical net measure (11.7), not from a rescaling of the physical Stokes action.*

Proof. The change of variables in (7.3) gives exactly one factor of r_j . A convergent geometric sequence of scales has finite sum, proving the assertion. $\square$

Remark 11.11 (Two realizations of the balance). The moving-cutoff realization takes $\mu_{\text{cov}} = \mathfrak{D}_{\text{NS}}$ and $\Delta_{\text{crit}} = \frac{1}{2}\mathfrak{d}_\nu$, where $\mathfrak{d}_\nu$ is the fixed inner Dirichlet functional (13.3). In this realization $\mathcal{F}^+ = 0$ after the integrable flux tail has been incorporated into the corrected energy. The more intrinsic realization in [Appendix F](#) retains the signed flux and uses $\mu_{\text{cov}} = \mathcal{D} - \mathcal{F}^+$. Both have the same saturation class and therefore the same target-relative gap.

12. COVARIANT ADJOINT GEOMETRY AND THE MOVING-CUTOFF ENERGY

The adjoint entropy $\mathcal{W}_{\text{ad}}$, the kinetic variance $\mathcal{E}_\Gamma^\circ$, and the ray defect $\mathcal{D}^{\text{ray}}$ have distinct roles. The first controls the localization geometry, the second detects the retained velocity core, and the third measures the component of viscous motion transverse to the target-gradient ray. None of them separately has the monotonicity and coercivity needed below.

12.1. The explicit nonnegative components. For a smooth positive adjoint density, define the Gaussian production

$$(12.1) \quad \mathfrak{D}_G(\rho) := 2\nu^2\theta \int_{B_R} \chi^2 \left| \nabla^2 f - \frac{1}{2\nu\theta} I \right|^2 \rho \, dy.$$

Define also the localized viscous production

$$(12.2) \quad \mathfrak{D}_{\nu,\Gamma}(V) := \nu \int_{B_R} m |\nabla V|^2 \, dy$$

and the instantaneous ray production

$$(12.3) \quad \mathfrak{D}_{\text{ray},\Gamma}(V) := \widehat{\mathfrak{D}}_{\text{ray},\Gamma}(V),$$

where the hatted, scale-critical density is defined by (9.29). At weak regularity $\mathfrak{D}_{\nu,\Gamma}$ is defined directly from the weighted Dirichlet integral. If the Gaussian or ray terms are retained in a weak production measure, their identification and compact stability require additional adjoint compactness beyond the localized energy argument; neither is inferred merely from local weak velocity convergence. The coercive floor used in the compactness argument is independent of the adjoint density:

$$(12.4) \quad \mathfrak{D}_{0,\zeta}(V) := \nu \int_{\mathbb{R}^3} \zeta^2 |\nabla V|^2 \, dy.$$

The Gaussian production remains a favorable nonnegative term in the smooth Perelman identity, but it may be discarded before taking a weak limit. The ray production is likewise an additional nonnegative covariant diagnostic on each finite-energy representative. Neither term is needed to separate retained compact limits from equality. Inserting the ray term into (12.4) would in addition require a localization theorem for the whole-space Helmholtz projection under merely local blow-up convergence.

Formula (8.17) contains a raw velocity-strain term with no separate sign. It becomes coercive after coupling it to the viscous and Gaussian terms in Proposition 12.2. The moving-cutoff kinetic energy below handles pressure and cutoff transport directly and dominates (12.4) because $\Phi = 1$ on $\text{supp } \zeta$.

12.2. The uncoupled entropy balance. The whole-space smooth calculation identifies every term that the correction must address. Put

$$(12.5) \quad \mathcal{L}_c F := \partial_\tau F - \nu \Delta F + c \cdot \nabla F, \quad c = V - ay - b.$$

For the kinetic density $e = |V|^2/2$, the repaired equation gives

$$(12.6) \quad \mathcal{L}_c e = -\nu |\nabla V|^2 + 2ae - \text{div}(PV).$$

Indeed, after moving the dilation and translation derivatives to the left, (5.4) becomes

$$\partial_\tau V + c \cdot \nabla V + \nabla P = \nu \Delta V + aV.$$

Taking its scalar product with V proves (12.6).

Proposition 12.1 (Scale-covariant kinetic–entropy balance). *Let (V, P) and a positive probability density ρ be smooth on $\mathbb{R}^3$, with sufficient decay, and suppose that ρ solves the unnormalized whole-space adjoint equation. Set*

$$\bar{V}_\rho(\tau) := \int_{\mathbb{R}^3} V \rho \, dy, \quad \mathcal{K}_\rho^c(\tau) := \frac{1}{2} \int_{\mathbb{R}^3} |V - \bar{V}_\rho|^2 \rho \, dy.$$

Then

$$(12.7) \quad \frac{d}{d\tau} \mathcal{K}_\rho^\circ = -\nu \int |\nabla V|^2 \rho + 2a\mathcal{K}_\rho^\circ - \int P(V - \bar{V}_\rho) \cdot \nabla f \rho.$$

Let θ solve (8.7). For every $\beta > 0$, define

$$(12.8) \quad \mathcal{G}_\beta := \theta \mathcal{K}_\rho^\circ - \beta \mathcal{W}_{\text{ad}}.$$

Then

$$(12.9) \quad \begin{aligned} \frac{d}{d\tau} \mathcal{G}_\beta = & -\mathcal{K}_\rho^\circ - \theta \nu \int |\nabla V|^2 \rho - 2\beta \nu^2 \theta \int \left| \nabla^2 f - \frac{1}{2\nu\theta} I \right|^2 \rho \\ & - \theta \int P(V - \bar{V}_\rho) \cdot \nabla f \rho + 2\beta \nu \theta \int (\nabla f)^T (\nabla V)_{\text{sym}} \nabla f \rho. \end{aligned}$$

Proof. Apply the whole-space form of Lemma 8.3 componentwise to the represented velocity equation. Since $\mathcal{L}_c V = aV - \nabla P$,

$$(12.10) \quad \bar{V}_\rho' = a\bar{V}_\rho - \int_{\mathbb{R}^3} \nabla P \rho \, dy.$$

Applying the same transport identity to (12.6) and integrating by parts gives

$$-\int \operatorname{div}(PV) \rho = \int PV \cdot \nabla \rho = -\int PV \cdot \nabla f \rho.$$

Subtracting $\frac{1}{2}|\bar{V}_\rho|^2$ and using (12.10) proves (12.7). In particular,

$$(\theta \mathcal{K}_\rho^\circ)' = -\mathcal{K}_\rho^\circ - \theta \nu \int |\nabla V|^2 \rho - \theta \int P(V - \bar{V}_\rho) \cdot \nabla f \rho,$$

because $\theta' + 2a\theta = -1$. Subtracting β times (8.17) proves (12.9). $\square$

Thus the covariant heat scale removes the scalar modulation from the kinetic variance as well as from the adjoint entropy. No sign or integrability condition on a is used in this cancellation. The strain term in (12.9) combines exactly with the viscous and Gaussian terms.

Proposition 12.2 (Strain–Gaussian square completion). *Under the hypotheses of Proposition 12.1, set*

$$S := \frac{1}{2}(\nabla V + (\nabla V)^T), \quad \Omega := \frac{1}{2}(\nabla V - (\nabla V)^T), \quad A_f := \nabla^2 f - \frac{1}{2\nu\theta} I.$$

Then

$$(12.11) \quad \int_{\mathbb{R}^3} \rho S : (\nabla f \otimes \nabla f) \, dy = \int_{\mathbb{R}^3} \rho S : A_f \, dy.$$

Consequently the entropy balance has the exact form

$$(12.12) \quad \frac{d}{d\tau} \mathcal{G}_\beta + \mathcal{K}_\rho^\circ + \mathfrak{Q}_{\beta,\theta}(V, f) = -\theta \int_{\mathbb{R}^3} P(V - \bar{V}_\rho) \cdot \nabla f \rho \, dy,$$

where

$$(12.13) \quad \begin{aligned} \mathfrak{Q}_{\beta,\theta}(V, f) := & \theta \nu \int_{\mathbb{R}^3} |\Omega|^2 \rho \, dy + \theta \left(\nu - \frac{\beta}{2} \right) \int_{\mathbb{R}^3} |S|^2 \rho \, dy \\ & + 2\beta \nu^2 \theta \int_{\mathbb{R}^3} \left| A_f - \frac{1}{2\nu} S \right|^2 \rho \, dy. \end{aligned}$$

If $0 < \beta \leq \nu$, then

$$(12.14) \quad \mathfrak{Q}_{\beta,\theta}(V, f) \geq \frac{\theta \nu}{2} \int_{\mathbb{R}^3} |\nabla V|^2 \rho \, dy.$$

In particular, the fixed choice

$$(12.15) \quad 0 < \beta \leq \nu$$

ensures (12.14). On every retained window, (8.6) further gives

$$(12.16) \quad \mathfrak{Q}_{\beta,\theta}(V, f) \geq \frac{\nu \kappa_* e^{-2M_*}}{2} \int_{\mathbb{R}^3} |\nabla V|^2 \rho \, dy.$$

Proof. Since $\rho = (4\pi\nu\theta)^{-3/2} e^{-f}$,

$$(12.17) \quad \nabla^2 \rho = \rho(\nabla f \otimes \nabla f - \nabla^2 f).$$

Incompressibility gives

$$\partial_i \partial_j S_{ij} = \Delta \operatorname{div} V = 0.$$

The decay hypotheses therefore justify two integrations by parts and give

$$\int_{\mathbb{R}^3} S : \nabla^2 \rho \, dy = \int_{\mathbb{R}^3} \rho \partial_i \partial_j S_{ij} \, dy = 0.$$

Using (12.17) and $\operatorname{tr} S = \operatorname{div} V = 0$ yields

$$\int \rho S : (\nabla f \otimes \nabla f) = \int \rho S : \nabla^2 f = \int \rho S : A_f.$$

This proves (12.11). Substitution into (12.9) gives (12.12). Finally,

$$\begin{aligned} & \theta\nu |\nabla V|^2 + 2\beta\nu^2 \theta |A_f|^2 - 2\beta\nu \theta S : A_f \\ &= \theta\nu |\Omega|^2 + \theta \left(\nu - \frac{\beta}{2} \right) |S|^2 + 2\beta\nu^2 \theta \left| A_f - \frac{S}{2\nu} \right|^2, \end{aligned}$$

because $|\nabla V|^2 = |S|^2 + |\Omega|^2$. This proves (12.13). When $\beta \leq \nu$, the first two terms on its right control $(\theta\nu/2)(|S|^2 + |\Omega|^2)$, proving (12.14). Combining this with the lower bound in (8.6) proves (12.16). $\square$

Lemma 12.3 (Localized strain transmutation). *Let V, ρ, f be smooth on B_R , with $\operatorname{div} V = 0$, let $\chi \in C_c^\infty(B_R)$, and assume $\rho = (4\pi\nu\theta)^{-3/2} e^{-f}$ and $\int_{B_R} \chi^2 \rho = 1$. With A_f as in Proposition 12.2,*

$$(12.18) \quad \int_{B_R} \chi^2 \rho S : (\nabla f \otimes \nabla f) \, dy = \int_{B_R} \chi^2 \rho S : A_f \, dy + \mathfrak{C}_\chi(S, \rho),$$

$$(12.19) \quad \mathfrak{C}_\chi(S, \rho) := - \int_{B_R} \rho S : \nabla^2(\chi^2) \, dy - 2 \int_{B_R} S : (\nabla(\chi^2) \otimes \nabla \rho) \, dy.$$

Writing $\mathcal{A}_\chi := \operatorname{supp}(|\nabla \chi| + |\nabla^2 \chi|)$, one has for every $\hat{\chi} \in C_c^\infty(B_R)$ satisfying $\hat{\chi} = 1$ on a neighborhood of $\mathcal{A}_\chi$,

$$(12.20) \quad \begin{aligned} |\mathfrak{C}_\chi(S, \rho)| &\leq \|\nabla^2(\chi^2)\|_\infty \int_{\mathcal{A}_\chi} \rho |S| \, dy \\ &\quad + 2\|\nabla(\chi^2)\|_\infty \left(\int_{\mathcal{A}_\chi} \rho |S|^2 \, dy \right)^{1/2} \left(\int_{B_R} \hat{\chi}^2 \frac{|\nabla \rho|^2}{\rho} \, dy \right)^{1/2} \end{aligned}$$

Moreover, with $m = \chi^2 \rho$, the localized quadratic form

$$(12.21) \quad \begin{aligned} \mathfrak{Q}_{\beta,\theta}^\chi(V, f) &:= \theta\nu \int_{B_R} m |\Omega|^2 \, dy + \theta \left(\nu - \frac{\beta}{2} \right) \int_{B_R} m |S|^2 \, dy \\ &\quad + 2\beta\nu^2 \theta \int_{B_R} m \left| A_f - \frac{S}{2\nu} \right|^2 \, dy \end{aligned}$$

satisfies, whenever $0 < \beta \leq \nu$,

$$(12.22) \quad \mathfrak{D}_{\beta, \theta}^{\chi}(V, f) \geq \frac{\theta \nu}{2} \int_{B_R} m |\nabla V|^2 dy = \frac{\theta}{2} \mathfrak{D}_{\nu, \Gamma}(V).$$

Proof. Identity (12.17) holds on B_R . Moreover $\partial_i \partial_j S_{ij} = 0$, as in the proof of Proposition 12.2. Testing this distributional identity against the compactly supported function $\chi^2 \rho$ gives

$$0 = \int_{B_R} S : \nabla^2(\chi^2 \rho) dy.$$

Expanding the Hessian and using the symmetry of S yields

$$\int_{B_R} \chi^2 S : \nabla^2 \rho dy = - \int_{B_R} \rho S : \nabla^2(\chi^2) dy - 2 \int_{B_R} S : (\nabla(\chi^2) \otimes \nabla \rho) dy.$$

Substitute (12.17), then use $\text{tr } S = 0$. This proves (12.18) and (12.19). The first term in (12.20) follows directly from support and the L^∞ norms of the cutoff derivatives. For the second, use

$$|S| |\nabla \rho| = (\sqrt{\rho} |S|) \left(\frac{|\nabla \rho|}{\sqrt{\rho}} \right)$$

and Cauchy–Schwarz on $\mathcal{A}_\chi$. The last integral over $\mathcal{A}_\chi$ is bounded by the displayed $\hat{\chi}$ -weighted Fisher information because $\hat{\chi} = 1$ there. The identity $|\nabla V|^2 = |S|^2 + |\Omega|^2$ and the same pointwise square completion as in Proposition 12.2 prove (12.21)–(12.22). $\square$

Thus velocity strain requires no independent correction in the interior smooth identity. The only localized discrepancy is the annular commutator $\mathfrak{C}_\chi$. The scalar scale term has disappeared exactly from (12.12). The sole unsigned interior term in the whole-space scale-covariant identity is the pressure transfer

$$-\theta \int P(V - \bar{V}_\rho) \cdot \nabla f \rho.$$

For a localized adjoint density there are, in addition, the explicitly displayed cutoff and normalization terms from (8.14). We now compare this structural identity with the local kinetic-energy balance, where pressure occurs as a signed cutoff flux.

In particular, the adjoint pairing $-\theta \int P(V - \bar{V}_\rho) \cdot \nabla f \rho$ is not identified with the moving-cutoff flux $\int PV \cdot \nabla \Phi$. The pressure-tail estimates apply to the latter. The proof therefore discards the adjoint balance before invoking monotonicity; no cancellation between these two distinct pressure terms is claimed.

12.3. The moving-cutoff kinetic energy. Fix a radial nonincreasing $\phi \in C_c^\infty(B_2)$, with $0 \leq \phi \leq 1$ and $\phi = 1$ on B_1 . Let $R : [\tau_0, \infty) \rightarrow [R_*, \infty)$ be locally absolutely continuous and nondecreasing, and put

$$(12.23) \quad \Phi(\tau, y) := \phi(y/R(\tau)), \quad \mathcal{E}_R(\tau) := \frac{1}{2} \int_{\mathbb{R}^3} |V|^2 \Phi dy.$$

Write $a_+ = \max\{a, 0\}$, $a_- = \max\{-a, 0\}$, and define

$$(12.24) \quad \tilde{\mathfrak{D}}_R(\tau) := \nu \int |\nabla V|^2 \Phi dy + a_+(\tau) \int |V|^2 \Phi dy.$$

The full unsigned boundary-flux density is

$$(12.25) \quad \begin{aligned} B_{\text{mov}}(\tau) := & \frac{1}{2} \left| \int |V|^2 \partial_\tau \Phi \right| + \frac{\nu}{2} \left| \int |V|^2 \Delta \Phi \right| + \frac{1}{2} \left| \int |V|^2 V \cdot \nabla \Phi \right| + \left| \int PV \cdot \nabla \Phi \right| \\ & + \frac{1}{2} |b(\tau)| \left| \int |V|^2 \nabla \Phi \right| + \frac{1}{2} a_-(\tau) \int |V|^2 \Phi + \frac{1}{2} a_+(\tau) \int |V|^2 (-y \cdot \nabla \Phi). \end{aligned}$$

Every integral in (12.24)–(12.25) is over $\mathbb{R}^3$. The pressure term is unchanged by adding a function of time to P , because $\int V \cdot \nabla \Phi = 0$.

Theorem 12.4 (Corrected moving-cutoff monotonicity). *Let (V, P) be a represented suitable weak solution of (5.4). If $B_{\text{mov}} \in L^1(\tau_0, \infty)$, then*

$$(12.26) \quad \mu_{\text{NS}}(\tau) := \mathcal{E}_R(\tau) + \int_{\tau}^{\infty} B_{\text{mov}}(s) \, ds$$

has a nonincreasing representative and, in distributions,

$$(12.27) \quad \frac{d}{d\tau} \mu_{\text{NS}} + \frac{1}{2} \tilde{\mathfrak{D}}_R \leq 0.$$

In particular $\mu_{\text{NS}} \geq 0$, and for $\tau_0 \leq s < t$,

$$(12.28) \quad \mu_{\text{NS}}(t) + \mathfrak{D}_{\text{NS}}((s, t]) \leq \mu_{\text{NS}}(s), \quad \mathfrak{D}_{\text{NS}}((s, t]) := \frac{1}{2} \int_s^t \tilde{\mathfrak{D}}_R(\tau) \, d\tau.$$

Proof. Testing the suitable local energy inequality with Φ gives

$$\begin{aligned} \frac{d}{d\tau} \mathcal{E}_R \leq & -\nu \int |\nabla V|^2 \Phi + \frac{1}{2} \int |V|^2 \partial_{\tau} \Phi + \frac{\nu}{2} \int |V|^2 \Delta \Phi + \frac{1}{2} \int |V|^2 V \cdot \nabla \Phi \\ & + \int P V \cdot \nabla \Phi - \frac{a}{2} \int |V|^2 \Phi - \frac{a}{2} \int |V|^2 y \cdot \nabla \Phi - \frac{1}{2} b \cdot \int |V|^2 \nabla \Phi. \end{aligned}$$

Add $\tilde{\mathfrak{D}}_R/2$. The viscous term leaves $-\frac{\nu}{2} \int |\nabla V|^2 \Phi \leq 0$, while $-aM/2 + a_+M/2 = a_-M/2$. Since ϕ is radial nonincreasing, $y \cdot \nabla \Phi \leq 0$; hence the scale-shell term is bounded above by the last term of (12.25). Taking absolute values of the other fluxes proves

$$\frac{d}{d\tau} \mathcal{E}_R + \frac{1}{2} \tilde{\mathfrak{D}}_R \leq B_{\text{mov}}.$$

The suitable energy-defect measure has the favorable sign and was discarded in this inequality. Differentiating the finite tail in (12.26) subtracts B_{mov} , proving (12.27). Both terms in (12.26) are nonnegative. $\square$

12.4. Canonical moving-cutoff hypothesis.

Analytic hypothesis 12.5 (Canonical moving-cutoff alternative). *For every putative terminal-loss orbit supplied by hypothesis 10.2, one of the following two alternatives holds.*

- (C1) *The orbit enters a local alternative already incompatible with a terminal singularity: pressure reconstruction or exterior leakage, multicore or same-point cascade, compact scale collapse, scale rigidity, dissipation divergence, thin negative drift, or the terminal Type I alternative.*
- (C2) *After restriction to a terminal tail $[\tau_0, \infty)$, there is a single nondecreasing locally absolutely continuous radius $R(\tau) \geq R_*$, with $R(\tau) \rightarrow \infty$, such that the cutoff Φ in (12.23) satisfies*

$$(12.29) \quad B_{\text{mov}} \in L^1(\tau_0, \infty).$$

The radius is chosen so that $\Phi = 1$ on B_{R_} , and hence on $\text{supp } \zeta$, for all $\tau \geq \tau_0$.*

Remark 12.6 (Construction required by the hypothesis). The moving monotonicity theorem itself is proved above. The hypothesis contains only the global scheduling and concentration–remainder statement. The pressure-tail lemma and tailored radius schedule must control the diffusion-shell, convection, pressure, and translation terms. Monotone scale interpolation removes the positive scale term, and Theorem 6.4 closes an ordinary compact nonpositive-drift recurrence. Appendix E records the remaining estimates. No adjoint reset law and no pressure correction to $\mathcal{W}_{\text{ad}}$ is required.

12.5. Equality states.

Definition 12.7 (Oscillation-removable equality state). A represented state is oscillation-removable when its velocity is spatially constant on the connected retained region $\{\zeta > 0\}$ for almost every represented time. The constant may depend on time. Such a state has zero Galilean-reduced oscillatory concentration. No assertion that a localized Dirichlet adjoint density equals the whole-space Gaussian is included in this definition; the Gaussian remains the equality model only for the whole-space identity (8.16).

Theorem 12.8 (Inner-Dirichlet and zero-production rigidity). *Let $\mathfrak{w} = (V, P, a, b; [0, 1])$ be a normalized window or a compact limit of normalized windows. If*

$$(12.30) \quad \int_0^1 \mathfrak{D}_{0,\zeta}(V(\tau)) \, d\tau = 0,$$

then $\mathfrak{w}$ is oscillation-removable and

$$(12.31) \quad \mathcal{C}_1^\circ(V; [0, 1]) = 0.$$

In particular, zero moving-localized dissipation on the window implies (12.31) whenever $\Phi = 1$ on $\text{supp } \zeta$.

Proof. Since $\nu > 0$, (12.30) gives $\nabla V = 0$ almost everywhere on the connected set $\{\zeta > 0\}$ for almost every τ . Hence $V(\cdot, \tau) = c(\tau)$ there. Its ζ^3 -weighted mean is the same vector $c(\tau)$, so the integrand in (10.2) vanishes. This proves (12.31). The final assertion follows from the bound $\mathfrak{D}_{\text{NS}}(I) \geq \frac{1}{2} \int_I \mathfrak{D}_{0,\zeta}$ and nonnegativity. $\square$

Proposition 12.9 (Moving-cutoff implication for the current alternative). *Assume the canonical moving-cutoff alternative hypothesis 12.5, including the PDE exclusion asserted in alternative (C1). Then the target-relative current-alternative hypothesis hypothesis 11.7 holds.*

Proof. Alternative (C1) is alternative (N1). In alternative (C2), let

$$\mu_{\text{cov}}((s, t]) := \mathfrak{D}_{\text{NS}}((s, t]) = \frac{1}{2} \int_s^t \tilde{\mathfrak{D}}_R(\tau) \, d\tau$$

on the terminal tail. The corrected localized energy balance (12.28) and $\mu_{\text{NS}} \geq 0$ give

$$\mu_{\text{cov}}([\tau_0, \infty)) \leq \mu_{\text{NS}}(\tau_0) < \infty.$$

Let $\mathcal{J}$ be the infinite set of indices for which the selected unit windows lie in $[\tau_0, \infty)$, and let I_j denote their pairwise disjoint half-open versions. The measure μ_{cov} is absolutely continuous with respect to represented time, so this endpoint convention does not change any integral. For a unit window translated to $[0, 1]$, set

$$\Delta_{\text{crit}}(\mathfrak{w}) := \frac{1}{2} \mathfrak{d}_\nu(\mathfrak{w}) = \frac{\nu}{2} \int_0^1 \int \zeta^2 |\nabla V|^2 \, dy \, d\tau.$$

Because $\Phi = 1$ on $\text{supp } \zeta$, one has $\mu_{\text{cov}}(I_j) \geq \Delta_{\text{crit}}(\mathfrak{w}_j)$. Lower semicontinuity is (A.8). If $\Delta_{\text{crit}} = 0$, Theorem 12.8 makes the state spatially constant on the retained connected core and hence oscillation-removable. These statements give (11.16)–(11.18). In the notation of Definition 11.2, take $\mathcal{K} = \mu_{\text{NS}}$, $\mathcal{D} = \mathfrak{D}_{\text{NS}}$, and $\mathcal{F}^+ = \mathcal{R}^+ = 0$. Then (12.28) is the critical localized-energy balance and flux domination is immediate. $\square$

13. MOVING-CUTOFF REALIZATION OF THE NORMALIZED GAP

This section identifies the abstract covariant dissipation of [Section 11](#) with the moving-localized dissipation. It is a concrete realization of the structural balance. No numerical estimate for the gap is needed: compactness upgrades the qualitative equality theorem to a uniform statement.

Lemma 13.1 (Compactness–rigidity gap principle). *Let $\mathcal{K}$ be a sequentially compact space, let $\mathcal{B} \subset \mathcal{K}$ be sequentially closed, and let $\mathfrak{d} : \mathcal{K} \rightarrow [0, \infty]$ be sequentially lower semicontinuous. If*

$$(13.1) \quad \{\mathfrak{d} = 0\} \cap \mathcal{B} = \emptyset,$$

then

$$(13.2) \quad \inf_{x \in \mathcal{B}} \mathfrak{d}(x) > 0.$$

Proof. If the infimum were zero, there would be $x_j \in \mathcal{B}$ with $\mathfrak{d}(x_j) \rightarrow 0$. Sequential compactness gives $x_j \rightarrow x_\infty$ after extraction. Closedness gives $x_\infty \in \mathcal{B}$, while lower semicontinuity gives $\mathfrak{d}(x_\infty) = 0$, contradicting (13.1). $\square$

For the Navier–Stokes application, $\mathcal{K}$ is the sequential closure of the selected windows from one canonical orbit in the full topology of clause (iii) of [hypothesis 10.2](#); $\mathcal{B}$ is the subset satisfying (10.6); and

$$(13.3) \quad \mathfrak{d}_\nu(\mathfrak{w}) := \int_0^1 \mathfrak{D}_{0,\zeta}(V(\tau)) \, d\tau.$$

Strong local L^3 convergence makes $\mathcal{B}$ closed, while (A.8) makes $\mathfrak{d}_\nu$ lower semicontinuous. Both statements are proved in [Proposition A.2](#). The first assertion of [Theorem 12.8](#) supplies (13.1). Thus the compactness gap uses no semicontinuity assertion about the corrected localized energy, the Gaussian Hessian, or the nonlocal ray distance.

Theorem 13.2 (Production gap on a canonical orbit). *Fix the canonical orbit generated by one putative terminal loss. Under alternative (C2) of [hypothesis 12.5](#), there exists $\delta_* = \delta_*(\text{orbit}) > 0$ such that every selected target-retaining window contained in the terminal tail satisfies*

$$(13.4) \quad \mathfrak{D}_{\text{NS}}(I_j) \geq \delta_*.$$

Proof. Apply [Lemma 13.1](#) to the closure of the selected windows with $\mathfrak{d} = \mathfrak{d}_\nu$, as described above. It gives a number $\delta_\nu > 0$, depending on this orbit, such that $\mathfrak{d}_\nu(\mathfrak{w}_j) \geq \delta_\nu$ for every selected window. Because $\Phi = 1$ on $\text{supp } \zeta$, the localized dissipation gives

$$\mathfrak{D}_{\text{NS}}(I_j) \geq \frac{\nu}{2} \int_{I_j} \int \zeta^2 |\nabla V|^2 = \frac{1}{2} \mathfrak{d}_\nu(\mathfrak{w}_j) \geq \frac{1}{2} \delta_\nu.$$

Set $\delta_* := \delta_\nu/2$. $\square$

Corollary 13.3 (Divergence on the retained target component). *For the infinite pairwise disjoint selected windows of the canonical orbit,*

$$(13.5) \quad \sum_{j=1}^{\infty} \mathfrak{D}_{\text{NS}}(I_j) = \infty.$$

More generally, any nonnegative countably additive measure μ satisfying (G.6) on those windows obeys $\sum_j \mu(I_j) = \infty$.

Proof. The first conclusion follows from (13.4). The general conclusion is [Corollary G.3](#). Only the component selected by the retained target is used; no estimate of target-null transport or material-frame motion enters the sum. $\square$

Corollary 13.4 (One-window entropy estimate). *Every selected target-retaining window $I_j = [\tau_j, \tau_j + 1]$ contained in the terminal tail of the fixed canonical orbit satisfies*

$$(13.6) \quad \mu_{\text{NS}}(\tau_j + 1) \leq \mu_{\text{NS}}(\tau_j) - \delta_*.$$

Proof. Apply (12.28) on I_j and then use (13.4). $\square$

14. COVARIANT EXCLUSION OF SINGULARITIES

Proof of Theorem 1.1. Fix a global finite-energy suitable weak solution (u, p) , and assume that $x_* \in \Sigma_u(T)$. The canonical critical-orbit hypothesis gives one forward orbit and normalized target-retaining windows with pairwise disjoint interiors $I_j = [\tau_j, \tau_j + 1]$, $j \geq 1$. Relabel them so that $\tau_j + 1 \leq \tau_{j+1}$.

Apply the target-relative current-alternative hypothesis. Alternative (N1) is incompatible with retention of the singular target. In alternative (N2), discard the finite prefix before τ_0 . The normalized rigidity theorem gives $\Delta_{\text{crit}}(\mathbf{w}_j) \geq \delta_{\text{rig}} > 0$, while (11.17) gives $\mu_{\text{cov}}(I_j) \geq \delta_{\text{rig}}$. This is excluded by Theorem 11.9. Alternative (N3) is excluded by Theorem C.26: a first failure belongs to one of the ordered local alternatives in (N1), whereas the bounded marked residual contradicts product-scale decay of quotient activity. Hence $\Sigma_u(T) = \emptyset$. No density assumption on the selected times is required; only the existence of infinitely many disjoint target-retaining windows matters.

For $u_0 \in H_\sigma^m(\mathbb{R}^3)$, $m > 5/2$, let T_* be the maximal classical existence time and suppose $T_* < \infty$. Choose a global finite-energy suitable Leray solution. It agrees with the classical solution on $[0, T_*)$ by weak-strong uniqueness and supplies a suitable continuation through T_* . Apply case (b) of hypothesis 10.2. It produces a canonical orbit even if the concentrating centers escape to spatial infinity. The three current alternatives give the same contradiction as above. Hence $T_* = \infty$. Standard parabolic regularity then gives smoothness on every compact positive time interval, and weak-strong uniqueness identifies the Leray solution with this global classical solution. $\square$

The proof uses each hypothesis once. The critical-orbit hypothesis turns terminal loss into infinitely many compact quotient states. The target-relative current alternative says that a retained state is reduced, charged to one finite critical net-dissipation measure, or carried by regenerated critical current. On the last output, ordered failures return to the first alternative and the bounded residual is excluded directly by finite-depth rebase composition. Compactness gives a positive lower bound in the finite measure on the second output, and countable additivity gives the contradiction. The adjoint entropy, pressure atlas, moving radius, and Type I/Type II routes are constructions of the two hypotheses; they do not appear in the final deduction.

Remark 14.1 (Remaining self-contained argument). The remaining equation-specific theorem is the finite-stage concentration-remainder and current alternative. It must construct the signed localized-energy balance in one coherent scale-space chart and prove $\mathcal{F}^+ \leq \mathcal{D}$ after true exterior input has been separated, unless the undominated current has the regenerated support described in Definition C.15. The latter case is now reduced to an ordered first-failure output or excluded by Theorem C.22; no critical-tail exclusion is assumed for the bounded marked residual. Cancellation alone does not prove the measure inequality. Appendix F records the exact flux statement; Section 6 and Appendix G give the target-null equality classification on the compact branch.

APPENDIX A. WEAK COVARIANT CALCULUS AND COMPACT STABILITY

This appendix records the standard weak covariant identities and proves the local stability statements used in the gap theorem. It does not supply the strong orbit compactness or the moving-cutoff schedule; those remain in the stated analytic hypotheses.

A.1. Regularization of the equation. Let $S_h v(t) = h^{-1} \int_t^{t+h} v(s) ds$ be a Steklov average on an interior interval, and let v_ε denote spatial mollification. After a diagonal choice,

$$(A.1) \quad u_{\varepsilon,h} \rightarrow u \quad \text{strongly in } L_{\text{loc}}^2 H_{\text{loc}}^1,$$

with weak-* convergence in $L^\infty L^2$. The nonlinear commutator is

$$(A.2) \quad \mathcal{C}_\varepsilon := \operatorname{div}((u \otimes u)_\varepsilon - u_\varepsilon \otimes u_\varepsilon).$$

Since $u \in L_{\text{loc}}^{8/3} L_{\text{loc}}^4$,

$$(A.3) \quad \mathcal{C}_\varepsilon \rightarrow 0 \quad \text{in } L_{\text{loc}}^{4/3} H_{\text{loc}}^{-1}.$$

On a compact repaired interval, spatial mollification commutes with the translation derivative, while

$$(A.4) \quad a[y \cdot \nabla, \varphi_\varepsilon *]V \rightarrow 0 \quad \text{in } L_{\text{loc}}^1 L_{\text{loc}}^2.$$

The spatial commutator is uniformly L^2 -bounded and converges strongly to zero; dominated convergence uses $a \in L_{\text{loc}}^1$ and the local $L_\tau^\infty L_y^2$ bound. Moreover,

$$\|(b_1 - b_2) \cdot \nabla V\|_{H^{-1}(B_R)} \leq c_R |b_1 - b_2| \|V\|_{L^2(B_R)},$$

with the analogous estimate for $(a_1 - a_2)(V + y \cdot \nabla V)$.

Proposition A.1 (Weak validity of the covariant identities). *For every suitable weak solution:*

- (i) $R(u) = 0$ in distributions, and $\mathfrak{D}_\tau^{X,\Lambda} V + \nu A_S V = 0$ on every repaired compact cylinder;
- (ii) the global energy balance is (3.11);
- (iii) the localized identity is (9.18) for every admissible smooth compactly supported weight;
- (iv) the Stokes-metric projection inequalities hold for almost every weak time.

Proof. For item (i), write the mollified equation with residual $\mathcal{C}_\varepsilon$. The linear and pressure terms pass by distributional convergence, the nonlinear residual vanishes by (A.3), and the modulation terms pass by (A.4) and L^1 -approximation of a, b . Item (ii) is the Leray–Hopf energy inequality in the measure form proved in Theorem 3.3. For item (iii), test the suitable local energy inequality by w_Γ , and test the weak momentum equation componentwise by the same scalar weight. The latter identity gives the distributional derivative of $M_\Gamma^{-1} |\int w_\Gamma u|^2 / 2$; subtracting it from the weighted kinetic-energy relation gives exactly (9.18), including the same nonnegative local defect measure. Approximation of temporal indicators gives the asserted one-sided representatives. Item (iv) is Cauchy–Schwarz in the solenoidal Gelfand triple. $\square$

A.2. Compactness of the variational data. Suppose, on compact cylinders,

$$(A.5) \quad V_j \rightarrow V \quad \text{strongly in } L_{\text{loc}}^2 \text{ and } L_{\text{loc}}^3, \quad \nabla V_j \rightharpoonup \nabla V \quad \text{in } L_{\text{loc}}^2,$$

with the corresponding pressure and modulation convergence. For the additional adjoint-weighted conclusions, assume

$$(A.6) \quad m_j \rightarrow m \quad \text{strongly in } L^\infty(B_R \times I), \quad m_j \geq m_* > 0 \quad \text{on } \operatorname{supp} \zeta \times I,$$

and, for normalized adjoint densities,

$$(A.7) \quad \inf_j \inf_\tau Z_j(\tau) > 0, \quad \sup_j \|\tilde{\rho}_j\|_{L^\infty L^2 \cap L^2 H_0^1} < \infty, \quad \sup_j \int_I \int_{B_R} \chi_j^2 |\nabla \sqrt{\rho_j}|^2 dy d\tau < \infty.$$

Proposition A.2 (Stability of the local variational data). *Under (A.5):*

- (i) *the fixed-cutoff means $\bar{V}_{j,\zeta}$ converge strongly in $L^3(I)$, and $C_{\Gamma_j}^\circ(V_j; I) \rightarrow C_\Gamma^\circ(V; I)$;*
- (ii) *the adjoint-weighted kinetic variances converge in $L^1(I)$ whenever $m_j \rightarrow m$ strongly on the full support of the probe;*
- (iii) *the fixed inner Dirichlet floor is lower semicontinuous:*

$$(A.8) \quad \int_I \mathfrak{D}_{0,\zeta}(V) \, d\tau \leq \liminf_{j \rightarrow \infty} \int_I \mathfrak{D}_{0,\zeta}(V_j) \, d\tau;$$

if in addition (A.6)–(A.7) hold, then the adjoint-weighted viscous production is also lower semicontinuous:

$$(A.9) \quad \int_I \mathfrak{D}_{\nu,\Gamma}(V) \, d\tau \leq \liminf_{j \rightarrow \infty} \int_I \mathfrak{D}_{\nu,\Gamma_j}(V_j) \, d\tau;$$

- (iv) *under (A.6), the positivity bound for m on $\text{supp } \zeta$ passes to the limit.*

No lower-semicontinuity claim for the Gaussian Hessian square, the varying-ray distance, or the corrected production is made here, and none is used in the gap theorem. Retaining such terms in a stronger weak-production statement would require identification of the adjoint logarithmic Hessian and all associated localization terms; that refinement is not part of the moving-cutoff argument.

Proof. The first assertion follows from strong L^3 convergence and Hölder’s inequality on the fixed support of ζ . The same argument, together with strong convergence of the weights, gives the second assertion. For the third, $\zeta \nabla V_j \rightharpoonup \zeta \nabla V$ weakly in L^2 , which proves (A.8). Under the additional weight convergence, $m_j^{1/2} \nabla V_j \rightharpoonup m^{1/2} \nabla V$ weakly in L^2 ; weak lower semicontinuity of the L^2 -norm gives (A.9). Uniform convergence of the weights gives the final assertion. $\square$

APPENDIX B. WEAK MATERIAL PULLBACK

The identities in Proposition 4.1 are classical for a smooth flow. Their use on a compact critical orbit requires a separate weak construction; it is not supplied by the Eulerian compactness statement (10.7).

B.1. Regularized material flows. For the smooth approximants used in Appendix A, construct the volume-preserving flows X_ε , deformation tensors $\mathbf{F}_\varepsilon$, and metrics $\mathbf{C}_\varepsilon$. Prove (4.3)–(4.7) before taking any limit. The approximation must preserve incompressibility and must record the commutator produced by mollifying convection.

B.2. Material compactness modulo relabeling. Specify a material relabeling gauge and a topology in which the metric, inverse metric, pulled-back velocity, and pulled-back vorticity have compact subsequences on retained cores. Establish enough convergence to pass the signed weak form of (4.6). No pointwise convergence of $\mathbf{F}_\varepsilon$ and no classical flow for an arbitrary suitable solution may be assumed. If only a regular Lagrangian flow is available, the proof must state the compressibility, composition, and chain-rule properties used in every pullback.

B.3. Compatibility with scale–translation charts. Prove that material relabeling commutes with the scale–translation composition law (10.9). Identify the transformation of $\mathbf{F}$, $\mathbf{C}$, and $\mathbf{C}^{-1}$ under a parabolic chart change, and show that the zero-dissipation equations are independent of the chosen representative. This hypothesis is required before material reduction can be invoked on a compact normalized window.

APPENDIX C. SINGULAR-GERM EXTRACTION

This appendix states the estimates needed to derive [hypothesis 10.2](#). The derivation is complete only when the following statements are proved directly from the suitable local energy inequality, pressure reconstruction, and parabolic compactness, without invoking global regularity.

C.1. Velocity concentration. The local concentration assertion was proved in [Theorem 2.1](#). For the global regularity problem one can start more directly from a maximal classical solution. The following record-point construction removes the existence of an ancient PDE profile from the list of assumptions.

Theorem C.1 (Record-point ancient-profile extraction). *Let u be a classical finite-energy solution of (1.1) on $[0, T)$, with divergence-free initial datum in $H^m(\mathbb{R}^3)$, $m > 5/2$, and suppose that $T < \infty$ is its maximal classical time. Then there are $t_j \uparrow T$, $x_j \in \mathbb{R}^3$, and $r_j \downarrow 0$ such that*

$$(C.1) \quad r_j^{-1} = M_j := \sup_{0 \leq t \leq t_j} \|u(t)\|_{L^\infty}, \quad |u(x_j, t_j)| \geq \frac{1}{2} M_j,$$

and the rescaled solutions

$$(C.2) \quad U_j(y, s) := r_j u(x_j + r_j y, t_j + r_j^2 s), \quad Q_j(y, s) := r_j^2 p(x_j + r_j y, t_j + r_j^2 s)$$

are defined on $\mathbb{R}^3 \times (-t_j/r_j^2, 0]$, satisfy

$$(C.3) \quad \|U_j\|_{L^\infty(\mathbb{R}^3 \times (-t_j/r_j^2, 0])} \leq 1, \quad |U_j(0, 0)| \geq \frac{1}{2},$$

and have a subsequence converging locally smoothly in velocity, and locally in pressure modulo functions of time, to a bounded mild ancient solution (U, Q) on $\mathbb{R}^3 \times (-\infty, 0]$. The limit satisfies

$$(C.4) \quad \|U\|_{L^\infty} \leq 1, \quad |U(0, 0)| \geq \frac{1}{2}.$$

Every further spacetime translation with translation time tending to $-\infty$ has, after pressure normalization and passage to a subsequence, a bounded mild entire Navier–Stokes limit.

Proof. The standard L^∞ continuation criterion implies $\limsup_{t \uparrow T} \|u(t)\|_\infty = \infty$; otherwise the mild solution can be continued a uniform positive time beyond T . Choose record times $t_j \uparrow T$ for which $M_j \rightarrow \infty$, and choose x_j as in (C.1). Such approximate maximizers are sufficient; for H^m -data the velocity is continuous and vanishes at spatial infinity at every time. Then $r_j = M_j^{-1} \downarrow 0$, and parabolic invariance gives the equation for (U_j, Q_j) . The record property gives (C.3). Moreover $t_j/r_j^2 = t_j M_j^2 \rightarrow \infty$.

Fix $R, L < \infty$. For all large j , the cylinder $B_{2R} \times [-2L, 0]$ lies in the domain of U_j . Bounded mild estimates, applied from a slightly earlier time, give uniform interior derivative bounds for the velocity on $B_R \times [-L, 0]$. Equivalently, one may combine the local energy inequality, the local Calderón–Zygmund/harmonic pressure decomposition, and interior parabolic regularity. Normalize Q_j by its spatial mean on each member of a countable nested cylinder exhaustion. The local pressure bounds and a diagonal Arzelà–Ascoli extraction give a smooth velocity limit and a compatible local pressure atlas. Passing to the mild formula proves that the limit is mild. The pointwise normalization passes to the limit and gives (C.4).

Finally, translate (U, Q) by times $s_j \rightarrow -\infty$. Every fixed compact time interval then lies in the translated domain. The same uniform bounded-mild estimates and diagonal argument give a solution on all of $\mathbb{R}^3 \times \mathbb{R}$. Spatial translations and constant parabolic rescalings commute with this extraction and preserve the assertion. $\square$

The nonzero value in (C.4) is not yet the Galilean-invariant target: a locally constant ancient limit can carry that normalization. The next selection must either produce a fixed positive oscillation after removal of the homogeneous mode or pass to the first larger parabolic scale on which such oscillation appears. Finite energy guarantees that the approximating physical fields are not nonzero constants on all of $\mathbb{R}^3$, but converting this fact into a compact oscillatory descendant is part of the concentration–remainder theorem below; it is not inferred from (C.4) alone.

C.2. Compactness selection. The endpoint of a finite descendant chain is controlled by an existing ancient-solution Liouville theorem. We record its exact form because it is the reason that only infinite or diffuse successor chains require the new covariant argument.

Theorem C.2 (Endpoint ancient Liouville theorem). *Let U be a mild ancient solution of the three-dimensional Navier–Stokes equations on $\mathbb{R}^3 \times (-\infty, 0)$. If there are times $s_k \downarrow -\infty$ such that*

$$(C.5) \quad \sup_k \|U(\cdot, s_k)\|_{L^3(\mathbb{R}^3)} < \infty,$$

then $U \equiv 0$.

Proof. This is the sequence- L^3 Liouville theorem of Albritton and Barker [1, Theorem 1.2]. Their statement requires mildness but no terminal L^3 hypothesis. $\square$

For a bounded ancient velocity define the unit quotient activity

$$(C.6) \quad \mathfrak{a}_U(z) := \inf_{c \in \mathbb{R}^3} \int_{Q_1(z)} |U - c|^3 dx dt, \quad z = (x, t), \quad t < -1.$$

The pressure is not part of this target. On a bounded mild family it is reconstructed after subtracting spatial means, and ordinary spacetime translations are compact on every fixed cylinder whose translated times tend to $-\infty$.

Proposition C.3 (Backward concentration–remainder alternative). *Let U be a bounded mild ancient solution on $\mathbb{R}^3 \times (-\infty, 0)$. After passage to subsequences, at least one of the following conclusions can be selected; the first applicable conclusion in the displayed order may be used.*

- (i) *There are $s_k \downarrow -\infty$ satisfying (C.5).*
- (ii) *After removal of one spatially homogeneous mild mode, there are $z_k = (x_k, t_k)$, $t_k \rightarrow -\infty$, and $\eta > 0$ such that $\mathfrak{a}_U(z_k) \geq \eta$. The translates about z_k have a locally smooth subsequence converging to a bounded mild entire solution W with $\mathfrak{a}_W(0, 0) \geq \eta$.*
- (iii) *There are $t_k \rightarrow -\infty$, $R_k \rightarrow \infty$, and measurable sets $E_k \subset \{|x| > R_k\}$ such that*

$$(C.7) \quad 1 \leq \int_{E_k} |U(x, t_k)|^3 dx \leq 2,$$

whereas every unit parabolic recentering meeting $E_k \times (t_k - 1, t_k + 1)$ has quotient activity tending to zero. This is the diffuse remainder.

- (iv) *The selected exterior fields converge locally, after subtraction of time-dependent constants, to a spatially homogeneous velocity. The momentum equation then pairs it only with an affine pressure, so the limit is a Galilean/parasitic quotient state and has zero target activity.*

In particular, a bounded ancient profile which is terminal under nontrivial spacetime descendants, has no diffuse remainder, and is not a homogeneous quotient state satisfies alternative (i) and is zero by Theorem C.2.

Proof. Put $A(t) = \int_{\mathbb{R}^3} |U(x, t)|^3 dx \in [0, \infty]$. If $\liminf_{t \rightarrow -\infty} A(t) < \infty$, choose a backward sequence on which A is bounded; this is (i). Otherwise $A(t) \rightarrow \infty$ as $t \rightarrow -\infty$. Since U is bounded, its L^3 -mass on every fixed ball is bounded uniformly in time. We may therefore choose $t_k \rightarrow -\infty$, radii $R_k \rightarrow \infty$, and, by absolute continuity of the integral, exterior sets E_k satisfying (C.7).

Cover $E_k \times (t_k - 1, t_k + 1)$ by a fixed bounded-overlap family of unit parabolic cylinders. If the supremum of the quotient activity on those cylinders has a positive limsup, choose one such cylinder for each k . Bounded mild interior estimates give uniform velocity derivative bounds on every compact subcylinder after recentering. The local Calderón–Zygmund/harmonic decomposition gives pressure compactness modulo time functions. Because $t_k \rightarrow -\infty$, the translated time domains exhaust $\mathbb{R}$. A diagonal extraction therefore gives an actual bounded mild entire solution. Strong local L^3 convergence preserves the positive quotient activity and proves (ii).

If the supremum tends to zero, retain the normalized exterior restrictions in (C.7). When their quotient oscillation remains visible only after aggregation over an increasing number of unit cylinders, they give exactly (iii). Otherwise the local Poincaré quotients vanish on every cylinder in each connected overlap component. The associated local constants agree up to an error tending to zero; diagonal passage gives a spatially homogeneous limit. Substitution in the momentum equation shows that its pressure is affine in space modulo a function of time. This is (iv). The four outcomes exhaust the covering dichotomy.

Under the terminal assumptions in the last sentence, alternatives (ii)–(iv) are unavailable, so (i) holds and Theorem C.2 applies. $\square$

The proposition is deliberately phrased in terms of actual translated PDE limits. The diffuse alternative is measure-valued at this stage and has not been declared an ordinary solution. To finish the independent proof one must show that its first scale of nonzero quotient oscillation produces an actual parabolic descendant, and that an infinite iteration of such descendants produces the stationary monotone ensemble of Definition 6.5. This is the remaining realization step; the next subsections formulate it without pressure-only or formal compactification outputs.

The first part of that realization is a stopping-time argument. For a suitable solution W , a center $z = (x_0, t_0)$, and a constant vector c , let

$$Q_r^c(z) := \{(x, t) : t_0 - r^2 < t < t_0, |x - x_0 - c(t - t_0)| < r\}$$

and write

$$(C.8) \quad C_W^\circ(z, r) := \inf_{c \in \mathbb{R}^3} r^{-2} \int_{Q_r^c(z)} |W - c|^3 dx dt.$$

This is the velocity criterion applied after an exact constant Galilean transformation; it is invariant under translations, Galilean changes, and parabolic scaling.

Proposition C.4 (First bad scale gives an ordinary descendant). *Fix $0 < \varepsilon_0 < \varepsilon_{\text{reg}}$, where ε_{reg} is the velocity regularity threshold used in Theorem 2.1. Let U be a bounded mild ancient solution and suppose that a diffuse remainder as in (C.7) contains points at which $U \neq 0$, after the homogeneous quotient has been removed. Then one of the following occurs.*

- (a) *A bounded-radius unit recentering carries a fixed positive quotient activity, and alternative (ii) of Proposition C.3 applies.*
- (b) *There are parabolic centers z_j , radii $\rho_j \rightarrow \infty$, and constant Galilean parameters c_j such that the exact rescalings*

$$(C.9) \quad W_j(y, s) := \rho_j (U(x_j + \rho_j y + c_j \rho_j^2 s, t_j + \rho_j^2 s) - c_j)$$

with the correspondingly transformed pressure are suitable on expanding backward cylinders, have a locally smooth subsequence modulo homogeneous velocity and affine-pressure modes, and converge to an ordinary bounded mild ancient solution W_∞ . On the nonhomogeneous branch,

$$(C.10) \quad \int_{Q_1} |W_\infty - \bar{W}_{\infty, Q_1}|^3 \geq c(\varepsilon_0) > 0.$$

Every compact subset of the limit is obtained by an exact parabolic rescaling of the original solution; no measure-valued state is used.

- (c) On the selected first-bad cylinders the local velocity bounds hold but the mean-subtracted pressures are not precompact in $L_{\text{loc}}^{3/2}$. In the local decomposition $P = P_{\text{loc}} + H$, the Calderón–Zygmund part is compact, so this failure is carried by a non-affine harmonic pressure tail sourced outside the selected cylinder.

If the rescaled forward time depth tends to infinity, the limit is entire. If it stays bounded, the standard terminal self-similar change of variables places the limit in (5.4) with a constant coefficient $a_* < 0$. Thus both time-depth alternatives have nonpositive scale drift.

Proof. For a smooth point z , one has $C_U^\circ(z, r) \rightarrow 0$ as $r \downarrow 0$. If $C_U^\circ(z, r) < \varepsilon_0$ on arbitrarily large backward cylinders, apply quantitative velocity ε -regularity after an almost minimizing Galilean transformation. Its derivative estimate gives

$$|\nabla U(z)| \leq \frac{C(\varepsilon_0)}{r^2}.$$

Thus a solution for which this happens at every point is spatially homogeneous. On the nonhomogeneous quotient branch there is consequently a point and a scale at which $C_U^\circ = \varepsilon_0$. Select the first such scale. If these first scales stay bounded, a bounded-radius cylinder has a positive oscillatory subsequence and gives (a).

Assume that the first scales tend to infinity. Apply the standard parabolic point-selection procedure. If a chosen cylinder contains a smaller cylinder at which the threshold is first reached and whose distance to the boundary, in its own units, is larger, replace the chosen cylinder by that one. Each replacement decreases the radius by a fixed factor while increasing the available normalized boundary distance. An infinite replacement chain in a fixed compact physical cylinder would have radii tending to zero and centers converging to a smooth point, contradicting $C_U(z, r) \rightarrow 0$. Thus, for the j -th selected cylinder, the procedure terminates with a radius ρ_j such that every subcylinder of radius at most ρ_j lying within normalized parabolic distance j has Galilean-reduced velocity quantity at most ε_0 . Absence of alternative (a) forces $\rho_j \rightarrow \infty$.

Rescale by (C.9). Cover a fixed compact cylinder in the new variables by finitely many subunit cylinders. On each of them the velocity criterion is below ε_{reg} , so interior regularity gives bounds for every velocity derivative on a slightly smaller cover. The local pressure decomposition

$$P = P_{\text{loc}} + H, \quad -\Delta P_{\text{loc}} = \partial_i \partial_j (\chi W_i W_j), \quad \Delta H = 0,$$

makes the local part compact after subtracting spatial means. If the harmonic parts are not compact modulo affine functions, alternative (c) holds. Otherwise the affine part is exactly the pressure paired with the removed homogeneous velocity, and diagonal compactness produces an ordinary suitable mild limit. The threshold equality at the selected cylinder passes by strong local L^3 convergence. If its value were carried only by the homogeneous mode, the limit belongs to the quotient alternative. Otherwise Poincaré’s inequality and the first-scale normalization give (C.10) after decreasing the constant once.

Let d_j^+ be the forward time depth of the rescaled physical domain. If $d_j^+ \rightarrow \infty$, the diagonal limit is entire. If d_j^+ is bounded, translate the terminal time to zero and use $t = -e^{-\tau}$, $\Lambda = e^{-\tau/2}$. The resulting equation is (5.8), up to a constant rescaling of τ , and hence has $a_* < 0$. These are the only two subsequential values of the time-depth ratio. $\square$

Proposition C.5 (Finite energy makes the non-affine pressure tail parabolically accessible). *Let u be a finite-energy solution on $(0, T)$, and let*

$$(C.11) \quad \widetilde{W}_j(y, s) := \lambda_j u(x_j + \lambda_j y, t_j + \lambda_j^2 s), \quad d_j := \frac{t_j}{\lambda_j^2} \longrightarrow \infty,$$

where $t_j \geq t_* > 0$ and $\lambda_j \downarrow 0$. A constant Galilean velocity may be subtracted when local oscillation is measured; the finite-energy representative $\widetilde{W}_j$ is retained for the whole-space pressure formula. Put $R_j = d_j^{1/4}$, and let

$$H_j^{>R_j} := \mathcal{R}_i \mathcal{R}_k (\mathbf{1}_{\{|y| > R_j\}} \widetilde{W}_{j,i} \widetilde{W}_{j,k}).$$

For every fixed $R < \infty$ and compact interval $I \Subset (-\infty, 0]$, the part of the pressure generated by $|y| > R_j$, after subtraction of its constant and linear Taylor terms on B_R , satisfies

$$(C.12) \quad \|\nabla^2 H_j^{>R_j}\|_{L^\infty(B_R \times I)} \leq C_{R, I, t_*, \|u_0\|_2} d_j^{-3/4} \longrightarrow 0$$

at every time at which the energy inequality holds. Moreover $R_j^2 = d_j^{1/2} = o(d_j)$. Consequently a failure of local pressure compactness modulo affine functions cannot be produced at spatial distances beyond the parabolic time depth. It must be sourced in the intermediate region

$$(C.13) \quad 2R < |y| < R_j, \quad -R_j^2 < s \leq 0,$$

up to a term converging to zero locally and an affine pressure mode.

Proof. The global energy inequality and (C.11) give, uniformly at admissible rescaled times,

$$(C.14) \quad \|\widetilde{W}_j(s)\|_2^2 \leq \lambda_j^{-1} \|u_0\|_2^2 = \sqrt{\frac{d_j}{t_j}} \|u_0\|_2^2 \leq t_*^{-1/2} \|u_0\|_2^2 d_j^{1/2}.$$

The pressure Poisson equation is unchanged by subtraction of a spatially constant velocity: the constant–constant term has zero second derivative, and the two cross terms vanish because the velocity is divergence free. Thus the tail may be estimated using the finite-energy representative.

The kernel of $\mathcal{R}_i \mathcal{R}_k$ is homogeneous of degree -3 , and its second spatial derivatives are bounded by $C|y|^{-5}$ away from the origin. Taylor’s formula on B_R therefore gives

$$\begin{aligned} \sup_{x \in B_R} |\nabla^2 H_j^{>R_j}(x, s)| &\leq C_R \int_{|y| > R_j} |y|^{-5} |\widetilde{W}_j(y, s)|^2 dy \\ &\leq C_R R_j^{-5} \|\widetilde{W}_j(s)\|_2^2 \leq C_{R, t_*, \|u_0\|_2} d_j^{-3/4}, \end{aligned}$$

because $R_j^5 = d_j^{5/4}$. The bound is uniform on a fixed compact rescaled time interval by the energy inequality. Interior estimates for harmonic functions control all lower local norms after the affine Taylor jet has been removed. Finally, $R_j^2 = d_j^{1/2} < d_j$ for large j , so the backward parabolic cylinders associated with the intermediate region lie inside the rescaled time domain. This proves the assertion. $\square$

Remark C.6 (What the pressure estimate does not prove). The accessibility estimate rules out a genuinely super-parabolic pressure source. It does not by itself turn the intermediate stress in (C.13) into a fixed positive Galilean-reduced L^3 oscillation on one cylinder. That implication requires a quantitative concentration–remainder lemma for the velocity. Treating the positive

dyadic L^2 tail in the pressure estimate as though it were already such an L^3 descendant would be invalid.

Remark C.7 (What the stopping time proves). [Proposition C.4](#) removes diffuse velocity mass as a terminal label: it becomes an actual velocity descendant or a homogeneous quotient state. It also isolates the only pressure obstruction as a non-affine harmonic tail with exterior ancestry; it does not silently call that tail compact. An infinite sequence of such actual descendants need not have a stationary law satisfying all invariances in [Definition 6.5](#). The marked construction in [Section C.3](#) retains the successor witnesses and their currents and routes the resulting recurrent path to a critical tail instead.

Definition C.8 (Actual target-retaining descendant). An actual target-retaining descendant of a bounded ancient represented solution is a locally smooth limit, with compatible local pressure gauges, of exact spacetime translations, constant Galilean changes, and constant parabolic rescalings of that solution, for which a fixed quotient-oscillation floor holds on Q_1 . A first-bad-scale limit is included only through [Proposition C.4](#). Thus every descendant is an ordinary Navier–Stokes solution; a diffuse measure, a failed pressure chart, or a formal boundary value is not a descendant.

Corollary C.9 (Finite ordinary descendant chains close). *Let U_0 be the record-point profile in [Theorem C.1](#), after removal of the homogeneous quotient. Iterate [Proposition C.3](#) and, on its diffuse branch, the ordinary compact alternative of [Proposition C.4](#). Assume in addition that every intermediate pressure source localized by [Proposition C.5](#) either yields an actual target-retaining first-bad-scale descendant or has only an affine local pressure limit. If the resulting target-retaining chain terminates after finitely many actual descendants, the last profile is impossible. Consequently, after this intermediate pressure-to-velocity implication has been proved, a putative terminal singularity can remain only through an infinite chain of actual target-retaining descendants.*

Proof. At a terminal node there is no further nonhomogeneous descendant and no diffuse remainder. The homogeneous alternative has zero quotient target. Therefore the last node satisfies the backward sequence- L^3 alternative of [Proposition C.3](#). It vanishes by [Theorem C.2](#), contradicting its retained quotient-oscillation floor. The same conclusion holds after a terminal self-similar change because its physical pullback is a bounded mild ancient solution and the critical L^3 -norm is invariant. $\square$

Lemma C.10 (Event law for an infinite realized chain). *Let K_η be a compact metric space of ordinary normalized represented solutions on which the quotient target $\mathcal{O}$ is continuous and $\mathcal{O} \geq \eta > 0$. Let $\mathcal{R} \subset K_\eta \times K_\eta$ be an actual-descendant relation. If*

$$(C.15) \quad w_0 \mathcal{R} w_1 \mathcal{R} w_2 \mathcal{R} \cdots ,$$

then the event-count measures

$$(C.16) \quad \nu_N := \frac{1}{N} \sum_{j=0}^{N-1} \delta_{w_j}$$

have weak- $$ limit points, every one of which is supported on $\{\mathcal{O} \geq \eta\}$. The closure of the shift orbit of the given one-sided sequence carries a shift-invariant probability measure whose zeroth marginal has the same property. No closedness of the profile-only relation is asserted.*

Proof. Compactness of K_η gives weak- $*$ compactness of $\mathcal{P}(K_\eta)$. Every measure in [\(C.16\)](#) is supported on the closed set $\{\mathcal{O} \geq \eta\}$, and the Portmanteau theorem preserves this support

condition under weak-* limits. The closure of the shift orbit of $(w_j)_{j \geq 0}$ is compact in $K_\eta^\mathbb{N}$. Weak-* limits of the shift empirical measures are shift invariant because, for every continuous F ,

$$\frac{1}{N} \sum_{j=0}^{N-1} (F(S^{j+1}w) - F(S^jw)) = \frac{F(S^Nw) - F(w)}{N} \longrightarrow 0.$$

Their zeroth marginals are limits of (C.16). The successor edge can be lost when its recentering witness escapes, so the conclusion deliberately concerns the event law and not a closed path relation. $\square$

Remark C.11 (The profile-only path space). If the unmarked relation happens to be closed, then the path space

$$(C.17) \quad \mathcal{P}_\eta := \{(v_j)_{j \geq 0} \in K_\eta^\mathbb{N} : v_j \mathcal{R} v_{j+1} \text{ for all } j\}$$

is nonempty and compact, is invariant under the left shift S , and carries an S -invariant Borel probability measure. Every coordinate marginal of that measure is target-retaining. The hypothesis is not available for the profile-only descendant relation: local convergence on fixed cylinders does not control observers whose centers escape. The marked construction below restores closedness by retaining those observers.

Proposition C.12 (A rooted target is not preserved by an invariant hull). *Let Q be a fixed bounded cylinder and define*

$$\mathcal{O}_Q(V) := \inf_{c \in \mathbb{R}^3} \int_Q |V - c|^3.$$

Suppose that spatial translates $T_{x_j}V$ converge strongly in $L^3(Q)$ to a spatially constant field as $|x_j| \rightarrow \infty$. Then

$$(C.18) \quad \mathcal{O}_Q(T_{x_j}V) \longrightarrow 0.$$

Consequently the closed covariant hull of a locally concentrated profile need not remain in $\{\mathcal{O}_Q \geq \eta\}$, even when the profile itself has $\mathcal{O}_Q(V) \geq \eta$. In particular, invariance of a probability law under the successor shift does not imply the spatial invariance required in Definition 6.5 while retaining the root target.

Proof. The map $V \mapsto \mathcal{O}_Q(V)^{1/3}$ is 1-Lipschitz in $L^3(Q)$: apply the triangle inequality before taking the infimum over constants, and then interchange the two fields. Its value on a spatially constant field is zero, which proves (C.18). The remaining assertions follow because a covariantly invariant hull contains all finite spatial translates and their local limit points, whereas the successor shift acts on a sequence of separately recentered profiles. $\square$

C.3. Marked descendant structure for regenerated current. The successor shift and the covariant translation action are different dynamical systems. The preceding proposition prevents replacing one by the other. We instead retain the geometric and local-energy data which realize each successor. The resulting relation is closed because an escaping witness is recorded as an exterior, pressure, diffuse, or scale-tail output rather than discarded.

Fix once and for all a countable family $\mathcal{T}$ of nonnegative smooth parabolic cutoffs which is dense in the test-function Fréchet topology on each member of a compact-cylinder exhaustion. The stopping radii and times are chosen from the full-measure set for which the endpoint energy traces and the coarea limits of every test in $\mathcal{T}$ exist. Equivalently, the boundary currents below may be defined as limits of smooth collars. The local energy distribution, including its nonnegative Radon defect, then defines for every physical parabolic cell C the coordinates

$$(C.19) \quad \mathbf{q}_C(\psi) := (K_C^-(\psi), K_C^+(\psi), D_C(\psi) + M_C(\psi), J_C(\psi)), \quad \psi \in \mathcal{T},$$

where the endpoint energy traces and the complete signed current are oriented with respect to C . The suitable-energy defect is included in the third coordinate, so the balance is an equality.

Definition C.13 (Marked successor record). Let $C_j = Q_{r_j}(z_j)$ and $C_{j+1} = Q_{r_{j+1}}(z_{j+1}) \Subset Q_{r_{j/2}}(z_j)$ be two actual cells cut from the same suitable solution. A marked successor record is

$$(C.20) \quad \mathbf{m}_j = (\mathbf{w}_j, \mathbf{w}_{j+1}, C_j, \lambda_j, \xi_j, \theta_j, \gamma_j, \mathbf{q}_j, \mathbf{s}_j),$$

where

$$(C.21) \quad \lambda_j = \frac{r_{j+1}}{r_j}, \quad \xi_j = \frac{x_{j+1} - x_j}{r_j}, \quad \theta_j = \frac{t_{j+1} - t_j}{r_j^2}.$$

Here γ_j is the additive pressure gauge on the overlap, $\mathbf{q}_j$ is the countable vector $(\mathbf{q}_{C_j}(\psi))_{\psi \in \mathcal{T}}$, and $\mathbf{s}_j$ is the inverse-limit support coordinate: for every finite block ending at j , every first finite family of cylinders and tests, and every $m \geq 1$, it records a terminal-sequence index at least m , a composed observer, and a realization error at most m^{-1} in the strong velocity and pressure coordinates. Compatibility is restriction of the $(m+1)$ -st record to the first m coordinates. Concatenation requires equality of the outgoing window, incoming window, pressure gauge, and oriented current trace on the common overlap.

The topology of the marked records is strong local L^3 for velocity, strong local $L^{3/2}$ modulo the recorded gauge for pressure, weak local L^2 for gradients, weak-* convergence for the suitable defect, Euclidean convergence for (λ, ξ, θ) , and coordinatewise convergence of $\mathbf{q}$ and $\mathbf{s}$. The countable coordinates make this topology metrizable on bounded sets.

Definition C.14 (Uniformly record-bounded terminal extraction). A terminal extraction (U_n, P_n) on domains $\mathcal{D}_n$ is *uniformly record-bounded* if each pair is an exact parabolic rescaling of the same physical solution, the domains exhaust every compact cylinder used by the extraction, and

$$(C.22) \quad \sup_n \|U_n\|_{L^\infty(\mathcal{D}_n)} \leq M_*$$

for one $M_* < \infty$. The record-point sequence in [Theorem C.1](#) has this property with $M_* = 1$ by [\(C.3\)](#).

Definition C.15 (Marked regenerated-current cascade). A marked regenerated-current cascade at level $\eta > 0$ is an infinite concatenated sequence $(\mathbf{m}_j)_{j \geq 0}$ with the following properties. Write V_j for the velocity component of $\mathbf{w}_j$.

(R1) The cells are strictly nested and

$$(C.23) \quad 0 < \lambda_j \leq \frac{1}{4}, \quad \mathcal{O}_{Q_1}(V_j) \geq \eta.$$

Every finite block is simultaneously realized in one member of the original terminal sequence by the recorded composed observers.

(R2) With $w_j = r_j^{-1}$, let $\mathcal{J}_j$ be the complete oriented local-energy current through the stopping surface between C_j and C_{j+1} , including its endpoint energy trace, and let $\mathcal{P}_j \geq 0$ be viscous production plus suitable defect in that shell. The scale-critical shell dissipation is

$$(C.24) \quad \mathcal{D}_j^{\text{crit}} := w_j \mathcal{P}_j.$$

It is invariant under Navier–Stokes scaling because physical local-energy production has homogeneity one in length. There are $N_k \uparrow \infty$ and $\delta_D > 0$ such that

$$(C.25) \quad \frac{1}{N_k} \sum_{j=0}^{N_k-1} \mathcal{D}_j^{\text{crit}} \geq \delta_D \quad (k \geq 1).$$

- (R3) Every nonzero sequence-realized support profile of the remaining weighted current regenerates a child satisfying (C.23).
 (R4) The marked records have compact closure; in particular there are $\lambda_* > 0$ and $B < \infty$ such that

$$(C.26) \quad \lambda_* \leq \lambda_j \leq \frac{1}{4}, \quad |\xi_j| + |\theta_j| \leq B.$$

- (R5) The original terminal extraction is uniformly record-bounded in the sense of Definition C.14. Every finite initial block of length N has a simultaneous realization whose ungauged root velocity $U^{(N)}$ satisfies

$$(C.27) \quad \|U^{(N)}\|_{L^\infty(\mathcal{Q}_N)} \leq M_*.$$

Here $\mathcal{Q}_N$ contains the images, in root coordinates, of all unit cylinders in the block. The normalized velocity at every descendant is the exact parabolic rebase of $U^{(N)}$, up to an additive constant Galilean velocity. The realization accuracy on the terminal descendant may be chosen so that its velocity $V_N^{(N)}$ satisfies

$$(C.28) \quad \|V_N^{(N)} - V_N\|_{L^3(Q_1)} \leq \varepsilon_N, \quad \varepsilon_N \downarrow 0.$$

Proposition C.16 (Extraction of the marked alternative). *Consider an infinite nested sequence of target-retaining current events furnished by one finite-stage concentration–remainder construction from a uniformly record-bounded terminal extraction. Suppose that each event is accompanied by its actual successor witness and the coordinates in (C.20). Apply the following tests in order:*

- (E1) *if the Cesàro mean of the scale-critical shell dissipations has a subsequence tending to zero, select shells whose normalized dissipation tends to zero;*
- (E2) *if a nonzero sequence-realized support profile of the remaining current has no target-retaining child, select that profile at its first failed compactness, pressure, exterior, diffuse, affine, or critical-tail test;*
- (E3) *if the radii fail to tend to zero, select the compact non-collapsing scale limit; otherwise retain a subsequence with $r_{j+1}/r_j \leq 1/4$;*
- (E4) *if, on this subsequence, the ratios have no positive lower bound, the normalized observer increments are unbounded, or the velocity, pressure gauge, suitable-defect, current, or realization coordinates have no compact closure, select the corresponding scale-tail, exterior, pressure/noncompact, defect, or realization-loss profile.*

If none of these selected outputs occurs, the remaining events contain a marked regenerated-current cascade in the sense of Definition C.15. Thus failure of the compact marked cascade is an explicit first-failure output; it is not an assumption hidden in the conclusion.

Proof. Put $A_N = N^{-1} \sum_{j < N} \mathcal{D}_j^{\text{crit}}$. If $\liminf_N A_N = 0$, choose $N_k \rightarrow \infty$ with $A_{N_k} \rightarrow 0$. Since these dissipations are nonnegative, for every k some $j_k < N_k$ satisfies $\mathcal{D}_{j_k}^{\text{crit}} \leq A_{N_k}$; this is test (E1). Otherwise choose $0 < \delta_D < \liminf_N A_N$; then (C.25) holds on every sufficiently long initial block.

Starting from the first retained event, follow a target-retaining child of each nonzero current-support profile. If no such child exists, the first failed test is exactly (E2). If a child always exists, recursive selection gives an infinite realized chain. If its radii do not tend to zero, a subsequence has a positive limiting radius; after normalization this is the compact non-collapsing scale output in (E3). If the radii tend to zero, choose successively the first later radius not exceeding one quarter of the preceding selected radius. Relabeling this subsequence gives the upper bound in (C.23); finite simultaneous realization is preserved under passage to a subsequence and composition of the intervening observers.

On the separated chain, failure of any lower scale bound, observer bound, or coordinate compactness has a first coordinate in the countable cylinder and test exhaustion on which compactness fails. Selecting that coordinate gives the output in (E4). Diagonal compactness gives the compact closure and the bounds in (C.26). The retained activity floor and the regeneration property are closed because velocity convergence is strong in the fixed local L^3 topology and because the complete successor witness is one of the marks. The record bound and the inverse-limit realization coordinate give (R5) by Lemma C.19 below. Thus properties (R1)–(R5) hold, so the remaining chain is a marked regenerated-current cascade. $\square$

Lemma C.17 (Exact scale-critical shell balance). *For every finite initial block of a marked regenerated-current cascade,*

$$(C.29) \quad \sum_{j=0}^{N-1} \mathcal{D}_j^{\text{crit}} = w_0 \mathcal{J}_0 - w_{N-1} \mathcal{J}_N + \sum_{j=1}^{N-1} (w_j - w_{j-1}) \mathcal{J}_j.$$

Common parent–child currents cancel with opposite orientations. Viscous production and suitable defect remain nonnegative on the disjoint shell interiors.

Proof. Apply the suitable local energy identity, with its nonnegative defect included, to the disjoint regions between consecutive stopping surfaces. The normal trace on a common surface occurs once with each orientation and cancels. Writing the resulting shell identity as $\mathcal{P}_j = \mathcal{J}_j - \mathcal{J}_{j+1}$, multiplying by w_j , and summing gives

$$\sum_{j < N} w_j (\mathcal{J}_j - \mathcal{J}_{j+1}) = w_0 \mathcal{J}_0 - w_{N-1} \mathcal{J}_N + \sum_{j=1}^{N-1} (w_j - w_{j-1}) \mathcal{J}_j.$$

This is (C.29). The sign assertion follows because the dissipation and defect measures are restricted to disjoint interiors; they are not obtained by subtracting a parent localization from its children. $\square$

Lemma C.18 (Finite simultaneous realization). *Every finite set of marked events and every finite set of compact normalized cylinders on which their activity and current coordinates are tested are realized, with compatible pressure gauges, by one member of the original terminal extraction.*

Proof. This is precisely the finite-block clause in (R1). More explicitly, the inverse-limit coordinate $\mathfrak{s}_j$ records, for each block length and each finite cylinder-and-test family, one terminal-sequence index and the composed observer realizing all of those coordinates. Restriction compatibility of $\mathfrak{s}_j$ preserves the same index when coordinates are discarded, and the overlap coordinate γ_j makes the pressure representatives agree up to their recorded additive gauge. Hence all members of the prescribed finite family occur in one realization. A diagonal choice over the countable exhaustion gives the final assertion. $\square$

Lemma C.19 (The record bound is inherited by finite marked blocks). *Let the marked events be generated from a uniformly record-bounded terminal extraction. If every finite block has the inverse-limit realization coordinate of Definition C.13, then clause (R5) of Definition C.15 holds with the same constant M_* as in (C.22).*

Proof. Fix N . The inverse-limit coordinate supplies terminal-sequence indices realizing the first N edges on the finite collection of cylinders needed for that block, with error tending to zero in every recorded strong local velocity coordinate. Choose one such index far enough along the sequence that the terminal velocity error on Q_1 is at most N^{-1} , and call the corresponding extraction member $U^{(N)}$. This is the ungauged root field: Galilean constants are retained in the

composed observer and are not added to $U^{(N)}$. Since every image cylinder in the finite block lies in the domain of this realization, $\mathcal{Q}_N \subset \mathcal{D}_{k(N)}$, and hence

$$\|U^{(N)}\|_{L^\infty(\mathcal{Q}_N)} \leq \|U_{k(N)}\|_{L^\infty(\mathcal{D}_{k(N)})} \leq M_*.$$

Exact composition of translations, constant Galilean changes, and parabolic dilations gives the required descendant rebases. Taking $\varepsilon_N = N^{-1}$ proves (C.28) and therefore (R5). $\square$

Lemma C.20 (Quotient activity under a bounded parabolic rebase). *Let Q be a bounded spacetime cylinder, let U be defined on the image of an affine parabolic embedding $\Psi : Q \rightarrow \mathbb{R}^3 \times \mathbb{R}$, and suppose $\|U\|_{L^\infty(\Psi(Q))} \leq M$. For $0 < \lambda \leq 1$ and $h \in \mathbb{R}^3$, put*

$$(C.30) \quad V(y, s) := \lambda U(\Psi(y, s)) + h.$$

Then

$$(C.31) \quad \mathcal{O}_Q(V)^{1/3} \leq |Q|^{1/3} M \lambda.$$

Moreover, for any $V_1, V_2 \in L^3(Q)$,

$$(C.32) \quad \left| \mathcal{O}_Q(V_1)^{1/3} - \mathcal{O}_Q(V_2)^{1/3} \right| \leq \|V_1 - V_2\|_{L^3(Q)}.$$

Proof. Choose the constant $c = h$ in the infimum defining $\mathcal{O}_Q(V)$. Then

$$\mathcal{O}_Q(V) \leq \int_Q |V - h|^3 = \lambda^3 \int_Q |U \circ \Psi|^3 \leq |Q| M^3 \lambda^3,$$

which proves (C.31). For every constant c , Minkowski's inequality gives

$$\|V_1 - c\|_{L^3(Q)} \leq \|V_1 - V_2\|_{L^3(Q)} + \|V_2 - c\|_{L^3(Q)}.$$

Take the infimum over c , and then interchange V_1, V_2 , to obtain (C.32). $\square$

Lemma C.21 (Composition of marked descendant rebases). *For a finite concatenated block of marked successors, let*

$$(C.33) \quad \Lambda_N := \prod_{j=0}^{N-1} \lambda_j.$$

In every simultaneous realization of that block, its terminal normalized velocity has the form

$$(C.34) \quad V_N^{(N)}(y, s) = \Lambda_N U^{(N)}(\Psi_N(y, s)) + h_N \quad ((y, s) \in Q_1),$$

where $U^{(N)}$ is the ungauged root representative, h_N is a constant Galilean velocity, and $\Psi_N(Q_1) \subset \mathcal{Q}_N$. In particular, if the block satisfies (R5), then

$$(C.35) \quad \mathcal{O}_{Q_1}(V_N^{(N)})^{1/3} \leq |Q_1|^{1/3} M_* \Lambda_N.$$

Proof. For one edge, the normalized child velocity is obtained from the normalized parent by an exact translation, parabolic dilation of ratio λ_j , and a constant Galilean change. It therefore has the form

$$V_{j+1}(y, s) = \lambda_j V_j(\Psi_j(y, s)) + g_j,$$

where Ψ_j is the affine parabolic embedding determined by (ξ_j, θ_j) and the Galilean spatial drift. Composition multiplies the amplitude factors and again gives an affine parabolic embedding; all spatially constant velocity terms combine into one constant vector. Induction proves (C.34), with amplitude factor Λ_N . The cells are nested, so the image of the terminal unit cylinder lies in $\mathcal{Q}_N$. Apply Lemma C.20 with $M = M_*$ to obtain (C.35). $\square$

Theorem C.22 (Record-bounded marked cascades are impossible). *There is no marked regenerated-current cascade generated from a uniformly record-bounded terminal extraction with a positive activity floor.*

Proof. Suppose that a cascade exists at level $\eta > 0$. For each N , use (R5) to realize the first N edges with terminal velocity $V_N^{(N)}$ satisfying (C.28). By (R1), $\mathcal{O}_{Q_1}(V_N) \geq \eta$. On the other hand, Lemmas C.20 and C.21 give

$$\begin{aligned} \eta^{1/3} &\leq \mathcal{O}_{Q_1}(V_N)^{1/3} \\ &\leq \mathcal{O}_{Q_1}(V_N^{(N)})^{1/3} + \varepsilon_N \\ &\leq |Q_1|^{1/3} M_* \Lambda_N + \varepsilon_N. \end{aligned}$$

Since $\lambda_j \leq 1/4$, $\Lambda_N \leq 4^{-N} \rightarrow 0$. Letting $N \rightarrow \infty$ gives $\eta^{1/3} \leq 0$, a contradiction. $\square$

Lemma C.23 (Compact closed marked path space). *Let $(\mathbf{m}_j)_{j \geq 0}$ be a marked regenerated-current cascade. The closure of its shift orbit is a compact shift-invariant space X_{mark} . Concatenation and the activity floor are closed there. Every weak-* limit of the shift empirical measures is shift invariant and has zeroth marginal supported on $\{\mathcal{O}_{Q_1} \geq \eta\}$.*

Proof. Property (R4) gives relative compactness of the complete marked records, not merely of their profile coordinates. Take the closure in the countable product of the compact coordinate closures. The inverse-limit support coordinate is closed by Lemma C.18, so every point of this closure is still sequence-realized.

Concatenation is equality of outgoing and incoming overlap coordinates and is therefore closed. Strong local L^3 convergence makes the activity floor closed. Hence the shift-orbit closure is a compact shift-invariant space of concatenated marked paths. The endpoint calculation in the proof of Lemma C.10 gives shift invariance of every empirical weak-* limit, and the Portmanteau theorem preserves the activity floor. $\square$

Definition C.24 (Sequence-realized logarithmic critical tail). A sequence-realized logarithmic critical tail is a compatible projective family of annular limits on

$$\{(y, s) : e^{-L} < |y| < 1, -1 < s < 0\}, \quad L = 1, 2, \dots,$$

obtained from one diagonal subsequence of the original terminal extraction. The family includes velocity, pressure modulo its compatible additive gauge, the suitable defect, and all current coordinates in $\mathcal{T}$. It is nonzero when one fixed annular cross-section has quotient oscillation bounded below by a positive constant. Loss of tightness as $L \rightarrow \infty$, survival of a pressure or defect coordinate, and strong coherent convergence are, respectively, its diffuse, defect, and coherent alternatives.

Theorem C.25 (Marked recurrence produces a realized critical tail). *Every marked regenerated-current cascade produces a nonzero sequence-realized logarithmic critical-tail profile. More precisely, there are a recurrent path $p \in X_{\text{mark}}$ and integers $n_k \rightarrow \infty$ for which the same non-affine marked profile returns through observers whose total scale tends to zero. The associated annular blocks are simultaneously realized by the original terminal sequence.*

Proof. Let ϖ be a shift-invariant weak-* limit on X_{mark} . Poincaré recurrence gives a recurrent path in the support of ϖ , still satisfying the activity floor, and integers $n_k \rightarrow \infty$ such that

$$(C.36) \quad S^{n_k} p \rightarrow p \quad \text{in } X_{\text{mark}}.$$

This convergence includes velocity, pressure gauge, defect, current, observer, and realization coordinates.

Composing the first n observer maps gives

$$(C.37) \quad \Lambda_n = \prod_{j=0}^{n-1} \lambda_j,$$

$$(C.38) \quad X_n - X_0 = r_0 \sum_{j=0}^{n-1} \left(\prod_{i=0}^{j-1} \lambda_i \right) \xi_j,$$

$$(C.39) \quad T_n - T_0 = r_0^2 \sum_{j=0}^{n-1} \left(\prod_{i=0}^{j-1} \lambda_i^2 \right) \theta_j.$$

The bounds (C.26) make the last two series convergent, whereas

$$(C.40) \quad 0 < \Lambda_{n_k} \leq 4^{-n_k} \rightarrow 0.$$

After recentering once at the limits in (C.38)–(C.39), (C.36) is a return of the same marked profile at logarithmic displacements tending to infinity.

Retain the complete blocks between scales 1 and Λ_{n_k} . Their component shells have disjoint logarithmic interiors. By Lemma C.18, every finite family of annular velocity, pressure, defect, and current tests is realized by one member of the original terminal extraction. A diagonal extraction therefore gives a sequence-realized logarithmic annular state. It is nonzero because the recurrent cross-section retains $\mathcal{O}_{Q_1} \geq \eta$.

Loss of logarithmic tightness defines the diffuse critical-tail component. Under logarithmic tightness, failure of strong annular compactness or survival of a pressure or suitable-defect coordinate defines the defect component. In the remaining case the annular convergence is strong, the pressure gauges are compatible, and the limiting state is a coherent sequence-realized critical tail. These alternatives are exhaustive by weak-* compactness of the logarithmic mass, local compactness on each fixed annulus, and the dichotomy between zero and nonzero defect coordinates. In every case the output is the nonzero critical-tail profile asserted in the theorem. $\square$

Theorem C.26 (Exclusion of persistent regenerated current). *Alternative (N3) of hypothesis 11.7 is impossible. Every failure encountered before bounded marked realization is selected is an ordered output of alternative (N1); the bounded marked residual is excluded inside this paper.*

Proof. First apply Proposition C.16. Its tests (E1)–(E4) either give one of the ordered local outputs in (N1), or produce a marked regenerated-current cascade. The latter is impossible by Theorem C.22. Hence no residual case of (N3) remains. $\square$

Remark C.27 (Scope of the marked closure). The finite-depth estimate closes the record-bounded regenerated-current row before a logarithmic invariant law is needed. The quotient observable removes the accumulated constant Galilean velocity, while exact composition retains the product scale Λ_N . The required root bound is inherited from (C.22) by Lemma C.19. Pressure created by motion of the root remains in $\mathbf{q}_j$ and in the exact shell balance (C.29); it is not used in the amplitude estimate.

C.4. A single forward orbit. Construct one absolutely continuous chart on $[0, \infty)$, rather than independent charts on selected windows. Prove the repaired equation, $\Lambda(\tau_j) \rightarrow 0$, the chart composition law, and the existence of infinitely many unit windows with pairwise disjoint interiors satisfying (10.6). The construction must show that every reselection represents the same physical solution and preserves the quotient state. It must work both for bounded centers and for the escape-to-infinity alternative in case (b).

C.5. Compactness and persistence. Derive (10.5), strong local L^3 compactness, pressure compactness modulo (10.4), and passage of the fixed-cutoff $\mathcal{C}^\circ$ to the limit. Prove sequential compactness of the entire closure of the selected windows, not only subsequential compactness of the original sequence. Together with Section D.3, obtain strong convergence and positivity of the fixed-radius local weights in (A.6) when the optional adjoint diagnostic is retained. The orbit conclusions must hold with constants depending only on the normalized class $(\eta_0, \mathbf{M}, \nu)$.

The synthesis required for the single-orbit statement, including oscillation retention, transition coherence, and escape to spatial infinity, is specified in Sections C.1, C.2, C.4 and C.5.

APPENDIX D. GAUGE, PRESSURE, AND ADJOINT ESTIMATES

D.1. Scale–center normalization. Use the four-component map (7.8) and Jacobian (7.9) to solve for scale and center on every nondegenerate retained window. Prove uniqueness on the local slice, absolute continuity of X, Λ , and $a, b \in L^1_{\text{loc}}$. At degeneracy, the ordered selection in Section C.2 must either supply a new nondegenerate slice or leave the retained class. No moving condition $\bar{V}_m = 0$ is to be constructed; Galilean invariance is already built into (7.8), (9.4), and (10.2). When differentiating the gauge, include the derivatives of both fixed-cutoff means in (7.7).

D.2. Local pressure decomposition. For nested inner balls prove

$$P = P_{\text{loc}} + H, \quad -\Delta P_{\text{loc}} = \partial_i \partial_j (\zeta V_i V_j), \quad \Delta H = 0,$$

with compatible spatial means. Establish uniform Calderón–Zygmund bounds for P_{loc} , harmonic interior estimates for H , stability under chart transitions, and convergence in $L^{3/2}_{\text{loc}}$.

D.3. Normalized adjoint solutions. For the optional Perelman diagnostic, strengthen Lemma 8.2 by proving a uniform lower bound for Z , the Fisher estimate in (A.7), and compactness of the normalized density on padded windows. Use the covariant scale (8.7) and verify (8.6). For every entropy cutoff χ , choose a nested $\hat{\chi}$ equal to one near $\mathcal{A}_\chi$ and prove the Fisher bound required by (12.20). These estimates refine the geometric interpretation but are not inputs to the moving-cutoff contradiction.

D.4. Integrable transport errors. Construct a nondecreasing radius $R(\tau) \rightarrow \infty$ on the coherent orbit. Derive (12.25) directly from the suitable local energy inequality and prove that the diffusion-shell, convection, pressure, and translation components are summable on this schedule. Pressure must be handled by the local/harmonic decomposition of Section D.2, uniformly for every finite-valued moving radius above the scheduled lower bound. No adjoint normalization or terminal-data reset occurs in this hypothesis.

The four subsections of Appendix D specify the local bounds and repaired-chart estimates required by the covariant diagnostic and the moving-cutoff energy.

APPENDIX E. THE MOVING-CUTOFF ALTERNATIVE

The monotonicity calculation is already Theorem 12.4. This appendix must prove the global alternative in hypothesis 12.5. The required ingredients are the canonical pressure decomposition, a retained pressure-tail estimate, the tailored radius construction, compatible signed transition currents, and the recurrent treatment of nonpositive scale drift developed above.

E.1. Exact moving-cutoff identity. Starting from the suitable local energy inequality in the repaired equation, derive every term of (12.25), including the $\partial_\tau \Phi$ term and the signs of the two scale contributions. Prove the gauge invariance of the pressure flux and the favorable sign of the suitable energy-defect measure. This recovers Theorem 12.4 directly.

E.2. Pressure tails and the choice of radius. On successive unit windows, use critical tightness to choose radii $R_n \rightarrow \infty$ with a prescribed summable exterior L^3 tail. Prove uniform vanishing of the pressure flux for every moving radius above R_n , using the Calderón–Zygmund local part and harmonic pressure atlas separately. Interpolate the R_n monotonically and show that the diffusion-shell, convection, pressure, and translation summands of B_{mov} lie in L^1 . A failure must be assigned to a specific pressure, tightness, exterior, or modulation alternative before the retained moving-cutoff case is declared.

E.3. Transition and positive-scale terms. Begin with the chronologically ordered fixed-scale active cylinders furnished by the singular-germ extraction. Apply [Proposition 6.1](#). In the resulting exact chart $a_+ = 0$ almost everywhere, while the represented fields on every active cylinder are unchanged. Thus the positive scale-shell summand is absent; it is not estimated by its absolute value. Interpolate the center only in the inactive gaps and include its signed current in the compatible local-energy partition. Internal transition and center currents cancel under [Proposition F.1](#); a current surviving at the outer boundary is, by definition, exterior input and must be controlled by the pressure-tail radius construction.

What remains to be proved here is the PDE extraction statement needed before the interpolation can be invoked: singular contact must furnish the ordered fixed-scale cylinders with the same oscillatory target floor and with the compact bounds used on their normalized unit representations. Loss of one of those properties must be resolved by the concentration–remainder argument, not relabeled as a positive-scale estimate. In particular, failure of compactness must either give a further target-retaining parabolic rebase or a quantified exterior loss. Once this statement is established, no separate positive-scale branch remains.

E.4. Negative scale drift. Use the monotone chart, so $a \leq 0$. On a compact sequence of persistent drift windows, pass to an autonomous coefficient $a_* \leq 0$ and a constant translation coefficient after the usual time translation and diagonal compactness argument. Enlarge the two-sided limit to its compact covariant hull. If the oscillatory target persists on every minimal component, [Corollary 6.7](#) gives a contradiction. This closes the ordinary compact negative-drift branch by the identity [\(6.12\)](#), without interpreting $a_- \int |V|^2 \Phi$ as a dissipative budget.

For diffuse drift blocks, show directly that $a_-(V + y \cdot \nabla V) \rightarrow 0$ in $L^1 H_{\text{loc}}^{-1}$ on selected unit windows and that its compact local-energy pairings vanish. Any compact autonomous recurrent limit then has $a_* = 0$ and is covered by the same rigidity theorem. The only remaining case is failure to obtain an ordinary compact recurrent hull while retaining the target. The concentration–remainder extraction in [Remark 6.10](#) then produces a further active parabolic rebase, quantified exterior input, target loss, or a marked regenerated-current cascade. The last output is routed by [Theorem C.26](#). After this extraction is proved, the negative-drift term contributes no independent terminal branch and the moving-cutoff alternative follows without resetting adjoint terminal data.

APPENDIX F. CRITICAL FLUX–DISSIPATION BALANCE

This appendix isolates the single structural theorem required from the moving-cutoff analysis. The local energy equation first separates gross dissipation from signed transport. Compatible smooth localizations give an exact refinement identity. Cancellation of internal transport is a separate oriented-trace statement for disjoint spacetime cells; it is not a consequence of the partition-of-unity identity alone. Once such a disjoint decomposition is available, the remaining question is whether its outer scale-space boundary can feed the retained component at the same rate at which viscosity dissipates it. Strict inequality yields net dissipation. Equality is the

saturation branch and is treated by compactness and rigidity rather than by sharper estimates of the individual flux terms.

F.1. Compatible nested parabolic cells. Let $\{\psi_\alpha\}_{\alpha \in \mathcal{T}_0}$ be a finite family of nonnegative smooth local-energy tests on one represented time interval. It is compatible when every nonterminal test is the sum of its children. The family may be obtained from a smooth partition of a finite nested collection of parabolic cells; only the partition identity is used below.

F.2. Summed local energy identity.

Proposition F.1 (Refinement identity for local-energy tests). *For a compatible finite family, the sum of the signed currents of the terminal tests equals the signed current of the root test. The same equality holds separately for kinetic energy, dissipation, and suitable-energy defect. Consequently replacing a parent by its children changes none of the terms in the balance and introduces no additional pressure or convection contribution. The identity is invariant under $P \mapsto P + c(\tau)$.*

Proof. Iterate (11.4) from the leaves to the root. Each term in (11.2) is linear in the test and its derivatives, so the asserted equality holds term by term. Pressure-gauge invariance follows from $\int V \cdot \nabla \psi = 0$ for every compactly supported test. $\square$

This proposition identifies the aggregate root flux, but it neither removes that flux nor asserts that its inward part is dominated by dissipation. A genuine cancellation theorem requires a disjoint spacetime-cell formulation and compatible normal traces; even then, the outer boundary flux remains. Domination of that outer inward flux is the analytic content of flux domination.

F.3. Target-relative flux domination in the moving-cutoff branch.

Corollary F.2 (Aggregate critical balance under the moving-cutoff alternative). *Assume hypotheses 10.2 and 12.5. On a terminal tail, one of the following occurs:*

- (P1) *the orbit enters an exterior, pressure, modulation, scale-collapse, scale-rigid, dissipation-divergence, or terminal Type I alternative covered by alternative (C1);*
- (P2) *the critical localized-energy balance has dominated inward flux and may be chosen as*

$$(F.1) \quad \mathcal{K} = \mu_{\text{NS}}, \quad \mathcal{D} = \mathfrak{D}_{\text{NS}}, \quad \mathcal{F}^+ = \mathcal{R}^+ = 0, \quad \mu_{\text{cov}} = \mathfrak{D}_{\text{NS}}.$$

Moreover, on every normalized window in the compact retained closure,

$$(F.2) \quad \Delta_{\text{crit}}(\mathfrak{w}) := \frac{\nu}{2} \int_0^1 \int \zeta^2 |\nabla V|^2 dy d\tau$$

is sequentially lower semicontinuous, $\mu_{\text{cov}}(I_j) \geq \Delta_{\text{crit}}(\mathfrak{w}_j)$, and $\Delta_{\text{crit}} = 0$ implies zero Galilean-invariant target oscillation.

Proof. Alternative (C1) is precisely (P1). In alternative (C2), the corrected moving-cutoff identity gives

$$\frac{d}{d\tau} \mu_{\text{NS}} + \frac{1}{2} \tilde{\mathfrak{D}}_R \leq 0,$$

which is (11.5) with the choices (F.1); flux domination is immediate. Because $\Phi = 1$ on $\text{supp } \zeta$,

$$\mathfrak{D}_{\text{NS}}(I_j) \geq \frac{\nu}{2} \int_{I_j} \int \zeta^2 |\nabla V|^2 = \Delta_{\text{crit}}(\mathfrak{w}_j).$$

Lower semicontinuity is (A.8). Finally, Theorem G.1 shows that vanishing of the right side forces a spatially constant limit on the retained connected core and hence zero target oscillation. This proves (P2). $\square$

The corollary is deliberately conditional on [hypothesis 12.5](#). The direct proof of that hypothesis is the concentration–remainder, pressure–tail, and recurrent–realization problem stated in [Appendix E](#).

Remark F.3 (Saturation and scale-relative limits). Suppose a saturated subsequence also has an autonomous represented limit (V_*, P_*, a_*, b_*) . Passing to the limit in (5.4) gives

$$-\nu \Delta V_* + (V_* \cdot \nabla) V_* + \nabla P_* = a_*(V_* + y \cdot \nabla V_*) + b_* \cdot \nabla V_*.$$

This is the scale–translation relative-equilibrium equation of the autonomous limit. Saturation also gives $\nabla V_* = 0$ on the retained connected core by (F.2). Hence its Galilean-reduced target oscillation vanishes. The proof needs only this local equality classification, not a whole-space classification of every solution of the displayed stationary equation.

F.4. Persistence under passage to smaller scales. The coherent-orbit hypothesis already supplies infinitely many ordered target-retaining windows on one represented orbit. Applying [Corollary F.2](#) gives alternative (N1) or the flux-dominated balance in alternative (N2). In the latter case [Proposition 11.3](#) proves finiteness, while (F.2) supplies the required window deficit. Thus [hypothesis 12.5](#) implies [hypothesis 11.7](#) without introducing a scale-independent bound for the physical Stokes action.

The precise remaining PDE theorem is the same dichotomy with $\mathcal{F}^+$ equal to the inward part of the aggregate signed scale-space current of [Proposition F.1](#). One must prove $\mathcal{F}^+ \leq \mathcal{D}$ after the exterior and modulation alternatives are removed and prove that equality has the zero-oscillation conclusion of (F.2). The finite-cell cancellation proposition supplies neither assertion by itself.

APPENDIX G. TARGET-VISIBLE COERCIVITY AND ZERO-DISSIPATION LIMITS

The dissipation need not control every material direction uniformly. The singular target detects only the Galilean-invariant oscillatory component of the velocity. Compactness and the local Dirichlet form are coercive on that quotient: a sequence on which the form vanishes converges to a spatially constant velocity and therefore disappears from the target. This gives the required coercivity-or-reduction alternative without estimating the rotation of the material eigenframe.

G.1. Zero-dissipation limits are target-null.

Theorem G.1 (Target-nullity of zero-dissipation limits). *Let $\mathfrak{w}_j = (V_j, P_j, a_j, b_j; [0, 1])$ be normalized windows in the compact closure specified by [hypothesis 10.2](#). If*

$$(G.1) \quad \int_0^1 \int_{\mathbb{R}^3} \zeta^2 |\nabla V_j|^2 dy d\tau \longrightarrow 0,$$

then, after passage to a subsequence,

$$(G.2) \quad V_j \longrightarrow c(\tau) \quad \text{strongly in } L^3(\text{supp } \zeta \times [0, 1])$$

for a spatially constant vector field c . Consequently

$$(G.3) \quad \mathcal{C}_T^\circ(c; [0, 1]) = 0.$$

In particular, no sequence satisfying the fixed target floor (10.6) can satisfy (G.1).

Proof. The compactness in (10.7) gives, after extraction,

$$V_j \rightarrow V_\infty \quad \text{strongly in } L^3(\text{supp } \zeta \times [0, 1]), \quad \nabla V_j \rightharpoonup \nabla V_\infty \quad \text{weakly in } L_{\text{loc}}^2.$$

Weak lower semicontinuity and (G.1) imply

$$\int_0^1 \int \zeta^2 |\nabla V_\infty|^2 dy d\tau = 0.$$

The set $\{\zeta > 0\}$ is connected, so for almost every τ there is a vector $c(\tau)$ with $V_\infty(y, \tau) = c(\tau)$ on that set. This proves (G.2). The weighted mean of a constant is the same constant, hence the Galilean-invariant oscillatory integrand vanishes and (G.3) follows. Strong L^3 -convergence passes the fixed-cutoff oscillatory functional to the limit by Proposition A.2. The positive target floor would therefore pass to a limit with zero target value, which is impossible. $\square$

G.2. Coercivity on the target quotient.

Theorem G.2 (Target-visible coercivity-or-reduction). *Let $\mathcal{K}$ be the compact closure of the selected windows of one canonical orbit, and let*

$$\mathcal{K}_{\eta_0} := \{\mathfrak{w} \in \mathcal{K} : \mathcal{C}_\Gamma^\circ(V; [0, 1]) \geq \eta_0\}.$$

Exactly one of the following alternatives holds:

- (i) $\mathcal{K}_{\eta_0} = \emptyset$, so the compact orbit closure has lost the singular target in the Galilean quotient;
- (ii) $\mathcal{K}_{\eta_0} \neq \emptyset$, and there exists $\delta_\nu > 0$ such that

$$(G.4) \quad \nu \int_0^1 \int \zeta^2 |\nabla V|^2 dy d\tau \geq \delta_\nu \quad (\mathfrak{w} \in \mathcal{K}_{\eta_0}).$$

For the canonical target-retaining orbit, alternative (i) is excluded by (10.6); hence (G.4) holds on every selected window.

Proof. The target functional is continuous under the strong local L^3 convergence, so $\mathcal{K}_{\eta_0}$ is compact. Assume first that it is nonempty. If the infimum of the left side of (G.4) were zero, a minimizing sequence would satisfy (G.1). By Theorem G.1, its limit would have zero Galilean-invariant oscillation, contradicting membership in $\mathcal{K}_{\eta_0}$. The infimum is therefore positive. If this set is empty, alternative (i) holds; otherwise the positive infimum gives alternative (ii). $\square$

The preceding theorem is the material reduction needed by the regularity argument. The null space consists of spatially constant velocities on the retained connected core, which are precisely the local Galilean directions removed by the target observable. Material stretching, eigenvalue crossing, and rotation of compressed directions may be arbitrarily large inside this null class; none remains visible to the singular target.

G.3. Divergence on a persistent target sequence.

Corollary G.3 (Divergence of normalized target-visible dissipation). *For the infinite pairwise disjoint target-retaining windows of a canonical orbit,*

$$(G.5) \quad \sum_{j=1}^{\infty} \nu \int_{I_j} \int \zeta^2 |\nabla V|^2 dy d\tau = \infty.$$

If a nonnegative measure μ satisfies

$$(G.6) \quad \mu(I_j) \geq c_0 \nu \int_{I_j} \int \zeta^2 |\nabla V|^2 dy d\tau, \quad c_0 > 0,$$

then μ cannot have finite mass on the terminal tail.

Proof. Every summand in (G.5) is at least δ_ν by Theorem G.2; there are infinitely many windows. If (G.6) holds, countable additivity on the disjoint half-open windows gives

$$\mu([\tau_0, \infty)) \geq \sum_j \mu(I_j) \geq c_0 \sum_j \nu \int_{I_j} \int \zeta^2 |\nabla V|^2 dy d\tau = \infty.$$

$\square$

For the corrected moving-cutoff production, (G.6) holds with $c_0 = 1/2$ because $\Phi = 1$ on $\text{supp } \zeta$. Thus this realization needs no estimate of the material condition number and no classification of rotating eigendirections. A different physical measure may carry scale-dependent coefficients; its lower-bound series must then be proved divergent separately. The critical localized-energy balance avoids that loss because its inward feeding and dissipation are compared before returning to physical Stokes units.

APPENDIX H. LOCALIZED EQUALITY AND THE ROLE OF THE GAUSSIAN

For the moving-cutoff realization, the proof of Theorem 12.8 uses only the coercive fixed-cutoff Dirichlet floor and connectedness of $\{\zeta > 0\}$. These properties are closed under the convergence already required by the orbit hypothesis. The self-contained signed-current realization has a larger apparent equality class; its reduction to target-null states is the content of Appendix G.

The Gaussian statement has a separate and more limited role. For a smooth positive whole-space probability density, vanishing of $\mathfrak{D}_G$ gives $\nabla^2 f = (2\nu\theta)^{-1}I$, and normalization and integrability then give the heat kernel (8.18) up to translation. A localized Dirichlet adjoint density is not that whole-space Gaussian. The manuscript therefore uses Gaussian rigidity only as the equality model for the formal adjoint entropy, not as a premise in the local oscillation contradiction.

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